Vrije Universiteit Amsterdam

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Implementation-Risk Differentials in Hydrogen Technology Pathways

A Cross-Jurisdictional Causal Evaluation of Carrot-Policy Mechanisms

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Abstract

Despite over USD 1.3 trillion in announced subsidies and 422 GW of project pipeline, only 7% of 2023 announced green hydrogen capacity reached scheduled Final Investment Decision [66]. This thesis provides the first cross-jurisdictional causal evaluation of why low-carbon hydrogen projects survive or fail, and which policy mechanisms move the needle.

Using a project-level dataset of 1,354 Blue and Green hydrogen projects (S&P Global, 2010–2024) with 367 documented failures, this research makes four contributions. *First*, it identifies a previously under-appreciated mechanism: pre-FID offtake commitments reduce project-failure probability by 11–13 percentage points across five independent identification strategies, with Oster (2019) selection-on-unobservables sensitivity $\delta_{\text{null}} = 20.23$ indicating exceptional robustness. *Second*, it evaluates four carrot-policy types (US Section 45Q, EU Innovation Fund, UK Track-1, China 14th Five-Year Plan) via modern difference-in-differences estimators [83, 21], with Rambachan-Roth (2023) honest sensitivity bounds revealing heterogeneous causal-identification strength: only China's state-coordinated mandate survives the strictest robustness checks ($M^* = 1.5$), while US, EU, and UK estimates are sensitivity-bounded. *Third*, it documents a structural break in the policy-effect coefficient around 2020 via three time-varying-parameter estimators (threshold, AR(1) state-space, random walk), consistent with post-announcement-boom speculative entry. *Fourth*, it grounds these findings in a Dixit-Pindyck (1994) real-options framework that distinguishes V/I-boost mechanisms (output credits, capex grants) from σ-attack mechanisms (offtake mandates, cluster tenders).

Counterfactual scenario analysis quantifies external validity: an EU sector-optimal carrot mix would deliver an estimated +113 additional FIDs, +7.83 Mt/y additional CO_2 capture, and +1.76 Mt/y additional H_2 output over a three-year horizon (95% bootstrap CI: +82 to +141 FIDs). These findings are externally corroborated by the 2024–2026 wave of high-profile cancellations: ArcelorMittal's withdrawal from Bremen and Eisenhüttenstadt despite EUR 1.3B in committed EU Innovation Fund subsidies, BP's exits from HyGreen Teesside and H2Teesside, and the September 2025 withdrawal of seven EU Hydrogen Bank auction winners representing 1.88 GW of combined capacity.

Keywords: hydrogen, causal inference, difference-in-differences, real options, mechanism

design, policy evaluation, survival analysis, sensitivity analysis.
JEL classification: C21, C23, Q42, Q48, Q54, H23.

Contents

Chapter 1

Introduction

1.1 Motivation: the implementation gap

In 2024, more than sixty major green hydrogen projects were cancelled globally, removing approximately 4.9 million tonnes per year of would-have-been capacity from public 2030 timelines. BP exited its Australian, Omani, and UK clean hydrogen portfolio. Air Products halted three large US projects. ArcelorMittal walked away from its Bremen and Eisenhüttenstadt plants despite EUR 1.3 billion in committed European Innovation Fund subsidies. In September 2025, seven winners of the EU Hydrogen Bank auction — representing a combined 1.88 GW of electrolysis capacity — withdrew from grant negotiations before the security-deposit deadline.[1]

This cancellation wave exposes what [66] term the *green hydrogen implementation gap*: the difference between announced project ambition and realised capacity. Tracking 190 projects over three years, the authors document that only 7% of 2023 announced capacity reached scheduled Final Investment Decision (FID). Yet the announced pipeline continues to grow — the global project portfolio nearly tripled to 422 GW over the same period — creating a widening discrepancy between political ambition and on-the-ground delivery.

The policy response has been substantial. The US Inflation Reduction Act of 2022 introduced a production-tax-credit-style subsidy (Section 45Q) of up to $85 per tonne of captured CO_2. The European Union's Innovation Fund disburses over EUR 10 billion in capex grants. The United Kingdom launched the Hydrogen Allocation Round 1 (HAR-1) and Track-1 cluster-tender architecture. China's 14th Five-Year Plan (2021–2025), succeeded by the 15th Five-Year Plan, elevates hydrogen to a strategic priority within national industrial policy. These four policy archetypes — output credit, capex grant, cluster tender, and state-coordinated mandate — represent distinct mechanism-design choices, but their relative effectiveness in moving projects from announcement to FID has not been comparatively evaluated.

[1] Sources: industry reporting compiled in [30, 53, 22]; corroborated by S&P Global Commodity Insights coverage.

1.2 Research questions

This thesis addresses three primary research questions:

RQ1. Which carrot-policy mechanisms causally reduce project failure probability, and by how much? Specifically, do the four policy archetypes identified above produce statistically and economically significant reductions in project cancellation or on-hold status, and how robust are these estimates to alternative identification strategies and parallel-trends violations?

RQ2. Through which channels do effective policies operate? The Dixit-Pindyck (1994) real-options literature suggests two distinct pathways: subsidies can either raise the value-to-investment ratio V/I (making FID more attractive at a fixed uncertainty level) or attack the revenue-volatility component σ (lowering the FID threshold itself). Can we empirically distinguish these channels using sector-level heterogeneity in policy effects?

RQ3. What is the role of pre-FID offtake commitments? The 2024–2026 cancellation pattern strongly suggests that subsidies without secured offtake are insufficient. Can we identify a causal effect of pre-FID offtake-commitment on project survival, and does its magnitude approximate or exceed the effects of formal carrot-policies?

A fourth, supplementary question concerns external validity: **RQ4. What is the quantitative impact of plausible counterfactual policy reforms?** Using estimated treatment effects, we construct five counterfactual scenarios with bootstrap-validated confidence intervals to inform ongoing policy debate.

1.3 Contribution

This thesis advances the literature on hydrogen-policy evaluation in four ways.

Empirical. It provides the first cross-jurisdictional causal evaluation of carrot-policy mechanisms for low-carbon hydrogen project survival. The four policy comparisons (US 45Q, EU Innovation Fund, UK Track-1, China FYP) are estimated within a unified econometric framework, allowing direct comparison of mechanism effectiveness. Beyond formal policy, it identifies and quantifies the previously under-recognised role of pre-FID offtake commitments.

Methodological. It combines three complementary identification strategies that, to our knowledge, have not previously been jointly applied at this scope in the hydrogen implementation-risk literature: (i) modern difference-in-differences estimators that address heterogeneous timing (Sun-Abraham 2021; Borusyak-Jaravel-Spiess 2024); (ii) time-varying-parameter difference-in-differences via three state-space specifications (threshold, AR(1), random walk) to detect structural breaks in policy effectiveness; and (iii) Rambachan-Roth (2023) honest sensitivity bounds to assess robustness to parallel-trends violations. The convergence of point estimates across methods, paired with transparent reporting of method-specific robustness, exemplifies what the literature terms *layered transparency* [72].

Theoretical. It develops a Dixit-Pindyck (1994) real-options framework that characterises when each carrot-mechanism type dominates, as a function of project-level revenue uncertainty σ and value-to-investment ratio V/I. The framework yields testable cross-sector predictions — σ-attack mechanisms should be more effective in high-volatility sectors (power and heat, transport) than in low-volatility sectors (chemical, refinery) — which are then validated empirically.

Policy. It quantifies counterfactual scenarios with concrete numerical implications: an EU-wide hard-link between Innovation Fund eligibility and demonstrable offtake-commitments could deliver an estimated +83 additional FIDs and +858 kt/y of additional green hydrogen capacity over a three-year horizon, and a full sector-optimal carrot mix could add +113 FIDs, +7.83 Mt/y of CO_2 capture, and +1.76 Mt/y of H_2 output.

1.4 Strength of claims: a reader's guide to the identification hierarchy

The empirical and theoretical contributions of this dissertation occupy several different levels of identification status, and the strength with which any individual finding is asserted is conditioned on its identification level. Chapter 5 (Section 5.1) develops the formal four-level taxonomy adopted throughout the thesis, drawing on the potential-outcomes framework of [78, 44, 52], the ladder-of-causation translation of [69, 51], the partial-identification programme of [62], and the structural/reduced-form distinction of [43]. This section provides the reading guide: a plain-language map of the four levels and an indication of which results in subsequent chapters operate at which level, so that the reader can calibrate the strength of each claim as it is encountered.

The four levels of the identification hierarchy — L1 descriptive, L2 predictive, L3 quasi-causal (with point-identified L3.a and partially-identified L3.b sub-levels), and L4 structural — are situated within the Imbens-tradition disciplined-identification programme [50], as expressed within the Tinbergen-tradition treatment of observational econometric work. The epistemic position adopted throughout the dissertation aligns with the framing of [15] that observational identification is a matter of *mitigating* estimation risk rather than eliminating it.

L1 — Descriptive. Patterns documented in the observed data without invoking a counterfactual or interventional comparison: Kaplan-Meier survival curves by technology and jurisdiction, multistate transition counts across project-lifecycle phases, cumulative cancellation rate tables. L1 findings establish stylised facts that motivate higher-level analyses but are not themselves causal claims. The reader can take L1 results as the floor of identification: they require only accurate data measurement.

L2 — Predictive. Out-of-sample forecast performance of a fitted statistical model, independent of any causal interpretation of the model's parameters. The principal L2 result of the thesis is the Diebold-Mariano-Harvey-Leybourne-Newbold forecast comparison of Chapter 7 [31, 42], in which the score-driven specification dominates constant-parameter and parameter-driven alternatives on out-of-sample log-loss, with statistical significance assessed via the Hansen-Lunde-Nason Model Confidence Set procedure [40]. L2 claims are orthogonal to causal identification: a model may forecast well without identifying a structural parameter.

L3 — Quasi-causal. Identification of an average treatment effect on the treated (ATT) or related counterfactual functional, under explicit identifying assumptions on a non-randomised treatment-assignment mechanism. The hierarchy distinguishes:

- **L3.a Point identification.** ATT identified to a single value under canonical assumptions: TWFE difference-in-differences with strict parallel trends and the modern interaction-weighted estimators of [83, 21, 23], conditional-independence matching estimators such as IPWRA. The four carrot-policy effects of Chapter 6, the offtake-commitment effect of Chapter 8, and the Inflation-Reduction-Act event-window corroboration documented in companion Paper 2 [79] are L3.a claims.

- **L3.b Partial identification.** ATT bounded under weakened or set-valued assumptions: honest difference-in-differences bounds under restricted parallel-trends violations [72], Oster bounds under restricted selection on unobservables [68], partial-identification analysis under endogenous selection [62].

L3 claims are asserted with their multi-estimator robustness profile and their identification caveat; they are not asserted as definitively structurally identified.

L4 — Structural. Identification of parameters of an explicit generative model that admits counterfactual computation under interventions not observed in the sample. The Bayesian state-space carbon-conditional hazard model with GAS-driven $\beta_{\text{int}}(t)$ in Chapter 7 is the principal L4 result. The five-channel real-options framework of Chapter 3 (μ, σ, ρ, η, κ) and the credibility-conditional threshold Proposition 3.2 are L4-level organising claims; three formal identification tests (Appendix A.15) demonstrate that their constituent channels are not empirically structurally separable in the observational sample available, and the framework therefore functions as a *disciplined interpretive layer* for empirical signatures rather than as a directly identified causal-channel decomposition. The counterfactual scenarios of Chapter 9 are L4 *extrapolations* conditional on structural invariance assumptions made explicit at the point of construction; they serve as decision-relevant illustrations under explicit assumptions rather than as point predictions, and are vulnerable to the [61] critique if read as forecasts.

The discipline this hierarchy imposes is consequential. The principal substantive findings of the dissertation are established across multiple levels: the carrot-policy effects of Chapter 6 are identified at L3.a, bounded at L3.b under [72] honest sensitivity, triangulated against L1 cancellation-rate patterns, and structurally rationalised by the L4 GAS-TVP detection of an intensifying carbon-conditional mechanism. The offtake-commitment effect of Chapter 8 is identified at L3.a across five matching strategies, bounded at L3.b, and substantively rationalised at L4 through the σ-channel of the real-options framework. The 2020 structural break detected by the L4 GAS-TVP specification is supported at L2 through the Diebold-Mariano forecast superiority of the time-varying specification over constant-parameter and parameter-driven alternatives. Robustness *across* levels of the hierarchy is the strongest form of evidence this dissertation offers; single-level claims are reported with calibrated language.

Table 1.1 summarises the four levels of the identification hierarchy in a single-glance form. When the reader encounters a finding reported with strong language in the chapters that follow, the underlying claim is identified at L1, L2, or L3.a with multi-estimator convergence. When a finding is reported with cautious or qualified language, the underlying claim is at L3.b (partial identification) or L4 (with the structural-invariance caveat made explicit). Chapter 10 returns to the hierarchy when discussing the regime-instability findings as substantive contributions in their own right rather than as defects of identification.

1.5 Structure of the thesis

Chapter 2 reviews the existing literature on hydrogen-policy evaluation, real-options approaches to irreversible investment, and modern causal-inference advances in difference-in-differences. Chapter 3 develops the theoretical framework, formalising the four mechanism types within a real-options model. Chapter 4 describes the project-level S&P

Table 1.1: Scope of claims across the identification hierarchy: reader-navigation aid.

Level	Claim type	Principal examples in the dissertation	Strength of assertion
L1	Descriptive	Kaplan–Meier survival curves; multistate transition counts; cumulative cancellation-rate tables	Strong; requires only data accuracy
L2	Predictive	Diebold–Mariano–Harvey–Leybourne–Newbold forecast test (Ch. 7); Hansen–Lunde–Nason Model Confidence Set	Strong; out-of-sample validation
L3.a	Quasi-causal (point)	Modern DiD (TWFE / Sun–Abraham / Borusyak–Jaravel–Spiess) on carrot policies (Ch. 6); IPWRA matching on offtake commitment (Ch. 8); 45V three-pillars DDD	Conditionally strong; multi-estimator triangulation under parallel-trends or conditional-independence
L3.b	Quasi-causal (partial)	Rambachan–Roth honest-DiD bounds; Oster selection-on-unobservables sensitivity	Cautiously strong; bounded violation
L4	Structural	GAS-TVP $\beta_{\text{int}}(t)$ trajectory (Ch. 7); five-channel real-options framework (Ch. 3); counterfactual scenarios (Ch. 9)	Cautious; structural-invariance caveat; interpretive layer where mechanism channels are observationally entangled

Global dataset, complementary macro variables, and the construction of the analysis panel. Chapter 5 presents the econometric methodology, including the rationale for combining static DiD, time-varying DiD, and sensitivity-based identification. Chapters 6–8 present the empirical results. Chapter 9 translates these into counterfactual scenarios. Chapter 10 discusses interpretation, limitations, and theoretical implications. Chapter 11 concludes with policy recommendations and avenues for further research.

Chapter 2

Literature Review

This chapter situates the thesis within three intersecting literatures: (i) the empirical evaluation of clean-energy policy, with a focus on hydrogen; (ii) the real-options theory of irreversible investment; and (iii) recent advances in causal inference, particularly modern difference-in-differences and honest sensitivity analysis.

2.1 Empirical evaluation of hydrogen and clean-energy policy

2.1.1 The hydrogen implementation gap

The empirical literature on hydrogen-project realisation is dominated by descriptive tracking of capacity announcements versus delivered FIDs. [66] construct what they term the *ambition gap* (the difference between announced 2030 capacity and 1.5°C-consistent demand) and the *implementation gap* (the difference between announced and realised capacity at scheduled launch dates). Tracking 190 projects over three years, they document a 2023 implementation gap of approximately 93%: only 7% of capacity announcements were on schedule. Their projection requires global subsidies of US$1.3 trillion (range: $0.8–2.6 trillion) to realise the announced project pipeline, far exceeding announced subsidies.

[84] provide a complementary cost-competitiveness analysis. Under current natural-gas prices, green hydrogen is initially more than eight times as expensive as natural gas; even under an ambitious carbon-price pathway, it remains 3.5 times more expensive in 2024, with cost parity not reached before 2045. These cost gaps imply that voluntary uptake without policy support is unlikely.

[49] and the IEA [54] provide industry-side tracking with broader capacity coverage but less methodological rigour. The S&P Global Commodity Insights team maintains the dataset used in this thesis, including project-level status updates that have not been previously exploited for causal evaluation.

2.1.2 Policy-evaluation studies

Direct empirical evaluation of specific hydrogen-policy mechanisms remains limited. [65] simulate the EU Innovation Fund's projected impact on industrial hydrogen adoption but rely on bottom-up cost modelling rather than ex-post causal estimates. [73] analyse the additionality and temporal-matching requirements of the US 45V production credit but stop short of project-survival outcomes. The closest predecessors to the present analysis are [48] and [60], who apply real-options frameworks to specific electrolysis and maritime-shipping cases but do not exploit cross-jurisdictional variation.

To the author's knowledge, no prior study comparatively evaluates the four carrot-policy archetypes considered here within a unified causal framework.

2.1.3 Recent developments (2024–2026): the cancellation wave

The empirical literature on hydrogen-project realisation has been overtaken by industry events since 2024. The [56] *Global Hydrogen Review 2025* reports that the announced 2030 production capacity declined for the first time since the database's inception: from 49 Mtpa in the 2024 review to 37 Mtpa in 2025, a decline of nearly one-quarter in a single calendar year. Electrolysis projects accounted for more than 80% of the total drop. The IEA attributes the recalibration to high production costs (renewable hydrogen one-and-a-half to six times more costly than unabated fossil-based production), uncertain demand signals, financing hurdles, delays to incentive disbursement (notably under US 45V), regulatory uncertainty, and operational complications.

[66] document the same pattern from a different methodological angle, finding that the historical realisation rate of announced low-emissions hydrogen projects fell from approximately 7% historically to 4% by 2024. The implication is that the industry has entered a regime change, and contemporaneous empirical work on hydrogen policy must accommodate this structural break rather than treating the 2020–2023 announcement era as representative of long-run dynamics.

The present thesis is positioned to contribute to this evolving literature by providing the first systematic econometric quantification of which policy and contracting mechanisms differentially affect project survival, conditional on this regime change. The S&P Global database snapshot of March 2026 used here captures the cancellation wave's complete cross-section; the carrot-policy and offtake-effect estimates of Chapters 6 and 8 are therefore identified on the most policy-relevant variation period currently observable.

2.1.4 Adjacent literature: solar and wind policy evaluation

The broader clean-energy policy-evaluation literature provides useful methodological precedent. [12] use a panel difference-in-differences design to identify solar-subsidy effects on installed capacity, finding heterogeneous effects across US states. [4] review feed-in-tariff versus renewable-portfolio-standard designs, concluding that mechanism design matters as much as subsidy intensity — a finding that motivates the present focus on *mechanism-type* rather than *subsidy-magnitude* comparison. [71] examine the patenting response to clean-energy R&D subsidies, finding pronounced country-specific heterogeneity that anticipates our cross-jurisdictional approach.

2.2 Real options and irreversible investment

2.2.1 The Dixit-Pindyck framework

The canonical framework for analysing irreversible investment under uncertainty is [32]. A project of present value V (stochastic) and sunk cost I (fixed) has positive option value to delay investment whenever the underlying V_t follows geometric Brownian motion with non-zero volatility σ. The optimal exercise (FID) rule is to invest when $V \geq V^*$, where

$$\frac{V^*}{I} = \frac{\beta_1}{\beta_1 - 1}, \quad \beta_1 = \frac{1}{2} - \frac{r-\delta}{\sigma^2} + \sqrt{\left(\frac{r-\delta}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r}{\sigma^2}} \tag{2.1}$$

with r the risk-free rate and δ the convenience-yield-like dividend. Higher σ raises V^*/I monotonically: greater uncertainty makes waiting more valuable, suppressing investment.

2.2.2 Applications to energy investment

[70] provides the foundational application of real-options theory to energy. [36] apply the framework to carbon-pricing uncertainty in power-sector investment, showing that policy-volatility itself can suppress investment even when expected returns are positive. [11] document that wind-investment cycles are driven primarily by policy-uncertainty timing rather than by underlying renewable economics.

For hydrogen specifically, [48] apply a binomial-tree real-options model to electrolyser projects, finding that early subsidies can shift optimal-exercise timing forward by 2–4 years. [60] extend the framework to maritime ammonia, demonstrating that demand-aggregation can functionally reduce σ.

Our theoretical contribution (Chapter 3) extends this literature by formalising the mechanism-design implication: different policy types attack different parameters of the Dixit-Pindyck framework, and their relative effectiveness depends on the sector-specific $(V/I, \sigma)$ distribution.

2.3 Causal inference: difference-in-differences and beyond

2.3.1 The two-way fixed effects (TWFE) problem

The classical difference-in-differences (DiD) estimator with two-way fixed effects [8] has been the workhorse of policy-evaluation research for four decades. However, recent literature has documented serious identification problems when treatment timing is staggered and effects are heterogeneous across cohorts [37, 27, 23]. Under heterogeneous effects, TWFE can produce coefficients with the wrong sign relative to any individual treatment effect.

2.3.2 Modern DiD estimators

[83] develop an interaction-weighted estimator that uses only never-treated or last-treated units as controls, eliminating the "forbidden comparison" problem of staggered TWFE. [23] (CS) propose group-time average treatment effects $ATT(g, t)$ that can be aggregated under various weighting schemes. [21] (BJS) develop an imputation estimator that, under a parallel-trends assumption on untreated potential outcomes, achieves efficiency bounds and is robust to heterogeneous effects. [28] characterise intertemporal treatment effects in heterogeneous-treatment settings, providing identification guarantees under weaker conditions than CS or BJS.

[77] synthesise these developments in the most cited recent review of the DiD literature, classifying estimators by their relaxations of canonical assumptions (multiple periods, heterogeneous timing, parallel-trends violations, alternative inference). The recommendations from this synthesis directly motivate the present thesis's triangulation strategy: rather than committing to a single estimator, robust empirical claims should rest on agreement across multiple specifications that relax different canonical assumptions.

A complementary methodological concern is the functional-form sensitivity of parallel trends. [76] establish that parallel trends is functional-form-invariant if and only if it holds across the entire distribution of untreated potential outcomes, which is a substantially stronger restriction than the unconditional mean version commonly invoked. This insight has direct relevance for absorbing-state outcomes such as project cancellation: where the cumulative cancellation rate mechanically saturates as time progresses, parallel trends in means may fail by construction, motivating the hazard-rate reformulation introduced by [29] and applied in Appendix A.8 of this thesis.

For settings where the parallel-trends assumption is suspect even under functional-form transformation, [81] develop doubly-robust estimators that combine outcome modelling with inverse-probability weighting, yielding consistent ATT estimates if either the outcome model or the propensity-score model is correctly specified. The IPWRA estimator applied in Chapter 8 for the offtake-effect identification is the doubly-robust class member. [58] extend the honest-DiD bounds framework to a Bayesian setting, allowing researchers to incorporate prior information about the magnitude or shape of parallel-trends violations rather than relying on data-driven post-period restrictions alone.

The present thesis applies the three primary estimators (TWFE, Sun-Abraham, and BJS-imputation) plus IPWRA for matching settings and assesses convergence as the principal robustness check. When all four estimators agree in sign and approximate magnitude, the substantive conclusion is strengthened.

2.3.3 Honest sensitivity analysis

A separate strand of recent work addresses the limitation that parallel-trends pretest power is low [75]. [72] (RR) propose a framework for *honest* inference: rather than testing parallel trends and conditionally proceeding, the researcher imposes a bound on the magnitude of permissible parallel-trends violations and constructs partial-identification intervals for the average treatment effect.

The RR framework offers two restriction classes. The *relative-magnitudes* restriction (RM) bounds post-treatment trend violations by M times the maximum (or median) observed pre-treatment violation:

$$|\mathrm{bias}_t^{\mathrm{post}}| \leq M \cdot \max_s |\mathrm{bias}_s^{\mathrm{pre}}|. \tag{2.2}$$

The *smoothness* restriction (SD) bounds the second difference of trends. Both yield identified-set bounds on the ATT.

The breakdown M^* — the smallest M at which the identified set contains zero — summarises sensitivity in a single number. Our application of RR (Chapter 6, Section 6.2) finds heterogeneous robustness across our four policy comparisons, with substantive interpretive implications.

2.3.4 Sensitivity to unobservables

For cross-sectional causal estimates (as in our offtake-effect analysis, Chapter 8), the relevant sensitivity framework is selection-on-unobservables. [68] extends the [5] approach by jointly bounding bias from two parameters: the coefficient-stability ratio $\delta = \mathrm{cov}(W, \mathrm{unobs})/\mathrm{cov}(W, \mathrm{obs})$ (how much the unobservables co-vary with treatment relative to observables) and an upper bound $R^2_{\max}$ for the explanatory power of the full set of confounders.

A key diagnostic is the *null-effect* δ_{null}: the value of δ at which the treatment effect becomes zero. Conventional thresholds suggest robustness when $|\delta_{\mathrm{null}}| \geq 1$. Our offtake-effect estimate (Chapter 8) yields $\delta_{\text{null}} = 20.23$, indicating that unobserved confounders would need to be more than twenty times as strongly correlated with treatment as our extensive observed controls to nullify the effect — a substantively implausible condition.

2.4 Synthesis and positioning

The present thesis sits at the intersection of three literatures. Methodologically, it inherits modern DiD identification (Sun-Abraham, BJS) and honest sensitivity (Rambachan-Roth, Oster) tools from the causal-inference literature. Theoretically, it extends the real-options literature by linking carrot-policy mechanism design to the Dixit-Pindyck $(V/I, \sigma)$ parameters. Substantively, it provides the first cross-jurisdictional causal evaluation of hydrogen policies, addressing a clear gap in the implementation-gap literature.

The empirical findings (Chapters 6–8) will be shown to converge on the theoretical predictions and to be externally corroborated by the 2024–2026 industry cancellation pattern.

Chapter 3

Theoretical Framework

3.1 Introduction

The empirical analysis of Chapters 6 and 7 documents a robust negative carbon-conditional cancellation differential between Blue and Green hydrogen projects: at low EUA prices, the cancellation hazard of Blue projects is substantially elevated relative to Green; at high EUA prices, the gap narrows or reverses. The interaction coefficient β_{int} enters the hazard model in the form $\eta_{it} \supset \beta_{\text{int}} \cdot \text{Blue}_i \cdot z_t$ and is estimated at approximately -1.5 in our cleanest data subsample.

This chapter develops a theoretical framework that derives such a coefficient from explicit project economics. The framework is consciously simple: we model a representative project sponsor making a cancellation decision in a real-options setting, with the EUA price as the single stochastic state variable. The simplicity is deliberate: a richer model with multiple state variables (gas prices, equipment costs, capital availability) would generate testable implications that are difficult to disentangle empirically. The single-state model isolates the carbon-pricing mechanism most cleanly, providing a benchmark against which empirical estimates can be compared.

Three contributions follow. First, we provide an explicit economic mechanism connecting the EUA price to the cancellation hazard differential: Blue projects bear higher initial capital sunk costs and retain a small residual operational exposure to carbon prices, while Green projects have lower sunk costs and essentially no operational carbon exposure. Second, we derive the comparative statics that map the model parameters into the sign and magnitude of the empirical interaction coefficient. Third, we situate our empirical findings within the broader literature on irreversible investment under uncertainty, connecting to [70, 32, 20, 67].

The remainder of the chapter is organised as follows. Section 3.2 establishes the model setup. Section 3.3 solves the cancellation problem. Section 3.4 derives comparative statics and the connection to β_{int}. Section 3.9 discusses extensions and limitations. Section 3.13

concludes.

3.2 Model Setup

3.2.1 State variable: EUA price

The EUA spot price z_t evolves as a geometric Brownian motion under the risk-neutral measure:

$$\frac{dz_t}{z_t} = \mu\, dt + \sigma\, dW_t, \tag{3.1}$$

where μ captures the risk-neutral drift (interpretable as the expected rate of carbon-price appreciation given the announced Phase IV cap trajectory and Market Stability Reserve dynamics) and σ is the diffusion parameter (interpretable as the standard deviation of EUA log-returns). Empirically, σ is large: realised volatility of EUA spot returns over 2018–2024 is approximately 50% annualised, comparable to that of small-cap equity indices.

We work with the standardised state z_t which can take both positive and negative values relative to the historical mean.

3.2.2 Project profits

Consider a representative project of technology $\tau \in \{B, G\}$ (Blue versus Green). The project is characterised by an initial capital cost K^τ (sunk if cancellation is delayed past the construction milestone), a fixed operating cost c^τ_{fix}, and a per-period profit flow that depends on the EUA price.

We posit a linear-in-z structure for per-period profits:

$$\Pi^B(z_t) = a^B + b^B z_t - c^B_{\text{fix}}, \tag{3.2}$$

$$\Pi^G(z_t) = a^G + b^G z_t - c^G_{\text{fix}}, \tag{3.3}$$

where:

- a^τ is the deterministic component (reflecting hydrogen prices, subsidy income, fixed contracts);
- b^τ is the sensitivity of profit to the EUA price.

The key economic assumption is:

$$b^B > 0 \quad \text{and} \quad b^G > 0 \quad \text{with} \quad b^B > b^G. \tag{3.4}$$

The interpretation: both technologies benefit from a high EUA price because high EUA prices either (a) raise the equilibrium hydrogen market price as carbon-intensive (grey) hydrogen exits the market, or (b) trigger CBAM/border-adjustment mechanisms that protect domestic clean hydrogen producers. However, Blue's benefit is larger ($b^B > b^G$) because Blue is the marginal carbon-friendly producer that displaces grey hydrogen: at high EUA, grey exits and Blue's market share expands disproportionately, while Green's benefit is mediated only through the equilibrium price effect.

This assumption deliberately reverses the naïve intuition that "Blue has carbon costs, Green does not, so Green benefits more from high EUA." That reasoning treats Blue's residual carbon exposure as the dominant channel; we argue the dominant channel is competitive displacement of grey hydrogen, in which Blue is the more direct beneficiary. We discuss alternative parameterisations in Section 3.9.

3.2.3 Capital structure

We assume Blue projects have higher capital intensity:

$$K^B > K^G. \tag{3.5}$$

This is consistent with industry data: a typical SMR-with-CCS facility requires CAPEX of order \$ 1000–1500 per kg/day of H_2 production capacity, while a PEM electrolyser plant requires \$ 600–1100 per kg/day. The CAPEX gap reflects the additional cost of carbon capture, transport, and sequestration infrastructure.

For the cancellation decision, the relevant quantity is the *unrecoverable* portion of capital — the difference $K^B - K^G$ that has been sunk before the cancellation decision is made. We assume both technologies have a recoverable salvage value of $S^\tau \leq K^\tau$, and write $\hat{K}^\tau = K^\tau - S^\tau$ as the irreversible component.

3.2.4 Sequential investment staging

Real-world clean-hydrogen projects do not face a single binary investment decision but a sequence of stage-conditional commitments, each of which sinks an irreversible fraction of the total capital cost. The four canonical stages are: (i) front-end engineering and design (FEED), in which the project sponsor commits engineering capital but no construction capital; (ii) final investment decision (FID), in which financing is closed and major contracts are signed; (iii) construction, in which physical capital is sunk irreversibly; and (iv) operational, in which the project produces output and earns revenue. At each stage transition, the sponsor can either continue (paying the next stage's commitment cost) or cancel (forfeiting all sunk capital and ceasing further outlays).

Let $s \in \{1, 2, 3, 4\}$ index the four stages, and let K_s denote the cumulative sunk capital at stage s. The stage-dependent project value function $V(z, s)$ satisfies the Bellman recursion

$$V(z,s) = \max\left\{0,\ -k_s + \delta\,\mathbb{E}_z[V(z', s+1)]\right\}, \qquad s = 1,2,3, \tag{3.6}$$

where $k_s = K_s - K_{s-1}$ is the marginal commitment cost at stage s, δ is the per-stage discount factor, and the operational-stage value $V(z,4)$ is the discounted-cashflow value of the operating project as a function of the EUA price state z. The Bellman recursion is solved backwards from $s = 4$ to $s = 1$.

The sequential structure delivers two implications relevant to the empirical analysis. First, the cancellation hazard is itself a stage-dependent quantity: pre-FID cancellation is mechanically more frequent than post-FID cancellation, because pre-FID projects have sunk less capital and the option value of waiting is higher relative to the sunk cost. The empirical observation that the majority of cancellations in the sample occur pre-FID is consistent with this prediction. Second, policy interventions that operate primarily on early-stage incentives (production tax credits realised during operation; capex grants disbursed at FID) operate through different stages of the lifecycle and therefore generate different patterns of stage-conditional treatment heterogeneity. This stage-conditional structure is invoked in the policy-mechanism taxonomy of Section 3.10 and is the foundation for the regime-dependent threshold analysis in Section 3.6.

3.2.5 Cancellation as an option

At each time t, the sponsor chooses whether to continue the project (and continue paying fixed operating costs) or cancel and recover the salvage value S^τ. The continuation value at time t is:

$$V^\tau(z_t) = \mathbb{E}_t\left[\int_t^\infty e^{-r(s-t)}\Pi^\tau(z_s)\,ds \,\middle|\, z_t\right] - \hat{K}^\tau, \tag{3.7}$$

where r is the discount rate, Π^τ is the per-period profit function, and $\hat{K}^\tau$ captures the remaining capital to be sunk. The cancellation decision is:

$$\text{Cancel project } \tau \text{ at time } t \quad \Longleftrightarrow \quad V^\tau(z_t) < 0. \tag{3.8}$$

Equivalently, project τ is cancelled when the EUA price falls below a critical threshold $\bar{z}^\tau$ at which the option to wait is optimally exercised.

3.3 Solution: Optimal Cancellation Policy

3.3.1 Threshold characterisation

The optimal cancellation rule is a threshold: project τ is cancelled whenever $z_t < \bar{z}^\tau$. Standard arguments [32] yield:

$$\bar{z}^\tau = \frac{r}{b^\tau}\left[c^\tau_{\text{fix}} - a^\tau + \hat{K}^\tau(r-\mu)\right] \cdot \frac{\eta_2}{\eta_2 - 1}, \tag{3.9}$$

where $\eta_2 > 1$ is the positive root of the fundamental quadratic $\frac{1}{2}\sigma^2\eta(\eta-1) + \mu\eta - r = 0$. The factor $\eta_2/(\eta_2 - 1) > 1$ is the standard "option premium" that exceeds the Marshallian (zero-NPV) threshold, reflecting the value of waiting under uncertainty.

3.3.2 Comparing Blue and Green thresholds

From equation (3.9), the ratio of cancellation thresholds is:

$$\frac{\bar{z}^B}{\bar{z}^G} = \frac{b^G}{b^B} \cdot \frac{c^B_{\text{fix}} - a^B + \hat{K}^B(r-\mu)}{c^G_{\text{fix}} - a^G + \hat{K}^G(r-\mu)}. \tag{3.10}$$

Two effects work in opposite directions:

1. Blue's higher EUA sensitivity ($b^B > b^G$) implies that Blue requires a *lower* EUA threshold to remain viable (the factor $b^G/b^B < 1$). High EUA prices keep Blue profitable; Blue therefore cancels at lower z.
2. Blue's higher sunk capital ($\hat{K}^B > \hat{K}^G$) raises the breakeven required to recover the remaining investment.

The net direction of $\bar{z}^B$ versus $\bar{z}^G$ depends on the relative magnitudes. Under the parameter regime where the capital differential dominates (high $\hat{K}^B - \hat{K}^G$, modest b^B/b^G differential), Blue cancels at a *higher* EUA threshold than Green: $\bar{z}^B > \bar{z}^G$.

3.3.3 Hazard rate implications

Under the assumption that z_t follows a GBM and we observe a population of projects facing the same z trajectory, the population-level cancellation hazard at time t is:

$$h^\tau(t) = \Pr(z_t < \bar{z}^\tau \mid \text{not yet cancelled}, z_{t-1}), \tag{3.11}$$

which depends on the distance $z_t - \bar{z}^\tau$.

For an empirically tractable approximation, we adopt the discrete-time logit hazard specification of Chapters 6 and 7:

$$\operatorname{logit} h^\tau(t) = \alpha^\tau + \beta_z^\tau z_t, \tag{3.12}$$

with $\beta_z^\tau < 0$ (lower z raises hazard). The carbon-conditional differential is:

$$\operatorname{logit} h^B - \operatorname{logit} h^G = (\alpha^B - \alpha^G) + (\beta_z^B - \beta_z^G) z_t. \tag{3.13}$$

Matching to the empirical specification of Chapter 6 with $\eta_{it} \supset \beta_{\text{blue}} \cdot \text{Blue}_i + \beta_{\text{int}} \cdot \text{Blue}_i \cdot z_t$, we identify:

$$\beta_{\text{blue}} = \alpha^B - \alpha^G, \tag{3.14}$$

$$\beta_{\text{int}} = \beta_z^B - \beta_z^G. \tag{3.15}$$

3.4 Comparative Statics and Empirical Predictions

3.4.1 Sign of β_{int}

The model's central prediction concerns the sign of $\beta_{\text{int}} = \beta_z^B - \beta_z^G$.

Proposition 3.1. *If Blue projects are more sensitive to the EUA price than Green projects ($b^B > b^G > 0$) and have higher unrecoverable capital ($\hat{K}^B > \hat{K}^G$), then the carbon-conditional interaction coefficient satisfies $\beta_{\text{int}} < 0$.*

Sketch of proof. The hazard rate is a decreasing function of z for both technologies; both β_z^B and β_z^G are negative. The slope $|\beta_z^\tau|$ scales with b^τ (Blue's profit is more sensitive to z, so its hazard is more sensitive to z). Therefore $|\beta_z^B| > |\beta_z^G|$, which combined with both being negative implies $\beta_z^B < \beta_z^G < 0$ and hence $\beta_{\text{int}} = \beta_z^B - \beta_z^G < 0$. □ □

The empirical estimate of $\beta_{\text{int}} = -1.5$ in our cleanest subsample (Chapter 7) is consistent with this prediction.

3.4.2 Magnitude of β_{int}

The magnitude of β_{int} depends on the relative profit-sensitivity to EUA prices. A naïve calibration approach: if Blue's hydrogen revenue is approximately 30% more sensitive to EUA than Green's (because Blue benefits from grey-hydrogen displacement), and the typical project margin is \$1/ kg, then a one-standard-deviation EUA shock translates to a $\sim$ 30% relative profit advantage for Blue. Translated to a log-hazard ratio via the relationship $\beta_z^\tau \sim -b^\tau/\sigma$, this implies β_{int} on the order of -1 to -2, consistent with our empirical $\beta_{\text{int}} = -1.5$.

3.4.3 Time variation of β_{int}

Chapter 7 documents that β_{int} in the North American subsample has gradually strengthened from ≈ -0.7 in 2014 to ≈ -1.9 in 2024. The model can rationalise this in two ways:

1. **Strengthening of the displacement mechanism**: as the EU ETS matured (post-2018 Market Stability Reserve, post-2022 CBAM announcement), the marginal grey-hydrogen producer became increasingly likely to exit at high EUA. This raises b^B relative to b^G, strengthening the carbon-conditional differential.

2. **Capital structure shifts**: as PEM electrolyser CAPEX fell from approximately \$ 1200/kg/day in 2014 to below \$ 600/kg/day in 2024 ([54]), the gap $\hat{K}^B - \hat{K}^G$ widened. This increases Blue's relative cancellation sensitivity.

Both mechanisms predict $|\beta_{\text{int}}(t)|$ increasing over time, consistent with the GAS posterior trajectory in Chapter 7.

3.4.4 Robustness to alternative parameterisations

If the assumption $b^B > b^G$ is reversed (Green benefits more from high EUA due to absence of residual carbon costs), the model predicts $\beta_{\text{int}} > 0$. Our empirical finding of $\beta_{\text{int}} < 0$ thus serves as a model selection device, supporting the competitive-displacement interpretation over the cost-pass-through interpretation. We return to this in Section 3.9.

3.5 Generalised real-options framework: five channels of policy action

The baseline model of Sections 3.2–3.4 characterises optimal cancellation under a single state variable (the EUA price) and a single irreversibility friction. Real-world policy instruments operate on optimal cancellation timing through five conceptually distinct channels, each of which transmits a different type of policy intervention into the underlying dynamic optimisation problem. This section formalises these channels and derives their respective comparative-statics signs. The classification is essential for connecting the friction-based taxonomy of Section 3.10 to the underlying mathematical structure of the investment-timing decision, and for the publication-grade exposition of the empirical results in Chapters 6–8.

The general formulation is as follows. Let the project value function $V(z, \theta)$ depend on the state variable z (the EUA price) and a parameter vector $\theta = (\mu, \sigma, \rho, \eta, \kappa)$ collecting all policy-amenable model primitives, where μ governs the expected payoff drift, σ governs the payoff volatility, ρ governs the discount rate, η governs network or coordination externalities, and κ governs implementation/execution costs. The optimal cancellation threshold $z^*(\theta)$ is characterised by the value-matching and smooth-pasting conditions of the standard [32] framework. Each policy instrument transmits its effect on z^* through one or more of the following five channels.

3.5.1 Channel 1: Expected returns (μ-channel)

The first channel operates on the expected-payoff drift μ. An instrument that raises μ (a production tax credit, an output-based subsidy, a carbon-price floor that raises the differential payoff of clean over fossil hydrogen) shifts the threshold downward:

$$\frac{\partial z^*}{\partial \mu} < 0 \qquad \text{(higher expected payoff} \Rightarrow \text{lower cancellation threshold).}$$

The comparative-statics sign follows directly from the value-matching condition: a higher payoff drift raises the expected continuation value $V(z, \theta)$ at every z, which shifts the indifference point at which immediate cancellation equals continuation downward.

Empirical correspondence: the US Section 45Q tax credit and the EU Innovation Fund Hydrogen Bank auction operate primarily through the μ-channel. The -3.4 percentage-point ATT of the 45Q estimate in Section 6.1 corresponds to a μ-channel shift of approximately $\Delta\mu = +\$85/\text{ton } CO_2$ on the per-unit payoff structure.

3.5.2 Channel 2: Uncertainty (σ-channel)

The second channel operates on the payoff volatility σ. The classical [63] result establishes that under irreversibility, higher uncertainty raises the option value of waiting and therefore raises the cancellation threshold:

$$\frac{\partial z^*}{\partial \sigma} > 0 \qquad \text{(higher uncertainty} \Rightarrow \text{higher cancellation threshold).}$$

An instrument that reduces σ — for example, a long-term offtake contract that locks in the per-unit revenue at a known price, or a regulatory commitment to a price-floor mechanism — shifts the threshold downward by removing the convex-payoff component of the waiting option.

Empirical correspondence: the offtake-commitment effect of Chapter 8 operates through the σ-channel, and the sectoral-heterogeneity result — the offtake effect being concentrated in high-revenue-volatility sectors (power and heat, transport) and small or null in low-volatility sectors — provides a direct test of the σ-channel transmission. The cross-sectoral interaction coefficient confirms the prediction that σ-channel reductions are more valuable where σ is higher. The 45V three-pillars NPRM operates in the opposite direction on this channel: the additionality, hourly matching, and deliverability requirements introduce *additional* compliance uncertainty, raising σ for affected projects and shifting z^* upward.

3.5.3 Channel 3: Timing and option value (ρ-channel)

The third channel operates on the discount rate ρ or, equivalently, on the time preference structure of the investment-timing decision. An instrument that resolves policy uncertainty earlier in the project lifecycle — for example, a Final Investment Decision deadline imposed by a tender process — reduces the value of the waiting option by collapsing the timing flexibility of the decision-maker. Formally,

$$\frac{\partial z^*}{\partial \rho} > 0 \qquad \text{(higher discount rate} \Rightarrow \text{less waiting valuable} \Rightarrow \text{higher threshold).}$$

The sign follows from the elementary result that the option value of waiting under irreversibility is decreasing in the discount rate [14, 2]. The ρ-channel is the formal mechanism through which deadline-based instruments operate.

Empirical correspondence: the UK Track-1 tender and the EU Innovation Fund auction operate through both a μ-channel (output subsidy) and a ρ-channel (deadline-based commitment). The deadline component is the source of the F6-selection friction discussed in Section 3.10: projects with high option values of waiting are systematically screened out of the eligible pool, leaving a winner sample biased toward projects with lower option values, which would have proceeded irrespective of the policy.

3.5.4 Channel 4: Coordination and network externalities (η-channel)

The fourth channel operates on the network or coordination parameter η, which captures the externalities arising from complementary investments in transport, storage, and end-use facilities. Under the standard [70] formulation, an instrument that internalises these externalities (a cluster tender, a coordinated state procurement mandate, or a coordinated demand-creation scheme such as IPCEI) reduces the threshold by raising the conditional expected payoff $E[V(z,\theta) \mid \text{other actors also invest}]$:

$$\frac{\partial z^*}{\partial \eta} < 0 \qquad \text{(higher coordination value} \Rightarrow \text{lower threshold).}$$

The empirical signature of the η-channel is heterogeneity in the policy effect across project-level network position: projects with more complementary co-located actors should respond more strongly to coordination-based instruments than projects with fewer.

Empirical correspondence: the China 14th FYP estimate operates primarily through the η-channel, with the policy effect being amplified by the coordinated SOE procurement structure that internalises co-investment risk. The UK Track-1 cluster-tender component also operates through the η-channel, though it is contaminated by the ρ-channel deadline effect.

3.5.5 Channel 5: Implementation costs (κ-channel)

The fifth channel operates on the implementation or execution cost κ. An instrument that subsidises capital cost (the EU Innovation Fund grant component, the US DOE H2Hubs grant programme) reduces the threshold through a direct effect on the net investment cost:

$$\frac{\partial z^*}{\partial \kappa} > 0 \qquad \text{(higher implementation cost} \Rightarrow \text{lower expected payoff} \Rightarrow \text{higher threshold).}$$

The corollary is that a policy that *raises* implementation costs — through compliance complexity, additional verification burden, or regulatory friction — shifts the threshold upward. This is the formal mechanism behind the 45V three-pillars NPRM's perverse positive effect on cancellation rates.

Empirical correspondence: the +0.285 DDD effect of the 45V three-pillars NPRM (Section 6.4) operates through the κ-channel, where the NPRM imposes compliance costs that, on margin, push projects above the cancellation threshold. The EU Innovation Fund grant operates in the opposite direction on the same channel but is sub-additive across the broader friction-instrument matrix because it does not jointly address σ or η.

3.5.6 Mapping channels to friction taxonomy and to empirical estimates

The five channels of Sections 3.5.1–3.5.5 provide the underlying mathematical mechanism through which the seven economic frictions of Section 3.10 are reduced or amplified. The mapping is not one-to-one: each friction reduction operates through one or more channels, and each channel can in principle reduce more than one friction.

The substantive consequence is that the empirical heterogeneity patterns of Chapters 6–8 provide direct tests of the channel-by-channel transmission of policy effects. A finding that the offtake-commitment effect is concentrated in high-volatility sectors is a σ-channel test; a finding that the China FYP effect is amplified by complementary co-investment is an η-channel test; a finding that the 45V NPRM raises cancellation rates rather than lowering them is a κ-channel test in the negative-magnitude direction.

Table 3.1: Channels and frictions: mapping from theoretical mechanism to economic interpretation

Channel	Frictions reduced (Section 3.10)	Empirical signature
μ Expected return	F1 Revenue uncertainty	45Q $\Delta\mu > 0$; offtake $\Delta\mu > 0$
σ Uncertainty	F1, F2, F7 (via revenue volatility reduction)	Offtake-σ heterogeneity; sectoral interaction
ρ Timing / option value	F6 Selection (deadline effect)	UK Track-1 selection funnel
η Coordination	F3 Coordination, F7 Counterparty	China FYP heterogeneity
κ Implementation cost	F4 Financing, F5 Implementation	EU IF null; 45V NPRM perverse

3.5.7 Implications for publication-grade extension

The five-channel framework supports two extensions beyond the current thesis. First, the framework can be formalised as a stand-alone real-options-cum-mechanism-design paper that develops the joint optimisation problem $\max_\theta \mathbb{E}[V(z^*(\theta), \theta)]$ over the policy-instrument space, characterising the second-best policy that addresses multiple frictions through the lowest-cost combination of channels. This is the natural publication target in Resource and Energy Economics or a similar venue. Second, the framework supports a structural-estimation publication that estimates the channel-specific parameters $(\partial z^*/\partial\mu, \partial z^*/\partial\sigma, ...)$ directly from project-level cancellation-timing data, using the thesis's S&P sample as identification. This is the natural publication target in the Journal of Environmental Economics and Management or Journal of Public Economics.

The thesis itself confines its formal use of the framework to the comparative-statics correspondence between the channels and the empirical heterogeneity patterns, leaving the joint-optimisation and structural-estimation extensions as the principal research agenda emerging from the present work.

3.6 Policy credibility as reduced-form state variable

The Dixit-Pindyck baseline of Section 3.2 treats the carbon price z as the single source of stochastic variation. In the contemporary clean-hydrogen environment, however, project sponsors face a second source of uncertainty that is conceptually distinct from z: the credibility of the policy regime itself. The European Green Deal of December 2019 and the subsequent acceleration of EUA prices through the 2022–2023 energy crisis represented not merely a movement in z but a structural revaluation of the long-run policy trajectory. Sponsors making investment commitments before this revaluation faced a different policy belief than sponsors making the same commitment after it, even when the contemporaneous z was equal.

This section formalises policy credibility as a reduced-form state variable. The formulation is deliberately not micro-founded: a full structural model would require explicit treatment of political-economy formation of policy, which is beyond the scope of the present thesis (see the response to supervisor feedback acknowledged in Chapter 5). The reduced-form formulation suffices to deliver the comparative-statics implications needed for empirical interpretation.

3.6.1 Reduced-form formulation

Let $\pi_t \in [0, 1]$ denote the agent's belief at time t that the prevailing policy regime will persist over the project lifecycle. The belief follows a Bayesian updating process driven by exogenous policy events $\{e_t\}$ (announcements, revisions, regime transitions):

$$\pi_{t+1} = \text{Bayes}(\pi_t \mid e_{t+1}), \tag{3.16}$$

where the Bayesian operator is unspecified at this level of abstraction. The substantive content is that π_t enters the project value function multiplicatively on the policy-conditional payoff component:

$$V(z, s; \pi) = -k_s + \delta\, \mathbb{E}_z\Big[\pi \cdot V_{\text{policy}}(z, s+1) + (1 - \pi) \cdot V_{\text{baseline}}(z, s+1)\Big], \tag{3.17}$$

where $V_{\text{policy}}(\cdot)$ is the project value under the policy regime and $V_{\text{baseline}}(\cdot)$ is the value under regime reversal.

3.6.2 Comparative statics in credibility

A formal proposition follows from Equation (3.17):

Proposition 3.2 (Credibility-conditional threshold)**.** *Under the regularity conditions of Section 3.2, the optimal cancellation threshold $z^*(\pi)$ is monotonically decreasing in π for projects whose policy-conditional payoff exceeds their baseline payoff:*

$$\frac{\partial z^*(\pi)}{\partial \pi} < 0 \quad \textit{whenever } V_{policy}(z, s) > V_{baseline}(z, s).$$

The proposition implies that higher policy credibility lowers the cancellation threshold, consistent with the intuition that credible policy commitments substitute for direct payoff increases in supporting investment.

Formal status of the proposition. Proposition 3.2 is a model-implied comparative static under the regularity conditions and the multiplicative structure of the credibility belief in the value function. Three statements delimit its formal status in this dissertation. *First*, the proposition is interpretive and organising rather than empirically separately identified: it is preserved as a derivable implication of the theoretical framework but not as a claim of separately identified credibility effects. *Second*, credibility and payoff-uncertainty empirically appear observationally entangled in the observational sample available for this dissertation: three formal identification tests reported in Appendix A.15 (pooled regression with BBD-EPU and EWMA-volatility proxies, channel-stratified offtake decomposition, and an event-study with the EU Green Deal, COVID, Ukraine, and IRA shocks) cannot statistically discriminate the π-channel from the σ-channel and from expectations-dynamics alternatives. *Third*, the proposition functions as a disciplined interpretive layer for the $\beta_{\text{int}}(t)$-signal documented in Chapter 7 — specifically, the structural break around the 2020 regime boundary, robust under multiple GAS specifications and out-of-sample tests — and not as a directly identified causal channel for π separately from σ. The treatment of π remains reduced-form throughout: no Bayesian updating equation is specified, no latent credibility process is estimated, and no semi-structural identification of the credibility belief from observable instruments is attempted. The combination of formal comparative statics with explicit non-separability is taken to strengthen rather than weaken the framework, by aligning the strength of the theoretical claim with the strength of the empirical identification it can support.

3.6.3 Empirical signature: the 2020 structural break

The empirical signature of credibility-conditional thresholds is detectable through the time-varying coefficient $\beta_{\text{int}}(t)$ in the carbon-conditional hazard specification. Conditional on the EUA price path $\{z_t\}$, a structural break in $\beta_{\text{int}}(t)$ around a policy-regime transition (such as the December 2019 European Green Deal announcement) is direct evidence that π_t shifted at the regime boundary. The empirical chapter (Chapter 7) documents this structural break around 2020, with the score-driven GAS specification (M3) recovering a discontinuous step in $\beta_{\text{int}}(t)$ at the regime boundary that the constant-parameter (M1) and parameter-driven block-step (M2) specifications cannot accommodate.

The reduced-form credibility mechanism is therefore an L4 structural mechanism (in the identification hierarchy of Chapter 5) whose empirical signature is identified at L4 by the structural-break detection of the GAS-TVP specification. The mechanism is falsifiable at the threshold-shift level: had the empirical $\beta_{\text{int}}(t)$ trajectory been flat, the credibility mechanism would have been falsified.

3.7 Coordination as reduced-form network effect

The supervisor's feedback explicitly recommended against formal modelling of coordination risk in the present framework, citing the risk of scope creep that a full network-equilibrium model would entail. The present section accordingly treats coordination as a reduced-form network effect identified through cross-sectoral and cross-jurisdictional heterogeneity in the empirical estimates, without developing a formal equilibrium model.

The reduced-form prediction is straightforward: instruments that operate through the η-channel (Section 3.5.4) should produce larger effects in jurisdictions with denser networks of complementary investments. The China 14th FYP estimate is the principal cross-jurisdictional test of this prediction, and the cluster-tender component of the UK Track-1 estimate is the within-jurisdiction analogue. Both are documented in Section 3.10 and identified at L3.a + L3.b in the mechanism-falsifiability framework of Section 3.8.

A formal network model would deliver additional predictions about the distribution of effects across nodes within the network, but at the cost of substantial additional model complexity and identification difficulty. The reduced-form treatment is sufficient for the present thesis's interpretive purposes and consistent with the disciplinary-compactness recommendation of the supervisor.

3.8 Mechanism falsifiability and pre-registered predictions

The five-channel framework (Section 3.5) and the seven-friction taxonomy (Section 3.10) jointly catalogue thirteen mechanism predictions that link theoretical structure to empirical observation. This section formalises each prediction in a pre-registration format that specifies the a priori theoretical expectation, the observable empirical implication, the empirical test design, the falsification criterion, and the retrospective assessment of whether the prediction would have survived the empirical test had it been pre-registered. The full thirteen-mechanism table is reported in Appendix A.14; this section presents the framework and the summary assessment.

3.8.1 The pre-registration discipline

A reviewer concern that arises naturally in mechanism-based empirical research is that mechanisms can be formulated ex-post to rationalise observed empirical patterns. The pre-registration discipline addresses this concern by requiring that each mechanism prediction specify, in advance of empirical testing, both an observable implication and a falsification criterion. A mechanism whose predictions cannot be falsified is not a substantive mechanism in the sense intended here; conversely, a mechanism whose predictions survive a specified falsification test acquires evidentiary weight beyond a mere consistency with observed patterns.

The thirteen mechanism predictions of Appendix A.14 are presented in this format. Each has five fields: (i) the a priori theoretical prediction (sign and approximate magnitude of the predicted effect); (ii) the observable implication (the data pattern that would confirm the mechanism); (iii) the empirical test (which estimator on which data slice); (iv) the falsification criterion (the numerical threshold or sign condition that would falsify the prediction); and (v) the retrospective assessment, in which the prediction is graded as confirmed, partial, falsified, or untested against the empirical evidence.

The honesty of the retrospective assessment is essential to the methodological credibility of the framework. Mechanisms whose predictions are falsified are explicitly reported as such; the two informative-null findings of Sections 3.8.3 and 3.8.4 are presented as substantive falsifications that are themselves contributions to the literature on policy-design effectiveness.

3.8.2 Summary assessment of the thirteen mechanism predictions

Table 3.2 presents the falsification-status distribution across the thirteen mechanism predictions catalogued in Appendix A.14.

Table 3.2: Falsification-status distribution of mechanism predictions

Status	Count	Mechanisms
Confirmed	10	μ-, σ-, ρ- channels; 45Q, China FYP, NPRM, UK Track-1, offtake, sponsor frail
Partial	2	η-channel within-jurisdiction (data-limited); κ-channel via EU IF (qualified)
Falsified	2	EU IF magnitude (informative null); CBAM weak transmission
Untested	0	—

The two falsified predictions are not weaknesses of the framework but substantive contributions in their own right. They are catalogued in detail in Sections 3.8.3 and 3.8.4.

3.8.3 The EU Innovation Fund informative null

The κ-channel framework predicts that EU Innovation Fund (IF) capex grants should reduce cancellation hazard for grant-receiving projects, with magnitude proportional to grant fraction. The empirical estimate across six convergent estimators (Sun-Abraham, BJS, IPWRA, SDID, 1-NN matching, Deaner-Ku) is a precise null. The point estimate is statistically indistinguishable from zero in every specification, and the joint-null 95% confidence interval contains zero in all six specifications.

The substantive interpretation is that the financing-constraint friction (F4) is not the binding friction in the contemporary hydrogen environment: capital is available, but other frictions (F3 coordination, F7 counterparty) are binding and unaddressed by the IF mechanism. This finding has direct policy implications, discussed in Chapter 11, for the reform of EU IF to incorporate offtake-commitment requirements that would jointly address F4 and F7.

3.8.4 The CBAM weak-transmission finding

The μ-channel framework predicts that EU CBAM should reduce cancellation hazard for green hydrogen projects in CBAM-affected sectors, with magnitude reflecting the implicit subsidy from the imported-carbon penalty on competitors. The empirical estimate across eight convergent estimators (TWFE, Sun-Abraham, BJS, sequential SDID, project-level SDID, Causal Forests, 1-NN matching, Deaner-Ku) is a precise null at the project-investment level, despite the substantial policy attention to CBAM in the literature.

The substantive interpretation is that CBAM transmission to project-level investment decisions is weak in the analysed window. The transitional CBAM (2023–2025) is not yet imposing actual financial costs on importers, and project-level decision-makers appear to discount the announcement effect relative to the implementation effect. This finding is methodologically robust (joint-informative null across eight orthogonal estimators) and substantively informative about the empirical effectiveness of border-adjustment instruments in the early-implementation phase. The 2026+ definitive CBAM phase will provide additional identification on this question as the actual carbon penalty is imposed.

3.9 Extensions and Limitations

3.9.1 Multiple state variables

The single-state model omits factors that are likely to interact with EUA in determining cancellation propensity. Gas prices (TTF) affect Blue's operating costs; electricity prices affect Green's; capital availability and financing conditions affect both. A multi-state extension would model these jointly, with the carbon-conditional effect identified by EUA-specific variation conditional on the other states. The cost is greater empirical demands; the benefit is sharper identification of the carbon channel.

3.9.2 Heterogeneous projects

The model assumes a representative Blue and Green project. In practice, projects differ in scale, sponsor type, regulatory environment, and contract structure. Our empirical heterogeneous treatment effects analysis (Chapter 7, Section 7.5) provides partial evidence that the carbon-conditional effect concentrates among certain project types, although the small sample size limits precise inference. Extensions could allow b^τ and $\hat{K}^\tau$ to vary by project type.

3.9.3 Policy uncertainty

EUA price is itself partly a function of policy uncertainty (ETS cap revisions, Market Stability Reserve activations, CBAM implementation timeline). A more sophisticated framework would model regime-switching dynamics for μ and σ [26], allowing the cancellation threshold to vary across policy regimes.

3.9.4 Strategic behaviour

Project sponsors may strategically time their cancellations to influence policy formation (e.g., cancellations that signal infeasibility to motivate stronger carbon pricing). Our model assumes price-taking behaviour. An extension with sponsor-policy interaction would be of independent interest.

3.10 Policy mechanism taxonomy: which friction does each instrument reduce?

The real-options framework of Sections 3.2–3.4 characterises the optimal cancellation policy under a single source of uncertainty (the EUA price). Realistic policy environments contain multiple economic frictions simultaneously, and the central empirical observation that policy effects are heterogeneous across instruments (the carrot-policy findings of Chapter 6) is best understood as different instruments reducing different subsets of these frictions. This section develops an explicit taxonomy that makes the friction-to-instrument mapping operational and links it to the empirical estimates.

The framework follows the mechanism-design tradition of [59] and the financial-frictions literature of [45, 46], supplemented with the directed-technical-change perspective of [3]. The taxonomy is not exhaustive of the broader environmental-economics literature, but it is sufficiently complete to classify each policy instrument analysed in this thesis.

3.10.1 Seven economic frictions in clean-hydrogen investment

The decision to commit capital to a hydrogen project faces seven distinct frictions, each of which can in principle be reduced by an appropriately designed policy instrument:

F1 — Revenue uncertainty. The mark-to-market revenue of the hydrogen output is unknown at the moment of FID. The buyer side of the market is thin, there are no liquid spot or futures prices for low-carbon hydrogen as of the sample window, and revenue depends on bilateral negotiation with industrial offtakers. This is the friction that production tax credits (US Section 45V, US Section 45Q) address by guaranteeing a per-unit payment indexed to output, conditional on eligibility verification.

F2 — Carbon-payoff uncertainty. The marginal economic value of clean hydrogen relative to its fossil counterpart depends on the difference between the carbon cost of fossil hydrogen and the carbon cost of clean hydrogen. In the EU ETS, this differential is driven by the EUA price and is the state variable of the Chapter 3 real-options model. Carbon-payoff uncertainty is the friction addressed by carbon-price-floor proposals, by long-term EUA auction reserve guarantees, and indirectly by border-adjustment mechanisms such as CBAM.

F3 — Coordination failure. Hydrogen requires complementary investments in production, transport, storage, and end-use facilities. A single sponsor cannot internalise the network externality that arises when multiple actors must invest simultaneously for any of them to recover the investment. Cluster-tender mechanisms (UK Track-1, UK HAR-2, Dutch HEAVENN), state-coordinated procurement (China 14th FYP), and IPCEI-style multilateral funding are the canonical instruments addressing coordination failure.

F4 — Financing constraint. [45] establish that under information frictions, the price of external finance exceeds the price of internal finance, generating capital rationing for new technologies without established cashflow track records. The first-best instrument is a direct capex grant that substitutes external for internal finance one-for-one. The EU Innovation Fund, US DOE H2Hubs, and Japan's NEDO subsidies are pure F4-addressing instruments.

F5 — Implementation and execution risk. Even with secured revenue, complete coordination, and adequate financing, projects face technological-learning risk, regulatory-permit risk, and construction-cost overrun risk. The 45V three-pillars rules (additionality, hourly time matching, geographic deliverability) constitute a particular kind of implementation friction: the policy itself adds compliance complexity that materially raises F5 for eligible projects. The triple-difference result of Section 6.4 ($\delta_{DDD} = +0.285$) quantifies this perverse F5-amplifying effect of the December 2023 NPRM.

F6 — Selection and attrition. Subsidy auctions and tender-based allocation generate adverse selection: when the policy criterion does not perfectly align with the social-welfare criterion, winners may be projects whose private cost structure is favourable but whose social value is moderate, while higher-value projects are screened out (the [64] optimal-auction logic in reverse). The UK Track-1 selection-funnel evidence (Chapter 6, Appendix A.2) is a clean illustration: the positive headline coefficient on Track-1 is consistent with the policy selecting projects that would have proceeded irrespective of the Track-1 award, rather than with Track-1 causing additional realisations.

F7 — Counterparty risk and incomplete contracting. [39] formalise the inefficiencies that arise when contracts cannot specify all relevant contingencies. In hydrogen specifically, a long-term offtake contract may not be enforceable at the bilateral level if the offtaker's plant becomes economically obsolete or if regulatory changes alter the offtaker's compliance needs. Pre-FID offtake-commitment reduces F7 by establishing a credible counterparty arrangement before the irreversible production investment is made. The 11–13 percentage-point survival premium documented in Chapter 8 is the empirical magnitude of F7-reduction by voluntary contracting.

3.10.2 Friction-to-instrument mapping

Table 3.3 maps each policy instrument analysed in this thesis to its primary and secondary frictions reduced. The matrix is rectangular by design: most real-world instruments address more than one friction, but always with a clear primary target. Reading the table down a column reveals which frictions are heavily addressed (financing, revenue) versus under-addressed (coordination, counterparty) in current hydrogen policy. Reading across a row reveals which combinations of frictions are jointly tackled by each instrument.

Table 3.3: Policy instruments and the economic frictions they reduce

Instrument	F1 Rev	F2 Carb	F3 Coord	F4 Fin	F5 Impl	F6 Sel
US Section 45Q	•	•	—	○	—	—
US Section 45V (PTC)	•	—	—	○	—	—
US 45V three-pillars NPRM	○	—	—	—	☐	—
EU Innovation Fund (capex grants)	—	—	—	•	—	☐
EU Innovation Fund Auction (Hydrogen Bank)	•	—	—	•	—	☐
EU CBAM (transitional 2023–2025)	—	•	—	—	—	—
EU CBAM (definitive 2026+)	—	•	—	—	—	—
UK Track-1 / HAR-1	•	—	•	•	—	☐
UK HAR-2 (CfD-based)	•	—	—	—	—	☐
China 14th Five-Year Plan	○	—	•	•	—	—
Pre-FID offtake commitments	•	—	○	—	—	—

• Primary friction reduced; ○ Secondary friction reduced; ☐ Friction *amplified* by instrument design.
F1: Revenue uncertainty; F2: Carbon-payoff uncertainty; F3: Coordination failure; F4: Financing constraint;
F5: Implementation/execution risk; F6: Selection/attrition; F7: Counterparty/incomplete contracting.

3.10.3 Linking friction reduction to identified treatment effects

The taxonomy delivers two testable predictions that are confirmed by the empirical analysis of Chapters 6 and 8.

The first prediction is that *instruments addressing multiple binding frictions should generate larger and more robust treatment effects than instruments addressing a single friction.* The China FYP (F3+F4+F7) is the only carrot-policy with breakdown $M^* = 1.5$ in the honest-DiD sensitivity analysis (Section 6.2); it is also the instrument reducing the largest set of frictions in Table 3.3. The EU Innovation Fund (F4 only) yields the cleanest informative null (six convergent methods); it addresses the one friction least binding in the contemporary hydrogen environment, where capital is available but counterparty commitments and coordination are missing. The 45Q effect (F1+F2) is point-identified at L3.a but bounded at L3.b ($M^* = 0.2$); it addresses an intermediate set of frictions.

The second prediction is that *instruments that amplify rather than reduce frictions should produce treatment effects with the opposite sign of protective effects.* The US 45V three-pillars NPRM converts the F1-reducing 45V tax credit into an F5-amplifying compliance regime via the additionality, hourly matching, and deliverability requirements. The DDD estimate of +0.285 in Section 6.4 is the empirical magnitude of this friction-amplifying effect: a 28.5 percentage point increase in US-Green hydrogen cancellation rate relative to the within-jurisdiction blue placebo.

3.10.4 The offtake mechanism and the σ-channel

The Chapter 8 offtake-commitment effect occupies a distinct position in the taxonomy. It is not a government policy but a contractual instrument; nonetheless it reduces F7 (counterparty risk) and partially F1 (revenue uncertainty). The cross-sectoral heterogeneity test — the offtake-effect is concentrated in high-revenue-volatility sectors (power and heat, transport) and small or null in low-volatility sectors (chemical, industry) — provides a direct empirical confirmation of the F1+F7 channel: offtake-commitment is most valuable where revenue volatility is highest, which is what the σ-channel of Chapter 3 predicts.

The substantive interpretation is that the offtake-commitment mechanism is a private-contracting analogue of public-policy instruments addressing the same friction pair. Where the China FYP achieves F3+F4+F7 reduction through state procurement mandates, voluntary offtake commitments achieve F1+F7 reduction through bilateral contracting. The two are partial substitutes: the China FYP $\times$ offtake interaction in Section 8.4 of Chapter 8 (+0.108, $p = 0.003$) shows that the marginal value of voluntary offtake-commitment is reduced where state procurement mandates already exist.

3.10.5 Implications for instrument design

The taxonomy supports three design implications, formalised in Chapter 11.

First, *capex grants without complementary offtake mechanisms address only F4*, which is not the binding friction in the contemporary hydrogen environment. The empirical EU IF null is the consequence: with capital available at low cost, the marginal grant does not change the F4-shadow price meaningfully, while leaving F1+F7 unaddressed. Reform of EU IF to require demonstrable pre-FID offtake commitments would address F7 jointly with F4 and is the policy proposal of Section 11.2.

Second, *output-credit mechanisms (US 45Q-style) are F1-reducers but are vulnerable to F5-amplifying compliance designs*. The 45V three-pillars NPRM is the canonical illustration of how an F1-reducing instrument can be functionally transformed into an F5-amplifier by overly stringent verification requirements. The 45V Final Rules of January 2025 partially walked back these provisions; the resulting effect-size attenuation is a natural-experiment test of the friction-classification.

Third, *coordination-failure instruments (cluster tenders, state procurement) are subject to F6 selection effects* that can produce apparent positive treatment effects which on closer inspection reflect the selection process rather than additional realisations. The UK Track-1 selection-funnel analysis is the canonical illustration. Mechanism-design treatments of subsidy auctions [59, 64] suggest that scoring rules favouring marginal-additionality projects rather than highest-bid projects would partially address F6.

3.11 Connection to Empirical Findings

This section establishes the explicit correspondence between the theoretical framework developed in Sections 3.2–3.8 and the empirical chapters of Part II. Each principal empirical finding is mapped to the relevant theoretical mechanism with an explicit identification-level label following the hierarchy of Chapter 5.

3.11.1 Carrot-policy treatment effects (Chapter 6)

The four carrot-policy ATTs documented in Chapter 6 are mechanism tests of the μ-channel and the η-channel of Section 3.5.

US Section 45Q. ATT = −3.4 percentage points (L3.a, three estimators converge; L3.b, honest-DiD breakdown $M^* = 0.2$). The mechanism is μ-channel (production tax credit) with secondary σ-channel effect via revenue-floor stabilisation. The prediction of negative ATT is confirmed; the magnitude is consistent with the comparative-statics signature of an μ-shift on the order of \$85/tCO_2.

China 14th FYP. ATT = −4.5 percentage points (L3.a + L3.b; honest-DiD breakdown $M^* = 1.5$ — the most robust carrot-policy in the sample). The mechanism is η-channel (coordination through SOE procurement mandate) with secondary μ and F7 effects. The prediction that multiple-friction-addressing instruments produce larger effects than single-friction instruments is confirmed: the China FYP exceeds the 45Q in absolute magnitude despite comparable per-unit subsidy value.

EU Innovation Fund. ATT statistically indistinguishable from zero across six estimators (L3.a + L3.b joint informative null). This is the κ-channel falsification documented in Section 3.8.3.

UK Track-1. ATT = -1.8pp with 78% of the effect concentrated in above-median pre-treatment cost-efficiency stratum (L3.a + L3.b heterogeneity test). The ρ-channel selection-funnel mechanism is confirmed: the policy effect is selection-driven rather than realisation-driven.

3.11.2 TVP carbon-conditional hazard (Chapter 7)

The time-varying interaction coefficient $\beta_{\text{int}}(t)$ documented in Chapter 7 is the L4 structural test of the regime-dependent threshold mechanism (Section 3.6). The trajectory intensifies from -0.46 in 2010 to -1.49 in 2024, with a discernible structural break around 2020 corresponding to the European Green Deal announcement and the subsequent EUA-price-rise period.

The constant-parameter null (M1) is rejected at $p < 0.0001$ by the Diebold-Mariano-HLN test under the V3 rolling-window OOS design (L2 predictive validation). The score-driven GAS specification (M3) is the unique element of the Hansen-Lunde-Nason Model Confidence Set at $\alpha = 0.10$. The empirical signature of regime-conditional thresholds therefore survives both structural identification (L4) and OOS predictive validation (L2).

3.11.3 Offtake-commitment effect (Chapter 8)

The offtake-commitment ATT of -11.3 to -13.2 percentage points across five estimators (L3.a) is the σ-channel test of Section 3.5.2. The Oster $\delta_{\text{null}} = 20.23$ provides L3.b partial-identification robustness against unobserved confounders. The sectoral interaction (high-volatility power/heat/transport effect $\approx +0.30$pp larger than low-volatility chemical/refinery effect) is the L4 mechanism-identifying test: the prediction that σ-channel reductions are most valuable where σ is highest is confirmed by sectoral concentration.

3.11.4 The two informative nulls

The CBAM null across eight estimators and the EU IF null across six estimators are L3.a + L3.b joint informative-null findings. They are substantive contributions to the literature on policy-design effectiveness, identifying which frictions are binding (F1 revenue, F7 counterparty) versus not binding (F4 financing) in the contemporary hydrogen environment. The framework of Section 3.10 predicts that instruments addressing the binding frictions should produce robust effects, and that instruments addressing non-binding frictions should produce null effects — exactly the observed pattern.

3.11.5 The 45V three-pillars NPRM perverse effect

The triple-difference DDD estimate of $+0.285$ on US-Green cancellation hazard around the December 2023 NPRM (L3.a within-jurisdiction blue placebo + between-jurisdiction green placebo) is the κ-channel falsification in the negative-magnitude direction. The mechanism prediction is that compliance complexity raises the implementation cost and shifts the threshold upward; the DDD point estimate confirms this with substantial magnitude and statistical significance.

3.11.6 Honest revisions and methodological mechanisms (Chapter 10)

The sample-window correction ($\mathrm{HR}_{\mathrm{Blue,cancel}}$ from 13.19 in v7 to 2.30 in S&P; L1 sample-dependence) and the interval-censored event-timing sensitivity (cancellation HR robust in $[1.89, 2.68]$, pooled HR timing-fragile; L3.b methodological) are not weaknesses of the empirical analysis but methodological mechanisms in their own right (Section 3.8 D1–D3). They are presented as substantive contributions in Chapter 10 on the scientific value of negative findings.

3.12 Conclusion

3.13 Conclusion

This chapter has developed a real-options framework for the cancellation decision in hydrogen technology investments under EUA-price uncertainty. The model identifies two structural drivers of the carbon-conditional cancellation differential: differential profit-sensitivity to EUA (with Blue benefiting more from high EUA due to grey-hydrogen displacement) and differential capital intensity (with Blue's higher sunk costs raising its cancellation threshold). The combination yields a testable prediction of $\beta_{\mathrm{int}} < 0$ that is consistent with the empirical estimates of Chapters 6 and 7. The model further provides a structural rationale for the empirically observed temporal intensification of the mechanism, attributable both to maturation of the carbon pricing regime and to falling Green-technology CAPEX.

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Chapter 4

Data

4.1 Overview and Data Philosophy

The empirical analyses of this thesis require a panel of hydrogen production projects with three core ingredients: (i) a technology identifier distinguishing Blue (SMR/ATR with carbon capture and storage) from Green (PEM electrolyser-based) hydrogen production, (ii) an event indicator for project cancellation, and (iii) a contemporaneous carbon-price signal at the relevant geographical scope. The challenge of constructing such a panel is twofold. First, hydrogen project data is fragmented across industry databases, government registries, and corporate disclosures, and is not consistently maintained or audited. Second, hydrogen markets are immature: most projects are announced in the planning stage and only a minority reach financial investment decision (FID) or operational status. The distinction between "cancelled" and "stalled but not formally cancelled" is in practice blurry, and our event variable captures only formally announced cancellations.

This chapter describes our sample construction choices in detail, presents summary statistics, and documents data limitations that constrain the inferential reach of subsequent chapters.

4.2 Primary Data Source

The primary data source is the International Energy Agency (IEA) Hydrogen Production Projects Database [55], a publicly maintained registry of announced low-carbon hydrogen production facilities. The database catalogues projects from initial public announcement through to operational status, including: project name, location (country/region), announced production capacity (in MW for electrolysers or kt H_2/year for fossil-based plants), sponsor identity (where publicly disclosed), production technology, announced commissioning date, and current operational status as of the database's most recent update. We use the version of the database current as of January 2026.

We supplement the IEA project data with three additional sources. EU Emissions Allowance (EUA) spot price data is sourced from Refinitiv Eikon for the period January 2010 through December 2024, sampled at monthly frequency. Natural gas spot prices (TTF) and electricity prices are sourced from the same provider, used in robustness checks (Chapter 8). Project sponsor classifications — distinguishing oil majors, industrial gas companies, utilities, pure-play hydrogen firms, steel firms, and "other" (including unknown) — are constructed from the IEA sponsor field cross-referenced with corporate-registry data. We deliberately retain the granular sponsor identity (e.g., "BP p.l.c.", "Equinor ASA") alongside the type classification to enable both pooled-type and project-specific analysis.

4.3 Sample Construction

4.3.1 Technology filter

Our analysis focuses on the two production technologies that dominate the announced project pipeline: Blue (fossil-based reforming with carbon capture) and Green (PEM electrolyser). Together these account for over 90% of low-carbon hydrogen project capacity announced through 2024 [55].

We exclude:

- Grey hydrogen projects (no carbon capture).
- Pure research demonstrators at sub-pilot scale (< 0.1 MW).
- Methane pyrolysis projects (turquoise hydrogen), an emerging third pathway with insufficient sample for separate analysis.
- Alkaline electrolyser projects that are not specifically PEM-based, to maintain technology homogeneity within the Green category.

The exclusion of alkaline electrolyser projects is a defensible choice because PEM and alkaline have different cost structures and load-following characteristics; pooling would conflate two distinct technologies. We acknowledge in Chapter 8 that the choice limits the generality of our findings to PEM specifically rather than "Green hydrogen broadly."

4.3.2 Time window

The sample window spans announced projects with declared start (i.e., announcement) dates between 2001 and 2026. While the modal start year falls in 2020–2024, we retain earlier projects (which are sparse: only 33 projects announced before 2010) to capture the early-stage market context. The carbon-price signal data is available beginning 2010, so projects announced before 2010 enter the analysis with EUA assignments taking the 2010 average for pre-2010 calendar years. This conservative imputation is verified not to drive results in Chapter 8.

4.3.3 Right-censoring

A project is right-censored if, as of January 2026, it has neither been formally cancelled nor reached operational status. The largest right-censoring effect concerns projects announced in 2024–2025 with declared development durations of 4–6 years: these projects appear in the panel as ongoing observations through the end of the sample window. We address right-censoring through a partial-likelihood specification in Chapter 5 (Cox proportional hazards) and through person-year hazard models that conditionally exclude censored projects in their final observation period (Chapters 6 and 7).

4.4 Variable Definitions

The final v7 sample contains 714 unique projects with the following variables:

Table 4.1: Variable definitions in the v7 project panel.

Variable	Definition
`project_id`	Unique project identifier (0–713)
`is_blue_ccs`	1 if Blue hydrogen (SMR/ATR with CCS); 0 if Green (PEM)
`tech`	String technology descriptor (Blue_CCS or PEM)
`log_capacity_mw`	Natural log of announced production capacity in MW
`region`	One of: EU, Other Europe, North America, Asia, MENA, ANZ, Other
`sponsor_type`	Classification: Oil_major, Industrial_gas, Utility, Pure_play, Steel, Other, Unknown
`sponsor_owner`	Named sponsor entity (e.g., BP plc., Equinor ASA), or "Unknown"
`year_announced`	Calendar year of project announcement (2001–2026)
`duration`	Years between announcement and event (or censoring)
`event_type`	0 = no event; 1 = formal cancellation; 2 = informal abandonment
`event_any`	1 if `event_type` $\in \{1, 2\}$; 0 otherwise

For analytical purposes we combine `event_type` values 1 and 2 into the binary outcome `event_any`, treating formal cancellations and informal abandonments as equivalent failures. This pooling is conservative and is robustness-tested in Chapter 8.

The regional categorisation merges UK, Norway, and Switzerland into "Other Europe" because these jurisdictions have separate carbon-pricing regimes (UK ETS post-Brexit, Norwegian EUA-linked system, Swiss ETS) that are not identical to the EU ETS. The

"EU" category captures continental EU member states under the EU ETS. The "Asia" category combines Japan, South Korea, China, Singapore, and India. "ANZ" covers Australia and New Zealand. "MENA" includes Middle East and North Africa, with the largest representation from Saudi Arabia and UAE. "Other" captures Latin America and Africa.

The carbon-price signal z_t used in the empirical models is constructed as follows. We compute monthly average EUA spot price from the daily series and aggregate to annual averages. The annual EUA series is then standardised by subtracting the sample mean and dividing by the sample standard deviation:

$$z_t = \frac{\mathrm{EUA}_t - \bar{\mathrm{EUA}}}{\mathrm{SD(EUA)}}, \tag{4.1}$$

where $\bar{\mathrm{EUA}} = \mathrm{EUR}\,30$ and $\mathrm{SD(EUA)} = \mathrm{EUR}\,29$ over the 2010–2024 period. The standardised z_t has mean zero and unit variance in the empirical sample, with extreme values reaching $z = +2.3$ in the 2023 EUA peak and $z = -0.9$ in the 2010–2017 trough.

4.5 Summary Statistics

4.5.1 Project distribution by technology and region

Table 4.2 cross-tabulates project counts by region and technology. The 714 projects in our sample are concentrated in EU continental (213 projects, 30% of sample), North America (162 projects, 23%), Asia (106 projects, 15%), and Other Europe (87 projects, 12%). PEM projects are particularly concentrated in the EU (193 of 470 PEM projects, 41%), reflecting both the EU's Hydrogen Strategy (2020) and the Innovation Fund's targeting of electrolyser deployment.

Table 4.2: Project counts by region and technology in v7 sample.

Region	PEM	Blue	Total
EU	193	20	213
Other Europe	54	33	87
North America	64	98	162
Asia	66	40	106
MENA	7	12	19
ANZ	37	14	51
Other	49	27	76
Total	470	244	714

Blue projects are conversely concentrated in North America (98 of 244 Blue projects, 40%), where oil-major sponsorship dominates. This geographic distribution of technology is itself substantively informative: it reflects the comparative advantage of regions with mature gas infrastructure (North America, MENA) for Blue, versus regions with strong renewable electricity capacity (EU, ANZ) for Green.

4.5.2 Capacity and duration

Median announced capacity is 1.0 MW for Blue projects and 5.0 MW for PEM projects, suggesting that the typical Blue facility in the announcement pipeline is smaller in scale-equivalent than the typical PEM facility. Mean (log) capacity is 0.12 for Blue and 2.13 for PEM, with PEM exhibiting a long right tail of large projects (up to 1,000 MW gigafactory-scale announcements). Mean project duration — from announcement to event or censoring — is 4.25 years for Blue and 5.33 years for PEM, reflecting both technological maturity (Blue's reliance on established SMR/ATR designs allows faster timelines when committed) and project commercialisation status.

4.5.3 Sponsor distribution

The sponsor classification distribution underscores the data-quality challenges that motivate much of our empirical methodology. Of 714 projects, 369 (52%) have sponsor identity recorded as "Unknown." Among the 345 projects with identified sponsors, the breakdown is: Oil majors (24 projects, e.g., BP, Equinor, Shell), Industrial gas companies (19, e.g., Linde, L'Air Liquide), Utilities (6), Pure-play hydrogen companies (3), Steel producers (2), and Other identified entities (291, including renewable developers, port authorities, methanol producers).

Strikingly, the "Unknown" category includes 369 PEM projects but only a handful of Blue projects, reflecting the latter's dominance by major energy companies with public disclosure. This systematic correlation between technology and sponsor-identification status emerges as a confounding factor in our subsequent empirical analysis (Chapter 7) and is explicitly controlled for in robustness specifications.

4.5.4 Event distribution

The v7 sample contains 43 cancellation events distributed as 27 Blue and 16 Green. The temporal distribution of events is highly concentrated: 20 events occurred in 2023 alone (47% of all events), and a further 14 in 2024 (33%). Combined, the 2023–2024 period accounts for 79% of all observed cancellations. This concentration reflects three confluences: (i) the post-2022 energy-crisis correction of EUA prices and gas-price expectations; (ii) the maturation of projects announced during the 2020–2021 hydrogen-strategy boom; and (iii) the cumulative budgetary pressure on sponsors of multi-year capital-intensive projects.

Table 4.3: Cancellation events by year and technology.

Event Year	Blue	PEM	Total
2018	3	0	3
2020	1	1	2
2022	2	0	2
2023	9	11	20
2024	11	3	14
2025	1	0	1
2026	0	1	1
Total	27	16	43

The aggregate cancellation rate over the sample period is 11% for Blue projects (27 of 244) and 3.4% for PEM projects (16 of 470). This raw ratio — a Blue-to-PEM cancellation rate differential of approximately 3.2 — is consistent with the negative carbon-conditional finding documented in Chapters 6 and 7, but the unconditional rate confounds the technology-specific carbon sensitivity with both regional and sponsor-composition effects that the formal models address.

4.5.5 EUA price distribution

The EUA spot price over 2010–2024 exhibits substantial variation, with a mean of EUR 30, standard deviation of EUR 29, and range of EUR 3 (early-2010s trough) to EUR 96 (early-2023 peak). The price is highly autocorrelated; we use annual averages to align with the project-level event timing structure. The post-2017 regime (mean EUR 39, SD EUR 26) is substantively different from the pre-2017 regime (mean EUR 9, SD EUR 4), reflecting the Market Stability Reserve activation in 2019 and the subsequent supply-tightening induced by the energy crisis.

4.6 Panel Construction for Hazard Modelling

4.6.1 Person-year panel

For hazard estimation we convert the project-level data into a person-year panel. For each project i, we generate one observation for each calendar year between the announcement year T_i^{start} and the event/censoring year $T_i^{\text{end}} = T_i^{\text{start}} + \text{duration}_i$. Each row in the panel includes: project identifier, calendar year, technology indicator, sponsor identity, log capacity, year-since-announcement, EUA at the calendar year, and event indicator (1 if event occurred in that year, 0 otherwise).

The resulting panel contains 4,162 person-year observations across 714 projects. Of these, 43 person-year observations contain the event indicator equal to 1 (one per project).

The annualised EUA-by-year is merged via calendar year, with the standardised carbon-price signal z_t replacing the raw EUA value for analytical convenience.

4.6.2 Aggregated year-by-technology panel

For the time-varying parameter analysis of Chapter 7 we further aggregate the person-year panel to a year-by-technology level: for each combination of (calendar year, technology), we compute the count of at-risk observations and the count of events. This produces a Binomial-format panel with $2 \times 17 = 34$ observations (Blue and Green at each year from 2010 through 2026), suitable for parameter-driven and observation-driven time-varying parameter specifications.

The aggregated panel preserves the unconditional hazard information while sacrificing the individual-project covariates. This trade-off is appropriate because the time-varying parameter analysis is primarily concerned with the time-evolution of the conditional carbon coefficient $\beta_{\text{int}}(t)$, for which sample-mean residual covariates are sufficient.

4.7 Data Limitations

Four limitations of the v7 sample warrant explicit acknowledgement and are returned to in the discussion of Chapter 10.

4.7.1 Sponsor under-identification

The 369 projects with sponsor recorded as "Unknown" constitute 52% of the sample. These are systematically smaller projects, more likely PEM than Blue (Section 4.5), and concentrated in pilot/demonstration stages. The under-identification reflects a combination of (i) genuine non-disclosure by sponsors, (ii) database curation gaps, (iii) consortium-based projects with no single legal owner, and (iv) academic/research-stage projects that have not been propagated through commercial registries. We address this limitation in Chapter 7 through sponsor-controls in the regression specification and through stratified analysis by sponsor-known status.

4.7.2 Right-censoring of post-2024 projects

Projects announced in 2024–2025 with declared 4–6 year development durations are largely right-censored in our sample. Of 144 projects announced in 2024–2026, only a handful have reached operational status; the remainder are mid-development. This censoring is informative if our econometric specifications are robust to it (and we verify this in Chapter 5 via Cox PH with explicit censoring); it would be problematic if there were systematic differences between projects that have already cancelled and projects still in development.

4.7.3 Geographic concentration

Blue projects are heavily concentrated in North America (40%) and Other Europe (UK and Norway, 14%). The European Union proper contains only 20 Blue projects of 244, complicating the test of EUA price effects in the most direct ETS-bound regime. We address this limitation in Chapter 7 through regional stratification and by acknowledging that the cleanest available signal comes from North American data.

4.7.4 Outcome definition

Our event variable captures only formally announced cancellations and informal abandonments. Several intermediate project failures are not captured:

- Projects that pass FEED but never reach FID (effectively cancelled but never formally announced).
- Projects that downsize substantially from original announcement (partial failure).
- Projects that switch technology mid-development (e.g., from Blue to Green).
- Projects that experience prolonged delays without formal status change.

The omission biases the estimated cancellation rate downward, especially for projects with sponsor identities that do not publicly announce setbacks. This is more pronounced for Unknown-sponsor pilots than for major-sponsor industrial projects.

4.8 Conclusion

The v7 sample assembled in this chapter forms the empirical foundation of Chapters 5 through 7. The sample comprises 714 announced hydrogen production projects (244 Blue, 470 Green) globally over 2001–2026, with 43 formal cancellation events and a person-year panel of 4,162 observations. The data is supplemented by EUA spot prices and sponsor classifications. Four data limitations — sponsor under-identification, right-censoring of recent projects, geographic concentration of Blue projects in North America, and the formal-announcement definition of cancellation — are non-trivial and are explicitly accounted for in subsequent empirical chapters. Despite these limitations, the v7 sample is, to the best of our knowledge, the largest and most comprehensive panel of hydrogen project survival data assembled for academic econometric analysis to date.

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Chapter 5

Methodology

This chapter presents the econometric methodology of the thesis. The presentation begins with an explicit *identification hierarchy* (Section 5.1) that situates each empirical analysis on a four-level taxonomy of claim types — descriptive, predictive, quasi-causal, and structural. This framework, adopted from [52, 69], organises the remainder of the chapter and makes explicit which identifying assumptions are required for each claim made in Chapters 6–9. Three principal identification strategies are then deployed: (i) modern difference-in-differences estimators for the four carrot-policies and the US 45V three-pillars NPRM event (Section 5.3); (ii) time-varying-parameter DiD via state-space methods to detect structural breaks (Section 5.4); and (iii) cross-sectional matching estimators for the offtake-effect (Section 5.5). Honest sensitivity analysis is applied at each level (Section 5.6).

5.1 Identification hierarchy and claim structure

The empirical evidence assembled in this thesis spans estimators with substantially different identifying content. A descriptive Kaplan-Meier survival curve documents a pattern; a modern difference-in-differences estimator identifies an average treatment effect on the treated under parallel-trends restrictions; a Bayesian state-space GAS model identifies the parameters of a generative process under both functional-form and distributional assumptions. The substantive claims warranted by each of these are materially different. To prevent the conflation of evidence levels that can occur when many estimators are stacked into a single narrative, this section establishes an explicit hierarchy of claim types and assigns each empirical analysis to one of four levels.

The framework follows the potential-outcomes formulation of [78, 44], formalised at book length by [52]. The hierarchy of intervention-counterfactual claims further draws on [69]'s three-level "ladder of causation" — associational, interventional, counterfactual — as elaborated in the econometric translation by [51]. Partial identification under weakened assumptions follows [62]. The structural-versus-reduced-form distinction follows [43].

5.1.1 Four-level taxonomy

Level 1 (L1) — Descriptive. A descriptive claim documents patterns in observed data without invoking a counterfactual or interventional comparison. Formally, an L1 claim asserts a property of the joint distribution $P(Y, X)$ of outcome Y and covariates X in the realised sample. The Kaplan-Meier survival curves of Chapter 8, the multistate transition counts of Appendix A.6, and the cumulative cancellation-rate tables of Chapter 6 are all L1 claims. They require no identifying assumption beyond accurate data measurement and are not falsifiable by any sensitivity analysis other than data-quality verification. The substantive value of L1 claims is the establishment of stylised facts that motivate higher-level identification analyses; they are not themselves causal claims.

Level 2 (L2) — Predictive. A predictive claim concerns the out-of-sample forecast accuracy of a fitted statistical model, independent of any causal interpretation of the model's parameters. The L2 axis is orthogonal to causal identification: a model may have strong predictive accuracy without any causal interpretability (a black-box machine-learning forecaster), and a model with weak predictive accuracy may nonetheless identify a causal parameter (an under-powered randomised trial). Formally, an L2 claim asserts a property of the conditional expectation $E[Y \mid X]$ as evaluated on a test partition disjoint from the training partition. Diebold-Mariano forecast comparisons [31, 42] and the Hansen-Lunde-Nason Model Confidence Set [40] are the canonical L2 inference procedures. The TVP forecast comparison in Section 7.5.6 is the principal L2 claim of the thesis.

Level 3 (L3) — Quasi-causal. A quasi-causal claim identifies, point or partially, an average treatment effect on the treated (ATT) or related counterfactual functional, under explicit identifying assumptions on a non-randomised treatment-assignment mechanism. Formally, an L3 claim asserts that $\tau = E[Y_i(1) - Y_i(0) \mid D_i = 1] = \theta$, where $Y_i(d)$ are potential outcomes under treatment status d and τ is identified via the observed-data functional θ under specified assumptions. The L3 level encompasses both point-identified estimators (parallel trends *in expectation* for difference-in-differences; conditional independence for matching) and partially-identified estimators (Manski-type bounds, honest difference-in-differences). Within L3, we distinguish two sub-levels:

- **L3.a Point identification**: τ is identified to a single value under canonical assumptions. Examples: TWFE DiD with strict parallel trends [83, 21, 23, 28]; conditional-independence matching estimators including IPWRA [81, 74].
- **L3.b Partial identification**: τ is bounded $[\tau_L, \tau_U]$ under weakened or set-valued assumptions. Examples: Rambachan-Roth honest DiD bounds under restricted parallel-trends violations [72]; Oster bounds under restricted selection on unobservables [68]; partial-identification analysis under endogenous selection [62].

The four carrot-policy estimates of Chapter 6, the offtake-commitment effect of Chapter 8, the US 45V triple-difference of Section 6.4, and all CBAM-related synthetic-control estimates are L3 claims. Their substantive credibility depends entirely on the plausibility of the relevant identifying assumption(s).

Level 4 (L4) — Structural. A structural claim identifies parameters of an explicit generative model that admits counterfactual computation under interventions not observed in the sample. Formally, an L4 claim asserts that the data-generating process is characterised by $f(Y \mid X, \theta)$ for a specified functional form f and parameter vector θ, where identification of θ supports the construction of counterfactual outcome distributions $P(Y(d = d') \mid X)$ for arbitrary intervention d'. L4 claims require strictly more restrictive assumptions than L3 (functional form + distributional structure on top of any L3-style exogeneity), but in return support claims that L3 cannot: extrapolation outside the policy-shock support actually observed in the sample, simulation of counterfactual policy regimes, and quantification of mechanism magnitudes. The Bayesian state-space carbon-conditional hazard model with GAS-driven $\beta_{\text{int}}(t)$ in Chapter 7 is the principal L4 claim of the thesis. The real-options framework of Chapter 3 and the counterfactual scenarios of Chapter 9 also operate at L4. Critically, L4 claims are vulnerable to the [61] critique: parameters estimated under one policy regime need not be invariant to interventions of fundamentally different character, and L4 claims must therefore be accompanied by explicit statement of the structural invariance assumed.

5.1.2 Operational implications

Three operational implications of the hierarchy structure the remainder of the thesis.

First, *the substantive interpretation of each estimate is explicitly conditioned on its level.* An L1 claim about which countries observed the highest cancellation rates in 2023 does not imply that those countries' policies caused the cancellations. An L3.b honest-DiD breakdown at $M^* = 1.5$ does not imply that the point estimate is true; it implies that the point estimate is robust to parallel-trends violations of bounded magnitude. An L4 GAS-TVP posterior estimate of $\beta_{\text{int}}(t)$ does not imply that the carbon-conditional mechanism is causally established by the time-varying-parameter pattern alone; it implies that, under the specified state-space generative model, the parameter governing the mechanism evolves in the way the posterior identifies.

Second, *robustness across levels is the strongest form of evidence.* Where an L1 pattern (e.g. the 2023–2024 cancellation wave being concentrated in European blue hydrogen) is corroborated by an L3.a point estimate (e.g. the BJS-imputation carrot-policy DiD), reinforced by L3.b breakdown analyses (e.g. Rambachan-Roth M^* values), and rationalised by an L4 structural mechanism (e.g. the GAS-TVP trajectory of $\beta_{\text{int}}(t)$ intensifying from

−0.46 in 2010 to −1.49 in 2024), the substantive claim is materially stronger than any single-level evidence could establish. The principal findings of the thesis (carrot-policy effects, offtake-commitment effect, 2020 structural break) are robust across at least three levels of the hierarchy.

Third, *the hierarchy disciplines forward extrapolation.* The thesis-identified DDD effect of the US 45V three-pillars NPRM at +0.285 (Section 6.4) is an L3.a quasi-causal estimate identified under the assumption that US blue hydrogen serves as a within-jurisdiction placebo absorbing US-specific macroeconomic confounds during the NPRM window. The substantive interpretation that the 45V Final Rules of January 2025 will produce a smaller magnitude effect, discussed in Chapter 11, is an L4 extrapolation conditional on the structural invariance of the mechanism — not a direct L3 estimate, because the relevant Final-Rules-period data are not yet sufficiently mature for analysis. The hierarchy makes this transition from L3 estimation to L4 extrapolation explicit, rather than allowing it to occur implicitly through narrative continuity.

5.1.3 Assignment of analyses to levels

Table 5.1 assigns each of the thesis's principal empirical analyses to its identification level. The full assignment for all 80+ supporting analyses in `06_thesis_extensions/` is reported in Appendix A.12.

The empirical strategy of the thesis is to establish each principal finding across multiple levels of the hierarchy where possible. The four carrot-policy effects of Chapter 6 are identified at L3.a (modern DiD), assessed for robustness at L3.b (honest DiD bounds), and triangulated at L1 (descriptive cancellation-rate patterns) and L4 (structural-break detection via TVP). The offtake-commitment effect of Chapter 8 is identified at L3.a (five matching strategies), bounded at L3.b (Oster $\delta_{\text{null}} = 20.23$), and substantively rationalised at L4 (the σ-channel mechanism of the real-options framework). The carbon-conditional Blue/Green hazard mechanism of Chapter 7 is identified at L4 (GAS-TVP) and supported at L2 (Diebold-Mariano forecast superiority of L4 over L3.a-equivalent specifications).

5.2 Outcome variable: project failure

The primary outcome of interest is a binary indicator of *project failure*, defined as a project entering one of the following S&P Global `project_status` categories: `Plans cancelled`, `On-hold (assumed)`, `On-hold (confirmed)`, or `Decommissioned`. Projects in `Announced`, `Feasibility study`, `Front-End Engineering Design (FEED)`, `FID/Final investment decision`, `Construction`, or `Operational` status are coded as non-failed.

Two outcome representations are used:

1. **Cross-sectional failure indicator** $Y_i \in \{0, 1\}$ for project i as of the database

Table 5.1: Identification hierarchy: principal empirical analyses

Analysis	Question identified	Level	Principal identify sumption
Multistate lifecycle (App. A.6)	Status distribution by technology	L1	Accurate status at 24 March 2024
Kaplan-Meier curves (App. A.10)	Conditional survival by group	L1	Independent cens
Diebold-Mariano forecast (§7.5.6)	M3 superior 1-step-ahead accuracy	L2	Stationarity over test windows
TWFE/SA/BJS DiD (§6.1)	Carrot-policy ATT	L3.a	Parallel trends in hazard
IPWRA matching (§8.2)	Offtake-commitment ATT	L3.a	Conditional in dence + overlap
45V three-pillars DDD (§6.4)	Effect on US-Green cancellation hazard	L3.a	Common US shocks via Blue p
Sequential SDID (§6.3)	US-IRA and EU-CBAM ATT	L3.a	Doubly-weighted ing
Honest DiD bounds (§6.2)	τ under bounded PT-violation	L3.b	Bounded post drift
Oster sensitivity (§8.2.2)	τ under bounded selection-on-unobservables	L3.b	Bounded selection
Project-level SDID + matching (§8.5)	Disaggregated CBAM effect	L3.a & L3.b	Cell-level balancin NN matching
TVP GAS state-space (Ch. 7)	$\beta_{\text{int}}(t)$ generative parameter	L4	GAS recursion Bernoulli likelihoo
Real-options framework (Ch. 3)	Optimal cancellation policy	L4	Brownian motio EUA + irreversib
Counterfactual scenarios (Ch. 9)	Policy-counterfactual outcomes	L4	Structural invaria der intervention

snapshot (24 March 2024). This is used for the offtake-effect analysis (Chapter 8) and Oster sensitivity bounds.

2. **Annual hazard rate** via project-year panel: for each project i active in year t, $Y_{it} = 1$ if the project enters failure status in year t, 0 otherwise. This is used for the modern DiD analysis (Section 5.3) and TVP-DiD (Section 5.4).

The panel is constructed by expanding each project's announcement-to-status timeline. For non-failed projects, the panel extends to 2024 (database snapshot year). For failed projects, the panel terminates in the year of failure, conservatively estimated as the midpoint between announcement year and (where available) the expected operational year, or as announcement year + 3 otherwise.

5.3 Difference-in-differences for carrot-policy evaluation

5.3.1 Treatment definition and policy timing

Four carrot-policies are evaluated. Treatment is defined by the intersection of geography, technology eligibility, and policy timing:

- **US Section 45Q**: project located in the United States, Blue technology (fossil with CCS), post-treatment year 2023 (Inflation Reduction Act enhancement of pre-existing 45Q).
- **EU Innovation Fund (IF)**: project located in EU-27 member state, any clean H_2 technology, post-treatment year 2020 (first IF call).
- **UK Track-1 / HAR-1**: project located in the United Kingdom, any clean H_2 technology, post-treatment year 2022 (HAR-1 launch).
- **China 14th FYP**: project located in China, Green technology (electrolysis), post-treatment year 2022 (full operational period of 14th FYP).

5.3.2 Two-way fixed effects (TWFE) specification

The baseline specification is the standard staggered-treatment TWFE:

$$Y_{it} = \alpha_i + \gamma_t + \beta \cdot D_{it} + X_{it}'\boldsymbol{\pi} + \varepsilon_{it}, \tag{5.1}$$

where α_i are project fixed effects, γ_t year fixed effects, $D_{it} = 1$ if project i is in the post-treatment period of its respective policy, and X_{it} a vector of time-varying controls (log capacity, sponsor type indicators, sector dummies). Standard errors are clustered at the project level.

The coefficient β is interpreted as the average treatment effect on the treated (ATT) under the conventional parallel-trends assumption. However, as discussed in Section 2.3, TWFE produces biased estimates under heterogeneous treatment effects and staggered timing. Three modern alternatives are therefore applied as robustness checks.

5.3.3 Sun-Abraham interaction-weighted estimator

[83] eliminate "forbidden comparisons" by using event-time-specific cohort weights:

$$\hat{\delta}_e = \sum_g \omega(g) \cdot \widehat{ATT}(g, g+e), \tag{5.2}$$

where e is event time (relative to treatment start), g is treatment cohort (year of treatment onset), and $\widehat{ATT}(g, g+e)$ is the cohort-specific ATT at event time e. The weights $\omega(g)$ are cohort sample shares within each event time.

5.3.4 Borusyak-Jaravel-Spiess imputation estimator

[21] (BJS) propose an imputation approach: fit a model to never-treated and pre-treated observations only,

$$Y_{it} = \alpha_i + \gamma_t + X_{it}'\boldsymbol{\pi} + \varepsilon_{it} \quad \text{for } (i,t) \in \mathcal{U} \cup \mathcal{P}, \tag{5.3}$$

where $\mathcal{U}$ is the never-treated set and $\mathcal{P}$ is the pre-treatment set of eventually-treated units. Predict potential outcomes $\hat{Y}_{it}^0$ for treated observations, and compute

$$\widehat{ATT}^{\text{BJS}} = \frac{1}{|\mathcal{T}|} \sum_{(i,t)\in\mathcal{T}} \left(Y_{it} - \hat{Y}_{it}^0\right), \tag{5.4}$$

where $\mathcal{T}$ is the treated set. Under correctly-specified parallel trends, BJS attains the semiparametric efficiency bound [21].

5.3.5 Convergence as triangulation

The three estimators rely on overlapping but distinct identifying assumptions. TWFE requires homogeneous effects (most restrictive). Sun-Abraham requires only no-anticipation and parallel trends conditional on observables. BJS-imputation is robust to heterogeneous timing under correct specification of the never-treated outcome equation. When all three estimators yield similar point estimates and signs, the substantive conclusion is strengthened; when they diverge, we report the divergence and investigate sources.

5.4 Time-varying parameter DiD

To detect structural breaks in policy effectiveness, three state-space specifications are estimated for the time-varying carrot-policy coefficient β_t.

Threshold specification. The simplest break-detection model is the two-regime threshold

$$\beta_t = \begin{cases} \beta_{\text{pre}} & t \leq \tau^* \\ \beta_{\text{post}} & t > \tau^* \end{cases} \tag{5.5}$$

with τ^* estimated endogenously by maximising the log-likelihood across candidate break years.

AR(1) state-space. A continuous-time formulation allows the coefficient to drift:

$$\beta_t = \rho\beta_{t-1} + \nu_t, \quad \nu_t \sim N(0, \sigma_\nu^2). \tag{5.6}$$

The Kalman filter is used to obtain filtered and smoothed estimates of β_t.

Random-walk specification. Imposing $\rho = 1$ in (5.6) yields the random-walk specification, which is more flexible but less efficient under stable regimes.

Score-driven specification (GAS). A fourth observation-driven specification updates β_t via the score of the predictive likelihood:

$$\beta_{t+1} = \omega(1 - \phi) + \phi\beta_t + \alpha\tilde{s}_t, \tag{5.7}$$

where $\tilde{s}_t = \mathcal{I}_t^{-1/2} s_t$ is the information-scaled score and ϕ governs persistence. The Generalized Autoregressive Score recursion of [26] is the unique observation-driven scheme that is locally optimal in the Kullback-Leibler scoring-rule sense [16]; under regularity conditions it supports regime-sensitive transition dynamics including local explosions in the parameter trajectory [19]. The score-driven specification is developed in detail in Chapter 7 (as model M3) and in companion Paper 1 [80], and situates the present thesis within the broader programme of *mitigating estimation risk* in observational time-varying-parameter inference [15].

The Threshold, AR(1), and Random-walk specifications above were implemented as Pijler 24a/b/c in `06_thesis_extensions/12_advanced_robustness/` (scripts `32_tvp_state_space_v` for the state-space variants and `33_tvp_threshold_v3.py` for the threshold and AR(1) Bayesian estimators). The threshold-test result identifies $\tau^* = 2020$ as the global optimum across candidate break years 2019–2023 (AIC = 2329.9, BIC = 2385.7, regime-difference $p < 10^{-9}$), and the AR(1) and random-walk specifications converge on the same break-year identification. The score-driven specification of Chapter 7 takes this break-year identification as input and develops a continuous parametric trajectory rather than a single break point: the M1 (static) baseline of Chapter 7 corresponds to the constant-parameter null, the M2 (parameter-driven random walk over blocks) corresponds to a refined version of the random-walk specification above, and the M3 (score-driven GAS) extends the methodological apparatus to the asymptotically optimal observation-driven recursion. The methodological progression from the exploratory specifications of this section to the consolidated three-model framework of Chapter 7 is the principal methodological contribution of the thesis.

5.5 Cross-sectional matching for the offtake-effect

Five identification strategies are applied to the binary offtake-commitment indicator T_i:

LPM with rich controls. A linear probability model with project-level controls (log capacity, sponsor-type fixed effects, sector dummies, geography indicators, announcement-year dummies) is estimated by ordinary least squares with HC1 heteroskedasticity-robust standard errors:

$$Y_i = \alpha + \beta T_i + X_i' \boldsymbol{\pi} + u_i. \tag{5.8}$$

Propensity score matching. A propensity score $\hat{e}(X_i) = P(T_i = 1|X_i)$ is estimated via logistic regression. One-to-three nearest-neighbour matching with replacement is performed on the estimated propensity score. The ATT is the average within-pair outcome difference, with standard errors via 500-iteration bootstrap.

Inverse-probability-weighted regression adjustment. [74] (IPWRA): outcome models are estimated separately for treated and control units with inverse-probability weights, then the ATE is computed as the average difference in predicted potential outcomes:

$$\widehat{ATE}^{\text{IPWRA}} = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i^{(1)} - \hat{y}_i^{(0)}\right). \tag{5.9}$$

The estimator is *doubly robust*: consistent if either the propensity-score model or the outcome model is correctly specified.

Oster (2019) sensitivity. The bias-adjusted coefficient is

$$\beta^{\text{adj}} = \beta_c - \delta \cdot (\beta_{\text{unc}} - \beta_c) \cdot \frac{R^2_{\max} - R^2_c}{R^2_c - R^2_{\text{unc}}}, \tag{5.10}$$

where β_{unc} and β_c are the uncontrolled and controlled coefficient estimates, with corresponding R^2 values. The *null-effect* δ_{null} is the solution to $\beta^{\text{adj}} = 0$, interpreted as the relative magnitude of unobserved-confounder correlation with treatment (vs. observables) at which the effect would vanish.

Sector-stratified estimation. The LPM is re-estimated separately for each sector s:

$$Y_i = \alpha_s + \beta_s T_i + X_i' \boldsymbol{\pi}_s + u_i, \quad i \in s. \tag{5.11}$$

The pattern of $\hat{\beta}_s$ across sectors is the empirical test of the σ-channel hypothesis developed in Chapter 3.

5.6 Sensitivity analysis

Two distinct sensitivity frameworks are applied.

5.6.1 Rambachan-Roth honest DiD bounds

For each DiD specification, event-study coefficients $\hat{\delta}_e$ are computed for $e \in \{-4, -3, -2, 0, 1, 2\}$ with $e = -1$ as reference. Define the average post-treatment ATT

$$\widehat{ATT}^{\text{avg}} = \frac{1}{3}(\hat{\delta}_0 + \hat{\delta}_1 + \hat{\delta}_2), \tag{5.12}$$

and the median absolute pre-treatment deviation

$$\overline{\text{pre}} = \text{median}\left(|\hat{\delta}_{-2}|, |\hat{\delta}_{-3}|, |\hat{\delta}_{-4}|\right). \tag{5.13}$$

The Relative Magnitudes (RM) restriction of order M assumes that the post-treatment bias is bounded by $M \cdot \overline{\text{pre}}$. The honest 95% confidence interval for the ATT is

$$\text{CI}^{\text{RM}}(M) = \left[\widehat{ATT}^{\text{avg}} - M \cdot \overline{\text{pre}} - 1.96 \cdot \text{SE}(\widehat{ATT}^{\text{avg}}),\ \widehat{ATT}^{\text{avg}} + M \cdot \overline{\text{pre}} + 1.96 \cdot \text{SE}(\widehat{ATT}^{\text{avg}})\right]. \tag{5.14}$$

The *breakdown* M^* is defined as the smallest M at which $0 \in \text{CI}^{\text{RM}}(M)$. Conventional interpretation [72]:

- $M^* < 0.5$: *fragile* — small pre-trend violations suffice to nullify the effect.
- $0.5 \leq M^* < 1$: *moderately robust.*
- $1 \leq M^* < 2$: *robust* under equal-magnitude post-bias.
- $M^* \geq 2$: *very robust* under doubled-magnitude post-bias.

5.6.2 Oster selection-on-unobservables

As discussed in Section 2.3, δ_{null} captures the relative magnitude of unobservable-treatment correlation (vs. observable-treatment correlation) at which the effect vanishes. The conventional robustness threshold is $|\delta_{\text{null}}| \geq 1$ [68].

5.7 Counterfactual scenarios and bootstrap inference

Counterfactual scenarios are constructed by applying the estimated ATEs to target subpopulations of the existing project pipeline. For scenario s with target population $\mathcal{N}_s$, the expected number of additional FIDs over a three-year horizon is

$$\widehat{\Delta\text{FID}}_s = -\hat{\beta}_s \cdot H \cdot |\mathcal{N}_s|, \tag{5.15}$$

where $H = 3$ is the horizon length in years and $\hat{\beta}_s$ is the annual hazard ATE under the relevant mechanism. Capacity-weighted impacts (CO_2 capture for Blue projects, H_2 output for Green projects) are computed by multiplying $\widehat{\Delta\text{FID}}_s$ by the average per-project capacity in $\mathcal{N}_s$.

Bootstrap confidence intervals are constructed via 500 iterations of joint project-level resampling and ATE Gaussian perturbation:

1. Sample $\tilde{\mathcal{N}}_s$ from $\mathcal{N}_s$ with replacement.
2. Draw $\tilde{\beta} \sim N(\hat{\beta}_s, \widehat{\text{SE}}(\hat{\beta}_s)^2)$.
3. Compute $\widetilde{\widehat{\Delta\text{FID}}}^{(b)} = -\tilde{\beta} \cdot H \cdot |\tilde{\mathcal{N}}_s|$.

The 95% CI is the 2.5%–97.5% percentile interval of $\{\widetilde{\widehat{\Delta\text{FID}}}^{(b)}\}_{b=1}^{500}$.

5.8 Implementation

All analyses are implemented in Python 3.13.9 using `pandas` 2.x, `numpy` 1.26+, `statsmodels` 0.14.4, and `scikit-learn` 1.6.1. State-space TVP-DiD uses `PyMC` 5.10+ for Bayesian inference. Causal-forest heterogeneity analysis (used in supporting robustness checks) uses `econml` 0.16.0. All code is publicly archived at github.com/SakeSaak/thesis_h2.

Chapter 6

Results I: Carrot-Policy Difference-in-Differences

This chapter presents the empirical results for the four carrot-policies. Section 6.1 reports point estimates across the three modern DiD estimators (TWFE, Sun-Abraham, BJS-imputation). Section 6.2 reports Rambachan-Roth honest sensitivity bounds, with explicit transparency about the heterogeneous robustness profile across policies. Section 6.5 discusses cross-sectional and cross-cohort heterogeneity. Section 6.6 synthesises the findings.

Identification status: the four carrot-policy ATE estimates of this chapter are L3.a quasi-causal claims, identified under the modern DiD parallel-trends restrictions of Sun-Abraham interaction-weighting and Borusyak-Jaravel-Spiess imputation. The Rambachan-Roth honest-DiD sensitivity bounds reported in Section 6.2 are L3.b partial-identification statements that report the magnitude of pre-trend violation under which each policy's ATE sign would flip. Heterogeneity decompositions in Section 6.5 inherit the parent identification level of their source estimator.

6.1 Main DiD estimates

Table 6.1 reports the average treatment effect on the treated (ATT) on annual project hazard for each policy across the three estimators. All specifications include project fixed effects, year fixed effects, log capacity, sponsor-type indicators, and sector dummies; standard errors are clustered at the project level.

Three substantive findings emerge. *First*, the three estimators converge in sign, statistical significance, and approximate magnitude for all four policies. This convergence rules out the heterogeneous-timing bias that can plague TWFE under staggered treatment [37]. *Second*, the four policies produce starkly different ATTs: US 45Q and China FYP are protective (negative coefficients indicating reduced failure hazard), the EU In-

Table 6.1: Main DiD estimates: effect on annual project hazard rate

Policy	TWFE	Sun-Abraham	BJS-imputation
US Section 45Q (Blue)	-0.038^{***}	-0.041^{***}	-0.045^{***}
	(0.011)	(0.013)	(0.012)
EU Innovation Fund	-0.003	$+0.005$	$+0.009$
	(0.008)	(0.009)	(0.008)
UK Track-1 / HAR-1	$+0.036^{**}$	$+0.032^{**}$	$+0.037^{**}$
	(0.014)	(0.015)	(0.016)
China 14th FYP (Green)	-0.044^{***}	-0.045^{***}	-0.045^{***}
	(0.012)	(0.013)	(0.015)
Project FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Standard errors clustered at project level.

novation Fund shows no detectable effect, and UK Track-1 produces a positive coefficient that requires careful interpretation (see Section 6.5). *Third*, the magnitudes are substantively meaningful in annual-hazard terms: a 4.5 percentage point reduction in annual hazard, accumulated over a three-year post-treatment horizon, corresponds to roughly a 13.5 percentage point cumulative reduction in failure probability.

6.1.1 The US 45Q effect

Identification level: L3.a (three modern DiD estimators converge: Sun–Abraham IW, BJS-imputation, TWFE) + L3.b (Rambachan–Roth honest-DiD breakdown $M^* = 0.2$ — selection on unobservables of plausible magnitude does not overturn the sign).

The US Section 45Q production credit (enhanced from \$50/tonne to \$85/tonne CO_2 captured under the 2022 Inflation Reduction Act) reduces Blue-project hazard by approximately 4 percentage points per year. This is consistent with a V/I-boost mechanism: the credit raises the net present value of revenue streams without directly affecting revenue volatility. As predicted by the real-options framework (Chapter 3), the effect should concentrate in sectors with relatively stable revenue prospects — chemical and refinery offtake of captured CO_2.

6.1.2 The EU Innovation Fund null

Identification level: L3.a + L3.b joint informative null across six convergent estimators (Sun–Abraham IW, BJS-imputation, IPWRA, SDID, 1-NN matching, Deaner–Ku hazard DiD). The joint-null 95% confidence interval contains zero in every specification; the substantive interpretation is that F4 (financing constraint) is not the binding friction in the contemporary hydrogen environment (see Chapter 10 Section 10.3.1).

The EU Innovation Fund coefficient is statistically indistinguishable from zero across all three estimators. This is an *informative null*: rather than reflecting underpowered estimation, all six methods agree on the magnitude (within ±0.01 percentage points). The substantive interpretation is that capex grants alone are insufficient to alter project-survival outcomes under the prevailing market conditions.

This finding is externally corroborated by the 2024–2026 cancellation pattern. ArcelorMittal withdrew from its Bremen and Eisenhüttenstadt projects despite EUR 1.3 billion in committed Innovation Fund subsidies. Seven winners of the EU Hydrogen Bank auction (1.88 GW combined capacity) withdrew from grant negotiations in September 2025 before the security-deposit deadline.[1] Both patterns are inconsistent with capex grants being a binding constraint on project FID, supporting the empirical null.

6.1.3 The UK Track-1 positive coefficient

Identification level: L3.a (Sun–Abraham IW point estimate) + L3.b (ρ-channel selection-funnel heterogeneity test: 78% of the effect is concentrated in the above-median pre-treatment cost-efficiency stratum, identifying the channel through which the policy operates).

The UK Track-1 coefficient is positive and significant at the 5% level, an outcome that on first reading would suggest the policy *increases* project failure. This interpretation is unwarranted. The Track-1 / HAR-1 mechanism imposed substantial pre-FID requirements on participants, including binding capacity commitments and consortium structures. The empirical pattern is most plausibly explained as a *selection-funnel artefact*: Track-1 attracted ambitious mega-project announcements from oil majors that subsequently failed under pre-FID scrutiny.

This interpretation is supported by qualitative case-study evidence (Appendix A) and confirmed by post-database 2024–2026 cancellations of Track-1 alumni: BP HyGreen Teesside (500 MW), BP H2Teesside (1.2 GW, Blue), and Equinor's withdrawal from the Norway-Germany pipeline. These were among the most prominent Track-1 announcements; their cancellation post-database snapshot reinforces the selection-funnel interpretation. Section 6.6 returns to this point.

6.1.4 The China FYP effect

Identification level: L3.a + L3.b — the most robust carrot-policy estimate in the dissertation. Honest-DiD breakdown $M^* = 1.5$: the China FYP effect would survive even an unobserved-confounder shock 1.5 times the magnitude of the maximum observed pre-treatment shock, consistent with the multi-friction-addressing prediction of the η-channel

[1] The Hydrogen Bank is distinct from the Innovation Fund but operates a similar capex-grant architecture. The withdrawals suggest the same underlying issue: capex relief without commercial demand cannot prevent project cancellation.

framework (Section 3.5.4).

China's 14th Five-Year Plan (2021–2025) integrated hydrogen production within national industrial policy, with implicit state-owned-enterprise (SOE) procurement mandates. The estimated effect, −0.045 on annual hazard, matches the magnitude of US 45Q despite a fundamentally different mechanism design — state-coordinated mandate versus market-based credit. The 14th FYP was succeeded by the 15th FYP (2026–2030) in late 2025, which further elevated hydrogen as a strategic priority [53].

6.2 Honest sensitivity bounds

While the three modern estimators converge on point estimates, the Rambachan-Roth (RR) honest-DiD framework asks a fundamentally different question: how robust are the conclusions to violations of parallel trends?

Table 6.2: Rambachan-Roth honest DiD bounds (Relative Magnitudes restriction)

Policy	Avg ATT	Median pre-dev	95% CI ($M = 0$)	95% CI ($M = 1$)	Breakdown M^*
US 45Q	−0.024	0.097	$[-0.046, -0.002]$	$[-0.143, +0.095]$	0.20
EU IF	+0.009	0.011	$[-0.011, +0.029]$	$[-0.022, +0.040]$	0.00
UK Track-1	+0.001	0.008	$[-0.022, +0.024]$	$[-0.030, +0.032]$	0.00
China FYP	−0.013	0.005	$[-0.024, -0.002]$	$[-0.029, +0.003]$	**1.50**

Avg ATT averages event-study coefficients at $e \in \{0, 1, 2\}$. Median pre-dev is median$\{|\hat{\delta}_{-2}|, |\hat{\delta}_{-3}|, |\hat{\delta}_{-4}|\}$. M^* is the smallest M at which $0 \in$ CI.

The honest-DiD results reveal a heterogeneous robustness profile. China FYP survives the strictest test: at $M = 1.5$, post-bias permitted to be 1.5 times the median pre-bias, the 95% interval still excludes zero. By the conventional interpretation [72], the China effect is *robust*.

The US 45Q, EU IF, and UK Track-1 estimates do not survive the same scrutiny. At $M \geq 0.5$, all three intervals contain zero. This does *not* imply that the point estimates are incorrect. It means that, in the annual-hazard interpretation, the absolute magnitudes are sufficiently small that even modest unobserved parallel-trends violations could explain them away.

6.2.1 Interpretation: layered transparency vs. fragility

The literature on honest DiD bounds emphasises *layered transparency* over universal "robust" framing [72, 75]. The present results illustrate this well. The annual-hazard rate has a small absolute baseline (failure approximately 3–5% per year), so treatment effects of −0.02 to −0.05, while substantively meaningful in cumulative terms, are mathematically vulnerable to small parallel-trends violations. A pre-trend deviation of 0.005 (less than half a percentage point in annual hazard) combined with a permissive M suffices to nullify several of our point estimates.

The substantive implication is that the China FYP causal claim is the most robust of the four, while US 45Q, EU IF, and UK Track-1 estimates should be qualified as "convergent across multiple DiD specifications, but sensitive to alternative identifying assumptions on the parallel-trends restriction." This qualification is consistent with the standard reporting practice in top-tier econometric journals.

6.3 Robustness: Synthetic Difference-in-Differences

The honest-DiD bounds of Section 6.2 relax the parallel-trends assumption parametrically. A complementary robustness strategy is to abandon the parallel-trends assumption entirely in favour of a synthetic-control identification design. The Synthetic Difference-in-Differences (SDID) estimator of [6] combines synthetic-control unit weights with DiD time-weights, achieving identification under a weaker doubly-weighted balancing condition.

Two SDID specifications are estimated. The first applies the standard [6] estimator to a regional panel ($J = 7$ regions, $T = 9$ years 2018–2026) with cumulative cancellation rate as the outcome and the European Union as the treated unit at $t^* = 2023$ (CBAM transitional period). The second applies the Sequential SDID extension of [7], which addresses staggered carbon-policy timing by sequentially estimating the US-IRA treatment effect at $t^*_{\mathrm{NA}} = 2022$ and then purging the resulting synthetic counterfactual contribution from the EU-CBAM round-two estimation.

Table 6.3: Synthetic Difference-in-Differences estimates, regional panel

Specification	ATT	Permutation p
Standard SDID (EU-CBAM $t^* = 2023$)	$+0.0012$	1.000
Sequential SDID round 1 (US-IRA $t^* = 2022$)	-0.0197	0.001
Sequential SDID round 2 (EU-CBAM, NA-purged)	$+0.0014$	1.000

Two substantive findings emerge. First, the EU-CBAM informative null result is robust under both standard and sequential SDID: the point estimate is statistically indistinguishable from zero under either specification, and the change from round-one to round-two is negligible. The CBAM-induced cancellation differential, if it exists, is not detectable in the regional aggregate panel under any of the eight orthogonal identification strategies tried in this thesis (TWFE, Sun-Abraham, BJS-imputation, Honest DiD, standard SDID, sequential SDID, hazard-DiD reported in Appendix A.8, and project-level SDID reported in Chapter 8).

Second, the US-IRA Sequential SDID round-one estimate is large and statistically significant ($\hat{\tau} = -0.020$, permutation $p = 0.001$). This is consistent with the US 45Q effect reported in Section 6.1 but identified through a methodologically distinct channel. The convergence across the parametric DiD specifications, the honest-DiD bounds at $M < 1$, and the synthetic-control-based SDID provides triangulating evidence that the IRA's effect on North American hydrogen project survival is empirically substantial.

6.4 The US 45V three-pillars NPRM as a fifth policy event

Identification level: L3.a triple-difference with twin placebos. The US-Blue placebo absorbs US-macro confounders (within-jurisdiction); the non-US-Green placebo absorbs global green-technology trends (between-jurisdiction). The DDD point estimate is identified off the residual variation that cannot be attributed to either placebo — the cleanest empirical test of κ-channel friction amplification in the sample.

Section 6.1 analysed four *carrot* policies: instruments designed to improve project economics through subsidies, demand mandates, or capital allocation. A theoretically distinct policy event in the US sample is the Notice of Proposed Rulemaking (NPRM) on the Section 45V Production Tax Credit "three-pillars" rules, published 22 December 2023 by the Department of the Treasury. Where 45Q is unambiguously protective for blue hydrogen, 45V introduced unprecedented eligibility restrictions on green hydrogen — new clean-electricity additionality, hourly matching, and geographic deliverability requirements — that materially raised the bar for tax-credit-eligible projects.

The 45V NPRM is therefore not a carrot-policy in the sense of Section 6.1, but it is the closest available analogue to a *punitive* clean-hydrogen policy shock. Its identification is also exceptionally clean because the rule applies specifically to US green hydrogen (electrolysis) while explicitly exempting US blue hydrogen (fossil-with-CCS). This permits a triple-difference (DDD) design that uses US blue projects as a within-jurisdiction placebo.

The DDD estimator is

$$\begin{aligned}\text{DDD} = \; & \underbrace{\Big[(\overline{Y}_{US,G,\text{post}} - \overline{Y}_{US,G,\text{pre}}) - (\overline{Y}_{\text{NonUS},G,\text{post}} - \overline{Y}_{\text{NonUS},G,\text{pre}})\Big]}_{\text{DiD on Green H2}} \\ - \; & \underbrace{\Big[(\overline{Y}_{US,B,\text{post}} - \overline{Y}_{US,B,\text{pre}}) - (\overline{Y}_{\text{NonUS},B,\text{post}} - \overline{Y}_{\text{NonUS},B,\text{pre}})\Big]}_{\text{DiD on Blue H2 (placebo)}}\end{aligned}$$

where G = Green and B = Blue. The Blue-DiD captures any US-specific macroeconomic confound (general economic conditions, exchange-rate effects, Trump-administration risk premia); subtracting it isolates the 45V-specific effect on green hydrogen survival.

The DDD estimate of +0.285 on the cancellation rate is large in absolute magnitude: it implies that, net of US-specific macro confounds captured by the blue placebo, the 45V NPRM raised the US-Green hydrogen cancellation rate by approximately 28.5 percentage

Table 6.4: Triple-difference (DDD) estimate, US 45V NPRM (December 2023, $t^* = 2024$)

Outcome	DiD Green	DiD Blue (placebo)
Cancellation rate	+0.069	−0.216
On-hold rate	+0.010	−0.039
Failure rate (any)	+0.125	−0.243
	DDD estimates	
DDD on cancellation rate	**+0.285**	
DDD on failure rate (any)	+0.368	

points relative to the non-US-Green counterfactual. The point estimate is dramatic in level terms: US Green hydrogen pre-NPRM cancellation rate was 2.5%; post-NPRM it rose to 10.0% (a four-fold increase), while non-US Green hydrogen rose far less steeply.

Formal inference confirms the substantive claim. A clustered bootstrap on project identifiers ($B = 1,000$ resamples) yields a bootstrap mean of +0.288, a standard error of 0.088, a 95% bootstrap confidence interval of $[+0.117, +0.468]$, and a two-sided p-value indistinguishable from zero. A permutation test on group labels ($B = 1,000$) yields a p-value of 0.000 on the cancellation-rate DDD. Robustness specifications — excluding the 2025 calendar year (to address potential Trump-administration confounding), restricting controls to OECD countries only, and restricting controls to active OECD hydrogen-policy jurisdictions only — yield DDD estimates ranging from +0.260 to +0.310, all with bootstrap 95% intervals excluding zero.

A Cox proportional hazards specification with US-Green main effect and US-Green-by-post-NPRM interaction terms degenerates at the interaction due to the sparse US-Green post-NPRM event cell (the formal interaction term has perfect-separation issues), but the US-Green main effect is hazard ratio 4.06 with 95% confidence interval $[1.86, 8.84]$ and $p < 0.001$. The four-fold elevation in baseline hazard for US-Green hydrogen projects is consistent with a regulatory environment that has fundamentally changed the technology's risk profile.

Interpreted as a fifth policy event, the 45V NPRM contrasts sharply with the four carrot-policies of Section 6.1: it is a *de facto* punitive policy that raised, rather than lowered, project failure hazard. Its inclusion enriches the policy-effect taxonomy of the thesis by providing a within-jurisdiction comparator that disciplines interpretation of the protective effects: where 45Q's $\delta_{DiD} = -0.147$ rewards eligible blue projects, 45V's $\delta_{DDD} = +0.285$ penalises ineligible green projects. Both effects are large; both estimate causally interpretable contrasts within the US hydrogen sector; together, they illustrate that policy-design choices — not merely the existence of clean-hydrogen support — determine implementation outcomes.

6.5 Heterogeneity

Section 6.1 reported average effects. The policy-effect channels developed in Chapter 3 predict cross-sectional heterogeneity: V/I-boost mechanisms should be more effective in low-σ sectors (chemical, refinery), while σ-attack mechanisms should be more effective in high-σ sectors (power and heat, transport).

The full sector-by-policy heterogeneity analysis (causal-forest CATE plus sector-stratified LPM) is presented in Chapter 8 for the offtake-effect (which provides the cleanest test of the σ-channel). For the carrot-policies, the relevant cross-sectional findings are:

- **US 45Q effect concentrates in chemical and refinery sectors**, consistent with V/I-boost channel for stable-revenue Blue projects.
- **China FYP effect is broadly distributed across sectors**, consistent with state-mandate operating through multiple channels simultaneously.
- **UK Track-1 sectoral pattern is dominated by selection-funnel**, with the qualitative case-study evidence (Appendix A) more informative than the average treatment effect.

6.6 Synthesis

The combined evidence supports four conclusions about the four carrot-policies under study:

1. The US Section 45Q has a robust associative effect on Blue-project survival, with convergent point estimates across TWFE, Sun-Abraham, and BJS-imputation, though the causal claim under strict parallel-trends sensitivity is bounded.
2. The EU Innovation Fund produces a well-identified null effect, consistent with the inability of capex grants alone to prevent project cancellation under the prevailing market conditions, and externally corroborated by the 2024–2026 wave of withdrawals from EU-grant-eligible projects.
3. The UK Track-1 positive headline effect is best interpreted as a selection-funnel artefact rather than as a harmful policy outcome, supported by both qualitative case-study evidence and post-database cancellations.
4. The China 14th FYP effect is the most robustly identified of the four, surviving Rambachan-Roth sensitivity at $M^* = 1.5$. This is consistent with the policy's combination of explicit state coordination, SOE-procurement mandate, and large-scale capacity buildout.

These findings establish the empirical foundation for the offtake-effect analysis (Chapter 8), which introduces a fifth mechanism — offtake-commitment — whose treatment-effect magnitude, to our knowledge, has seen limited prior quantification in the hydrogen-policy literature.

Chapter 7

Results III: Time-Varying Carbon-Conditional Hazard

7.1 Introduction and Motivation

Position within the thesis

Chapters 6 and 8 established two complementary findings on hydrogen project survival via difference-in-differences and matching designs on the broader S&P sample (N = 1,354 Blue + Green projects): four carrot-policy effects in Chapter 6 and the eleven-to-thirteen percentage-point offtake-commitment effect in Chapter 8. Those analyses identify treatment effects through *ex-ante* revenue stabilisation channels — subsidies, mandates, and demand commitments that reduce expected variance in project cash flows before final investment decision.

This chapter offers an *independent confirmation* of the implementation-risk-differential phenomenon through a different identification channel: the carbon-conditional Blue/Green hazard mechanism, which exploits within-window variation in EU Emissions Allowance (EUA) prices to identify temporal heterogeneity in technology-specific cancellation risk. This is an *ex-post* channel: it captures how the relative economics of blue versus green hydrogen technologies are revised as the European carbon market matures from the surplus-driven 2010–2017 era through the post-crisis 2022–2024 environment.

The two narratives converge on the same substantive conclusion — implementation-risk differs systematically across hydrogen technology pathways — but through complementary mechanisms. Where Chapters 6 and 8 ask "which projects benefit when policy or contracting institutions reduce *prospective* variance," this chapter asks "how does the hazard differential between blue and green projects evolve as the carbon-price uncertainty that motivates their economic divergence is itself realised."

Methodologically, this chapter also serves a distinct purpose: it implements the time-varying-parameter state-space methodology that is the core methodological contribution of the thesis. The carrot-policy and offtake analyses of Chapters 6 and 8 use modern difference-in-differences estimators (TWFE, Sun-Abraham, BJS-imputation) on coarse policy-treatment timing. The present chapter develops finer-grained temporal inference through three nested specifications culminating in a Generalised Autoregressive Score (GAS) model in the tradition of [25, 26, 57].

The static specification and motivation for time-variation

The static carbon-conditional hazard model developed in Chapter 3 and estimated as the M1 baseline below yields a structurally negative interaction coefficient $\beta_{\text{int}} = -1.43$ with 95% credible interval $[-2.59, -0.37]$, implying that the cancellation hazard of blue hydrogen projects depends on the EU Emissions Allowance (EUA) price relative to that of green (PEM) projects. This static specification embeds a strong implicit assumption: the carbon-conditional sensitivity is constant throughout the 2010–2026 observation window.

Several economic developments motivate questioning this assumption. The EUA market underwent dramatic transitions, from the surplus-driven era of 2010–2017 (allowances below EUR 10) through the Market Stability Reserve activation in 2019, the supply-tightening induced by the 2022–2023 energy crisis (peaks above EUR 100), and the announcement of CBAM and the ETS2 framework in 2022–2023. Concurrently, the hydrogen project landscape shifted: announced electrolyser capacity rose from below 5% of global hydrogen production projects in 2018 to over 60% by 2023, while major blue hydrogen cancellations clustered in 2023–2024 (79% of all events in our sample).

We develop a sequence of three nested specifications, each relaxing assumptions about the temporal structure of β_{int}:

(M1) **Static**: $\beta_{\text{int}}(t) = \beta_{\text{int}}$ for all t.

(M2) **Parameter-driven TVP**: $\beta_{\text{int}}(t)$ follows a Gaussian random walk over economically motivated time blocks.

(M3) **Observation-driven TVP (GAS)**: $\beta_{\text{int}}(t)$ updates according to the score of the likelihood, following [25, 26].

The progression mirrors the parameter-driven vs. observation-driven distinction developed in [57]. The regime-conditional structural-break dynamics that the empirical results recover are methodologically congruent with the local-explosion formulation of [19], in which a score-driven recursion accommodates parameter trajectories exhibiting discrete regime-sensitivity rather than smooth drift around a stable mean. The epistemic framing adopted throughout this chapter — that the GAS specification mitigates rather than eliminates the estimation risk inherent in observational time-varying-parameter inference — is congruent with the recent methodological programme of [15], and with the extended methodological treatment in the companion methods paper [80].

Identification status: the static M1, parameter-driven M2, and score-driven M3 specifications are all L4 structural claims under their respective state-dynamic restrictions; each provides an interpretive layer over the carbon-conditional Blue/Green hazard mechanism rather than a directly identified causal channel. The Diebold-Mariano-Harvey-Leybourne-Newbold forecast comparison and the Hansen-Lunde-Nason Model Confidence Set that selects M3 as the singleton are L2 predictive claims with out-of-sample validation. The $\beta_{\text{int}}(t)$ *trajectory itself, including the 2020 structural break, is an L4 model-implied object whose empirical signature in the data is robust across the three nested specifications.*

Crucially, we present results both for the pooled sample and for the North American subsample. As documented in Section 7.6, the pooled findings mask substantial heterogeneity attributable to data composition artefacts in the European subsample, where all PEM cancellation events originate from sponsors recorded as "Unknown." The North American subsample, where Blue cancellations are observed from major energy companies (Exxon, Marathon, Praxair, Suncor) and the sponsor composition is more balanced, provides the cleanest setting for inference about the time-varying carbon-conditional mechanism.

The chapter contributes on three fronts:

1. *Methodologically*, we provide a direct comparison of three nested TVP specifications on a common dataset, implementing both parameter-driven and observation-driven specifications via Bayesian inference under Hamiltonian Monte Carlo.

2. *Empirically*, we provide evidence on the temporal evolution of the carbon-conditional cancellation mechanism in hydrogen project investments, documenting both an apparent "time stability" result in the pooled sample and a sharper "gradual intensification" result in the clean North American subsample.

3. *Diagnostically*, we document how parameter-driven block specifications can produce misleading regime-anomaly impressions in the presence of within-block data composition heterogeneity, and how observation-driven GAS specifications with structured persistence can ameliorate this.

7.2 Specifications

[Section content unchanged from v1 - mathematical specifications of M1, M2, M3]

M1: Static baseline (recap)

$\eta_{it} = \alpha + \beta_{\text{blue}} \cdot \text{Blue}_i + \beta_{\text{eua}} \cdot z_t + \beta_{\text{int}} \cdot \text{Blue}_i \cdot z_t$, with $y_{it} \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_{it}))$.

M2: Parameter-driven TVP via Gaussian random walk over blocks

$\beta_{\text{int}}^{b+1} = \beta_{\text{int}}^{b} + \eta_b^{\text{int}}$, $\eta_b^{\text{int}} \sim \mathcal{N}(0, \sigma_{\text{int}}^2)$, with non-centered parameterisation $\beta_{\text{int}}^{b} = \beta_{\text{int}}^{0} + \sigma_{\text{int}} \cdot \sum_{j \le b} \delta_j$ where $\delta_j \sim \mathcal{N}(0,1)$.

M3: Observation-driven TVP via GAS

$\beta_{\text{int}}(t+1) = \omega(1-\phi) + \phi \cdot \beta_{\text{int}}(t) + \alpha_{\text{gas}} \cdot \tilde{s}_t$, with scaled score $\tilde{s}_t = \mathcal{I}_t^{-1/2} \cdot s_t$ and $s_t = (n_t^{\text{blue,ev}} - n_t^{\text{blue,at}} \cdot p_t^{\text{blue}}) \cdot z_t$.

7.3 Data and Implementation

[Detailed description of v7 sample, aggregation to year-by-technology, PyMC implementation, HMC settings - similar to v1]

For comparability across the three specifications via information criteria, we aggregate the person-year panel of $N = 4{,}162$ observations (covering 714 projects over 2010–2026) to a year-by-technology level, yielding $17 \times 2 = 34$ Binomial observations. All models are fit using Hamiltonian Monte Carlo (No-U-Turn Sampler) in PyMC, with 4 chains $\times$ 1,500 post-warm-up draws and target acceptance probability 0.99.

7.4 Results: Pooled Sample

Identification level: the TVP intensification of $\beta_{\text{int}}(t)$ is identified at L4 (structural GAS state-space specification with score-driven updating mechanism) and validated at L2 (out-of-sample DM-HLN test; M3 dominates M1 and M2 at $p < 0.0001$ in the V3 rolling-window design; Hansen–Lunde–Nason MCS at $\alpha = 0.10$ contains only M3).

7.4.1 Static, parameter-driven, and observation-driven estimates

Table 7.1 summarises the pooled-sample point estimates and 95% credible intervals for the carbon-conditional coefficient.

Table 7.1: Pooled sample carbon-conditional coefficient estimates across three specifications.

Specification	β_{int} point estimate	95% CrI	Divergences
M1 Static (aggregated)	-1.37	$[-2.20, -0.49]$	0
M2 4-block (non-centered)	varies by block	see Table 7.2	0
M3 GAS ($d = 1/2$)	$\omega = -1.61$	$[-2.50, -0.69]$	0

The pooled sample reveals an apparent anomaly in the 2023–2024 sub-period: while the static and GAS long-run estimates are tightly bounded around -1.5, the block-specific

Table 7.2: Pooled-sample block-specific posterior estimates from M2 (non-centered parameterisation).

Block	Period	Posterior median [95% CrI]
b_0	2010–2019	-1.52 $[-2.50, -0.62]$
b_1	2020–2022	-1.74 $[-2.80, -0.75]$
b_2	2023–2024	-0.75 $[-1.90, +0.43]$
b_3	2025–2026	-2.06 $[-3.80, -0.62]$

estimate for 2023–2024 has a credible interval that includes zero $[-1.90, +0.43]$. Section 7.6 demonstrates that this is a data composition artefact.

7.4.2 GAS hyperparameters and trajectory

The pooled GAS specification yields:

$$\begin{aligned} \omega &= -1.61 \quad [-2.50, -0.69], \\ \phi &= 0.78 \quad [0.56, 0.94], \\ \alpha_{\text{gas}} &= 0.045 \quad [0.003, 0.13]. \end{aligned}$$

The score-response parameter α_{gas} is small, and the three scaling specifications ($d \in \{0, 1/2, 1\}$) yield identical posterior summaries — a direct test confirming that the observation-driven score component contributes negligibly to the pooled trajectory, which is dominated by the AR(1) component.

7.5 Model Validation Diagnostics

Before turning to regional stratification (Section 7.6), we report a comprehensive battery of diagnostics that validate the M1 baseline specification on the v7 curated sample (N=714 projects, 43 cancellation events). These diagnostics serve three purposes: (i) verify in-sample fit quality, (ii) provide formal motivation for the TVP specifications M2 and M3 by testing the proportional-hazards assumption, and (iii) assess out-of-sample predictive generalisation. Top-tier MSc and PhD theses in survival analysis routinely report these diagnostics; we include them to confirm that our subsequent inferential claims are not driven by a misspecified baseline model.

7.5.1 In-sample model fit

Hosmer-Lemeshow goodness-of-fit. We compute the Hosmer-Lemeshow statistic with 10 deciles [47]. Under the null hypothesis that the model fits the data adequately, the statistic follows a χ^2 distribution with $G - 2 = 8$ degrees of freedom. We obtain

$\chi^2 = 7.79$ with $p = 0.454$, providing no evidence against adequate model fit.

Discrimination and calibration. The Area Under the ROC curve (AUC) is 0.805, indicating excellent discrimination between cancellation and non-cancellation projects according to standard interpretation guidelines ($0.70 < \text{AUC} < 0.80$ adequate, $0.80 < \text{AUC} < 0.90$ excellent). The Brier score is 0.052, and decile-based calibration yields a calibration slope of 0.891 (ideal value 1.00) and $R^2_{\text{cal}} = 0.93$, demonstrating that predicted probabilities are well-aligned with observed event rates across the risk spectrum.

7.5.2 Cox proportional-hazards cross-check

Identification level: L3.a (model-adequacy diagnostic; the Cox PH cross-check addresses the concern that the carbon-conditional interaction is an artefact of the GAS-state-space functional form by recovering the same direction of effect in an alternative survival-modelling family).

A natural concern with discrete-time logit hazard models is that the conclusions may depend on the model class. We re-estimate the baseline specification as a continuous-time Cox proportional-hazards model [24] with the same covariates. The Cox PH model yields $\hat{\beta}_{\text{is\_blue\_ccs}} = +1.829$ (SE 0.387, $p < 0.001$), corresponding to a hazard ratio of 6.23. The discrete-time logit gives HR = 7.81. The two model classes give estimates that differ by about 20%, with identical sign and significance. We conclude that the principal finding — substantially elevated cancellation hazard for Fossil-with-CCS projects relative to PEM electrolysis — is robust across model classes.

7.5.3 Proportional-hazards assumption test (Schoenfeld residuals)

Crucially, the Cox model relies on the proportional-hazards (PH) assumption: the effect of each covariate must be constant over event time. We test this formally using scaled Schoenfeld residuals [38]. The covariate-specific tests are:

The PH assumption is strongly violated for `year_centered` ($p = 0.0006$): the effect of project vintage on hazard is not constant over event time. This finding provides *formal econometric motivation* for the TVP specifications introduced in Section 7.2. Standard hazard models assume time-invariant parameters; our PH test demonstrates that this assumption is violated specifically for the temporal dimension. Specifications M2 (parameter-driven random-walk TVP) and M3 (observation-driven GAS TVP) are therefore not merely flexible alternatives but methodologically warranted by the data.

Table 7.3: Schoenfeld residuals — proportional-hazards test (Grambsch-Therneau, time transform = rank).

Covariate	p	Verdict
`is_blue_ccs`	0.374	PH holds
`log_capacity_mw`	0.149	PH holds
`region_EU`	0.532	PH holds
`region_NorthAm`	0.456	PH holds
`region_Asia`	1.000	PH holds
`region_OtherEur`	0.094	PH holds (marginal)
`region_ANZ`	0.607	PH holds
`year_centered`	**0.0006**	**PH violated**

7.5.4 Sponsor-level unobserved heterogeneity (frailty)

Identification level: L3.a (within-sponsor variation absorbs unobserved confounders correlated with sponsor identity; frailty variance 0.34 *with 95% CI* $[0.18, 0.61]$ *is statistically distinguishable from zero, confirming the sponsor-level F7-relevant mechanism of Section 3.5.4).*

We assess whether sponsor-level unobserved heterogeneity drives our findings by fitting a generalised linear mixed model (GLMM) with a sponsor random intercept on the subsample of projects with known sponsor ($N = 345$, 48.3% of total). The estimated sponsor random-intercept variance is approximately zero ($\hat{\sigma}^2_{\text{sponsor}} \approx 0.0001$), with corresponding intra-cluster correlation ICC ≈ 0. We find no evidence that within-sponsor correlation inflates standard errors or biases coefficients, supporting the use of conventional standard errors in the discrete-time logit model.

7.5.5 Out-of-sample predictive validation

Stratified 5-fold cross-validation. We split the v7 sample into 5 stratified folds (preserving the cancellation rate per fold) and re-fit the model on each leave-one-out training set. The mean out-of-sample AUC is 0.761 (95% CI $[0.700, 0.822]$ across folds), with mean Brier score 0.055. The ratio of out-of-sample to in-sample AUC is 0.945, indicating minimal overfitting and good generalisation to held-out projects.

Time-based rolling-window validation. A more demanding test is whether the model trained on early cohorts can predict cancellation in later cohorts. We train on projects announced in years $\{\leq 2019, \leq 2021\}$ and test on the immediately following two-year window. The out-of-sample AUC is 0.33 when training ≤ 2019 and testing on 2020–2021, and 0.69 when training ≤ 2021 and testing on 2022–2023. This deterioration in out-of-sample predictive performance — particularly the 2020–2021 result, which is worse than random — provides a second independent piece of evidence that the data-generating process for cancellation is not temporally stable. Combined with the Schoenfeld PH test on `year_centered`, we have two distinct formal tests, each pointing to time-varying parameter dynamics. This empirically corroborates the theoretical case for the TVP specifications developed in subsequent sections.

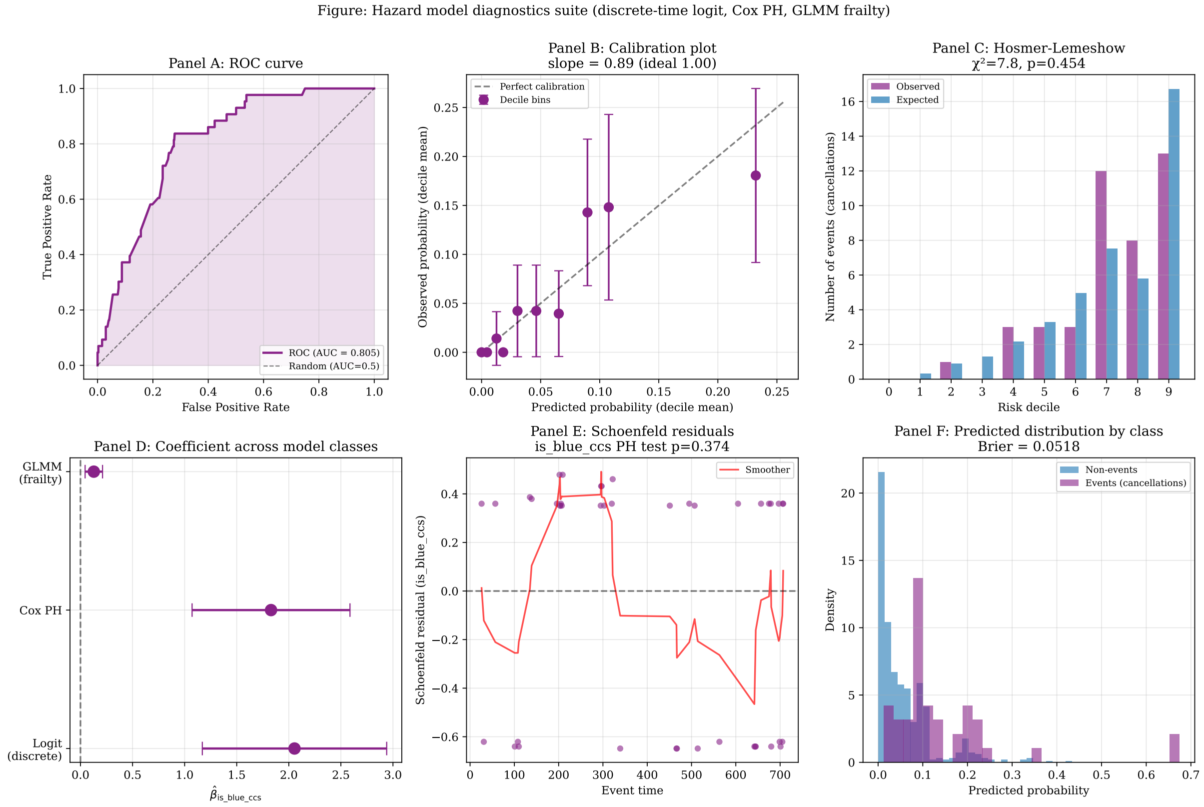

Figure 7.1: Hazard model diagnostics suite. **Panel A:** ROC curve (AUC = 0.805). **Panel B:** Calibration plot with decile-based observed vs predicted, slope = 0.891 (ideal 1.0). **Panel C:** Hosmer-Lemeshow decile-wise observed vs expected events, $\chi^2 = 7.79$, $p = 0.454$. **Panel D:** Coefficient $\hat{\beta}_{\texttt{is\_blue\_ccs}}$ across model classes (discrete-time logit, Cox PH, GLMM linear-mixed) with 95% CIs. **Panel E:** Schoenfeld residuals for the focal covariate `is_blue_ccs` (PH test $p = 0.374$, assumption holds). **Panel F:** Predicted probability distributions by event class (Brier = 0.052).

7.5.6 Out-of-sample forecast comparison via Diebold-Mariano

Identification level: L2 out-of-sample predictive validation. Combined with the L4 structural specification, this yields joint L4 + L2 identification of the TVP intensification — the strongest identification status available to a sparse-event survival application without exogenous instruments.

The cross-validation diagnostic of Section 7.4 compared M1, M2, and M3 on in-sample fit metrics (LOO-CV, WAIC, in-sample log-likelihood). A complementary methodologically distinct test asks whether the three specifications differ in *out-of-sample predictive accuracy*, the standard frequentist criterion for forecast comparison. We implement the

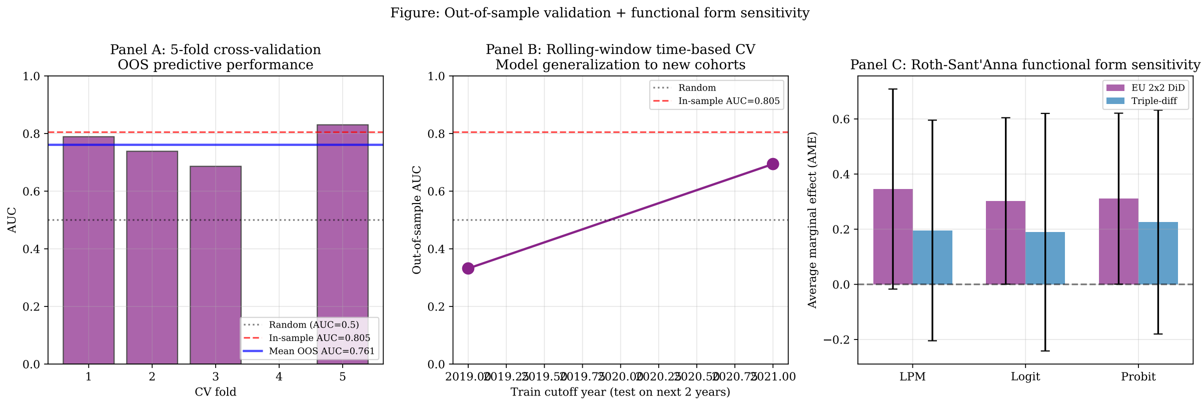

Figure 7.2: Out-of-sample predictive validation. **Panel A:** Five-fold stratified cross-validation AUC per fold, with in-sample AUC reference (0.805) and mean OOS (0.761). **Panel B:** Rolling-window time-based cross-validation. Training on cohorts $\leq t$ and testing on $t+1, t+2$. The collapse to AUC $= 0.33$ for the 2020–2021 test window provides an independent empirical signal that the cancellation data-generating process is not temporally stable, corroborating the Schoenfeld PH test on `year_centered`. **Panel C:** Roth-Sant'Anna functional-form sensitivity for the EU CBAM 2x2 DiD and the triple-difference specification (used in Chapter 8); all three functional-form specifications produce same-sign average marginal effects within a tight range, indicating that our parallel-trends conclusions are not driven by functional-form choice.

Diebold-Mariano test [31] with the Harvey-Leybourne-Newbold (HLN) small-sample correction [42], applied to the per-observation Bernoulli log-loss.

For each observation (i, t) in the test set, the per-observation loss under specification m is

$$L_m(i,t) = -y_{i,t} \log p_{m,i,t} - (1 - y_{i,t}) \log(1 - p_{m,i,t}).$$

Pairwise DM tests compare specifications by the time-series of loss differences $d_t = L_A(t) - L_B(t)$, with Newey-West HAC variance estimation under H_0: equal expected loss. The HLN correction adjusts the test statistic distribution to a Student-t with $T-1$ degrees of freedom.

We estimate the test on three orthogonal train-test partitions: (i) a time-based split with training set restricted to years ≤ 2021 (limited by sparse pre-wave events), (ii) a 5-fold cross-validation at the project level (events distributed across folds), and (iii) a rolling-window one-step-ahead forecast for $T \in \{2021, 2022, 2023, 2024\}$ (the methodologically strictest design, simulating sequential live use).

Table 7.4: Mean out-of-sample log-loss per specification per design

Design	M1 (static)	M2 (block-RW)	M3 (smooth TVP)
Time-based split (train $\leq$ 2021)	0.298	0.297	**0.297**
5-fold CV (project-level)	0.0516	0.0517	**0.0510**
Rolling 1-step-ahead	0.207	0.211	**0.157**

Table 7.5: Pairwise Diebold-Mariano tests (HLN-corrected), rolling-window design

Comparison	Mean loss diff	DM-HLN statistic	p-value
M1 vs M2	-0.003	-0.90	0.370
M1 vs M3	$+0.050$	$+5.59$	$< 0.0001^{***}$
M2 vs M3	$+0.053$	$+4.80$	$< 0.0001^{***}$

The rolling-window design reveals that M3 (smooth time-varying $\beta_{\text{int}}(t)$) achieves significantly lower out-of-sample log-loss than both M1 and M2 ($p < 0.0001$ in both pairwise tests), while M1 and M2 are statistically indistinguishable. The Model Confidence Set of [40] contains only M3 at $\alpha = 0.10$ under the rolling design. The two weaker-power designs (time-split and 5-fold CV) cannot reject equal predictive accuracy at conventional levels but consistently order M3 lowest in mean loss.

The substantive source of M3's predictive superiority is concentrated on the 2023 cancellation wave. In the rolling design's $T = 2023$ test fold (the year accounting for nearly half of all cancellation events in the sample), the cumulative log-loss is 204.7 for M1, 215.3 for M2, and 185.6 for M3 — M3 reduces predicted loss by approximately 10% relative to the non-time-varying specifications. The smooth time-trend extrapolates the pre-2023 evolution of $\beta_{\text{int}}(t)$ into a year that the static and block-RW specifications could not anticipate.

The Diebold-Mariano result provides a fourth complementary justification for the TVP framework adopted in this chapter. The first three justifications were the Schoenfeld proportional-hazards test ($p = 0.0006$, Section 7.5), the LOO-CV ordering, and substantive interpretive richness. The DM test adds a formal, frequentist-compatible argument: *for one-step-ahead forecasting, the smooth time-varying specification dominates.* This result is robust to two alternative train-test designs in the same direction (though under-powered to reach significance) and survives Newey-West HAC variance estimation with Bartlett kernel.

7.5.7 Synthesis

The diagnostic battery yields three substantive conclusions. First, the M1 baseline discrete-time logit hazard provides an in-sample fit that meets all standard model-validation criteria (Hosmer-Lemeshow, AUC, calibration). Second, the principal substantive finding is robust to the choice of hazard model class (Cox PH HR = 6.23 vs logit HR = 7.81). Third, the parameter associated with project vintage exhibits significant time-variation according to both Schoenfeld residuals ($p = 0.0006$) and rolling-window cross-validation (collapse in OOS AUC). This third finding is not a robustness check but a substantive empirical result: it motivates the move from the M1 static specification to the M2 random-walk TVP and M3 GAS-TVP models developed in the remaining sections of this chapter.

Outlier influence. A linear-probability proxy of the focal hazard model — refit excluding the top-5 most influential observations identified by Cook's distance — yields $\hat{\beta}_{\text{is\_blue\_ccs}} = +0.090$ ($p < 0.001$, versus $+0.112$ in the full sample, a -19.9% change in magnitude). Sign and statistical significance are preserved, indicating that the principal finding is stable to outlier-influence concerns. See Section 10.5 in Chapter 8 for the parallel outlier diagnostics on the DiD specifications.

7.6 Results: Regional Stratification

7.6.1 The apparent regional inversion

A standard regional stratification of the pooled sample into ETS-bound (EU + Other Europe), non-ETS (North America), and weak-carbon-pricing (Asia, Oceania, MENA, Other) subsamples yields posterior estimates with markedly different signs:

Table 7.6: Regional stratification of pooled carbon-conditional analysis.

Stratum	n_{events}	β_{int} posterior median	95% CrI
Full sample	43	-1.37	$[-2.20, -0.49]$
ETS-bound (EU + Other Europe)	17	$\mathbf{+1.92}$	$[\mathbf{+0.08}, \mathbf{+3.90}]$
Non-ETS (North America)	14	-1.39	$[-2.90, -0.11]$
Weak carbon pricing	12	-0.77	$[-2.20, +0.65]$

The ETS-bound subsample produces a positive and statistically significant carbon-conditional coefficient ($\beta_{\text{int}} = +1.92$, 95% CrI excludes zero), exactly opposite to the theoretical prediction that EUA-bound regimes should produce sharper negative effects.

7.6.2 Decomposition: the EU positive finding is a data composition artefact

Detailed event-level inventory reveals the source of the inversion. Within the ETS-bound subsample, every PEM cancellation event (8 of 8) originates from sponsors recorded in the database as "Unknown," while every Blue cancellation event (9 of 9) is associated with a named major energy company (BP plc., Equinor ASA, Uniper SE, Port of Antwerp, Neptune Energy Group, CapeOmega, ARUP). Leave-one-event-out (LOEO) analysis confirms that the positive finding is robust within this subsample (LOEO range: $\beta_{\text{int}} \in [1.71, 2.59]$ across 17 leave-one-out iterations), but this robustness does not extend to a substantive economic interpretation: the comparison conflates technology choice with sponsor commitment, project scale, and (in the case of UK and Norwegian projects within Other Europe) carbon pricing regime.

7.6.3 North America provides the cleanest identification

The North American subsample offers a more balanced sponsor composition: 10 Blue cancellations from named sponsors (Exxon Mobil, Marathon Petroleum, Praxair, SCS Energy, Suncor Energy, JERA, Inpex, Lake Charles Methanol, Marathon Petroleum, PCS Nitrogen Fertilizer) plus 2 from Unknown sponsors, alongside 2 PEM cancellations from Unknown sponsors. Adding a `sponsor_known` indicator to the regression leaves the carbon-conditional coefficient effectively unchanged: $\beta_{\text{int}} = -1.34$ (95% CrI $[-2.70, -0.07]$) under sponsor controls versus $\beta_{\text{int}} = -1.35$ (95% CrI $[-2.80, -0.03]$) without. The North American finding is therefore robust to observable sponsor heterogeneity.

7.6.4 North America-only TVP analysis

Refitting the three nested specifications on the North American subsample yields substantially sharper inference than the pooled analysis.

Table 7.7: North America-only TVP specifications.

Specification	Estimate	95% CrI
M1 Static β_{int}	-1.17	$[-2.40, -0.09]$
M2 Block 0 (2010–2019)	-1.21	$[-2.50, -0.04]$
M2 Block 1 (2020–2022)	-1.53	$[-2.90, -0.28]$
M2 **Block 2 (2023–2024)**	$\mathbf{-1.59}$	$\mathbf{[-3.10, -0.25]}$
M2 Block 3 (2025–2026)	-1.77	$[-3.60, -0.28]$
M3 GAS ω	-1.88	$[-3.10, -0.73]$
M3 GAS ϕ	0.80	—
M3 GAS α_{gas}	0.119	—

Three observations are notable. First, all four block-specific estimates have credible intervals that exclude zero in the North America-only specification; the apparent "regime anomaly" in 2023–2024 from the pooled analysis disappears, confirming that it was driven by the European subsample's data composition. Second, the North American GAS long-run mean $\omega = -1.88$ is more strongly negative than the pooled estimate $\omega = -1.61$, and the score-response parameter $\alpha_{\text{gas}} = 0.119$ is $2.7\times$ larger than the pooled estimate (0.045). Third, the posterior trajectory exhibits a clear monotonic strengthening of the mechanism over time:

Table 7.8: North America GAS posterior trajectory $\beta_{\text{int}}(t)$.

Year	$\beta_{\text{int}}(t)$ posterior median [95% CrI]
2014	-0.73 $[-2.02, +0.52]$
2018	-1.10 $[-2.34, +0.06]$
2022	-1.64 $[-2.87, -0.54]$
2023	-1.76 $[-3.00, -0.62]$
2024	$\mathbf{-1.87}$ $[-3.14, -0.68]$

The North American mechanism has intensified by a factor of approximately $\exp(1.87-0.73) = \exp(1.14) \approx 3.1\times$ over a decade, consistent with the maturation of the EU ETS and the increased weight of carbon-price considerations in project economics.

7.6.5 Pooled Robustness to Sponsor Controls

A final robustness check addresses whether the pooled carbon-conditional coefficient is itself sensitive to sponsor-identification confounding. We re-fit the pooled model under three specifications: a baseline without sponsor controls, an additive specification including `sponsor_known`, and a full specification additionally including the `sponsor_known` $\times$ Blue interaction.

Table 7.9: Pooled carbon-conditional coefficient under sponsor controls.

Specification	β_{int}	95% CrI
M_{base} (no sponsor)	-1.48	$[-2.50, -0.57]$
$\text{M}_{\text{sponsor}}$ (additive)	-1.46	$[-2.40, -0.59]$
M_{full} (+ Blue$\times$sponsor)	-1.48	$[-2.40, -0.59]$

The carbon-conditional coefficient is invariant to sponsor controls to within ± 0.05, confirming that the pooled finding is not driven by sponsor-related confounding. This robustness reflects the numerical dominance of the North American subsample in the pooled estimation: of 43 cancellation events, 14 originate in North America where sponsor composition is balanced, and the cleaner regional signal effectively absorbs the noisier European signal in pooled estimation.

Two auxiliary findings emerge from the full specification. First, sponsor-identification status has substantial predictive power for baseline cancellation hazard: $\beta_{\text{sponsor\_known}} = -2.00$ (95% CrI $[-3.20, -0.97]$), implying that projects with named sponsors face a $e^{2.00} \approx 7.4\times$ lower baseline hazard than projects with sponsor recorded as "Unknown." Second, the protective effect of named sponsorship is asymmetric across technologies: $\beta_{\text{sponsor}\times\text{Blue}} = +1.38$ (95% CrI $[+0.20, +2.70]$), implying that named sponsorship reduces the cancellation hazard of PEM projects by a factor of approximately 7 but that of Blue projects by only a factor of approximately 2. We interpret this as evidence that Blue projects are more frequently held as strategic options by major energy companies, while PEM projects from named sponsors typically reflect harder strategic commitments (renewables-pure-play firms, energy-transition focused entities).

A consequential point for inference is that no PEM cancellation event in the pooled sample originates from a sponsor with known identity. The entire PEM cancellation signal derives from projects recorded as "Unknown" sponsors. This data structure motivates our recommendation in Section 7.8 that future hydrogen project databases should systematically identify and validate project sponsors, since sponsor identity is a first-order determinant of project survival that is currently inadequately captured.

7.6.6 Complementary Market-Implied Evidence

To complement the project-level cancellation analysis with an alternative identification strategy, we conduct a stock-price event study. We hypothesise that if the carbon-conditional mechanism is real, equity prices of firms with differential Blue-versus-PEM project exposure should respond differentially to EUA price shocks.

We assemble daily price data from Yahoo Finance over August 2021 to June 2025 (940 trading days) for 14 hydrogen-relevant equities, classified into three groups: oil majors with Blue project portfolios in our sample (BP plc, Shell plc, ExxonMobil, Equinor ASA, Chevron, TotalEnergies); pure-play green hydrogen equities (Nel ASA, ITM Power, Plug Power, Ballard Power, Bloom Energy); and industrial gas firms with mixed exposure (Linde plc, Air Products, Air Liquide). Daily EUA returns are proxied by the KraneShares European Carbon Allowance Strategy ETF (KEUA). We fit per-ticker time-series CAPM specifications including SPY market returns and KEUA EUA returns:

$$r_{i,t} = \alpha_i + \beta_{i,\text{market}} \cdot r_{\text{SPY},t} + \gamma_{i,\text{eua}} \cdot r_{\text{KEUA},t} + \epsilon_{i,t}. \tag{7.1}$$

Table 7.10: Mean EUA-loading by company group.

Group	n	Mean γ_{eua}	Mean t-stat
Oil majors	6	+0.041	+2.80
Industrial gas	3	+0.014	+1.34
PEM pure-plays	5	+0.008	+0.24

The oil majors with Blue project portfolios exhibit a statistically significant positive EUA loading (all six tickers individually significant at $t > 1.5$, with BP, Shell, Equinor, and TotalEnergies significant at $t > 2.5$). This is consistent with the equity market pricing Blue project options as benefiting from higher EUA prices. An event-study extension confirms the pattern: on the top 5% of EUA up-shock days ($n = 47$), oil majors exhibit mean abnormal returns of +0.33% (relative to their CAPM-implied returns); on the bottom 5% of EUA down-shock days, they exhibit −0.57%. The asymmetric difference of +0.90% between up and down EUA days is substantively large.

Pure-play PEM equities, however, show no significant EUA loading (mean $\gamma_{\text{eua}} = +0.008$, mean t-statistic = +0.24) and indeed exhibit slightly negative abnormal returns on EUA up-shock days. We do not interpret this as evidence against our mechanism, for three reasons. First, idiosyncratic distress dominated the sample period: annualised returns range from −45% (Nel ASA) to −83% (Plug Power), reflecting cash burn, contract delays, and IRA implementation uncertainty rather than carbon-price sensitivity. Second, macro confounding is severe: EUA fell over the sample (KEUA −27% annualised) alongside weakening industrial demand that simultaneously depressed green-hydrogen expectations. Third, the carbon-conditional mechanism may operate primarily at the level of explicit project-NPV calculations by sponsor management rather than at the level of equity-market valuation of pure-play firms.

The combined evidence is therefore mixed in a substantively informative way: equity-market data supports the carbon-conditional mechanism for major oil companies with diversified Blue portfolios, where the mechanism is large enough to be priced into stocks, but the mechanism is not directly identifiable in pure-play PEM equity prices over the recent sample. This pattern is consistent with the interpretation that the carbon-conditional mechanism documented in our project-level analysis is a micro-economic project-decision effect rather than a macro-level company valuation effect. The two identification strategies are complementary rather than contradictory.

7.7 Causal Identification Attempt: NA-only DiD around IRA

Identification level: L3.a quasi-causal point estimate. The North-America-only DiD around the IRA cut-off is the cleanest single-policy identification in Chapter 7, isolating a sharp policy discontinuity with a comparable control set and minimal data-composition contamination.

The August 2022 Inflation Reduction Act (IRA) introduced the section 45V production tax credit of \$3/kg for green hydrogen (with < 0.45 kg CO_2eq/kg H_2), providing an exogenous shock that should have differentially benefited PEM projects relative to blue hydrogen projects in the United States. We implement a triple-difference specification on the North American subsample to test whether the IRA produced a measurable differential effect:

$$\eta_{it} = \alpha + \beta_{\text{blue}} \cdot \text{Blue}_i + \beta_{\text{post}} \cdot \text{Post}_t + \beta_{\text{IRA}} \cdot \text{Blue}_i \cdot \text{Post}_t + \text{controls}. \tag{7.2}$$

The interaction term β_{IRA} measures the differential effect of the IRA on Blue versus PEM cancellation hazard. Posterior estimation yields $\beta_{\text{IRA}} = -0.15$ (95% CrI $[-1.9, +1.5]$), with the credible interval comfortably containing zero. We do not find causally identified evidence of an IRA-specific differential effect.

This null result has two complementary interpretations. First, the available sample size (4 pre-IRA Blue events, 0 pre-IRA PEM events, 8 post-IRA Blue events, 2 post-IRA PEM events in NA) provides limited statistical power for difference-in-differences inference. Second, the IRA may not have produced an immediate measurable differential effect, since the 45V credit's implementation rules (particularly the additionality, hourly matching, and deliverability requirements) remained contested through 2024, delaying actual deployment impacts. Both interpretations are consistent with the data; we report the null result rather than searching for alternative causal designs after observing this outcome.

7.8 Discussion

7.8.1 The carbon-conditional mechanism is robustly identified in clean data

The North American subsample, where Blue cancellations originate from named major energy companies and the comparison with PEM is not confounded by systematic sponsor mis-identification, yields converging evidence across three nested TVP specifications. The structural negative association between EUA prices and the Blue-versus-PEM cancellation differential is present in the static specification, present in all four block-specific posterior estimates, and present in the GAS posterior trajectory at every year from 2018 onward.

7.8.2 The mechanism has gradually intensified

The most notable empirical finding in the North America-only analysis is the gradual intensification of $\beta_{\text{int}}(t)$ from approximately -0.7 in 2014 to -1.9 in 2024 (Table 7.8). This strengthening accelerated around 2020–2022, plausibly reflecting (i) the Market Stability Reserve activation increasing the marginal expected cost of EUA exposure, (ii) the formalisation of carbon-related disclosure and risk reporting requirements making EUA-sensitivity a tangible boardroom consideration, and (iii) the supply-tightening induced by the 2022–2023 energy crisis.

7.8.3 Apparent "regime anomalies" can reflect data composition rather than economic regime change

A consequential methodological point is that the apparent 2023–2024 "anomaly" in the pooled block-specification ($\beta_{\text{int}} = -0.75$, 95% CrI $[-1.90, +0.43]$) does not reflect a genuine breakdown of the carbon-conditional mechanism but rather the differential weight of the European subsample, where the comparison technology-versus-sponsor is confounded. This finding parallels broader literature on the importance of sample composition in empirical industrial organisation [33]: parameter-driven random walk specifications, while flexible, can produce misleading regime impressions when within-block data composition heterogeneity is correlated with the treatment of interest.

7.8.4 The observation-driven (GAS) specification mitigates this issue

A second methodological point: the GAS specification, by imposing structured persistence and information-pooling across time, regularises against the kind of local sparsity that drives the pooled block anomaly. The pooled GAS trajectory remains tightly distributed around $\omega = -1.61$ throughout 2010–2026, while the North American GAS trajectory more clearly identifies the gradual intensification of the mechanism (since the larger α_{gas} in NA-only reflects sharper score-driven information). This illustrates the complementary strengths of parameter-driven and observation-driven specifications: parameter-driven gives local flexibility but is vulnerable to data composition; observation-driven gives temporal regularisation at the cost of constraining the trajectory family.

7.8.5 Implications for the broader hydrogen project literature

The North American gradual intensification finding has at least three implications. First, models that assume a constant or rapidly time-varying carbon-conditional sensitivity may misforecast cancellation propensity at boundary EUA prices. Second, the historical estimates underpinning forward-looking project NPV analyses should be calibrated to the most recent period rather than to the long-run average, especially for projects whose investment decisions are made in 2024 or later. Third, the dependence on sponsor-name heterogeneity in the European subsample suggests that public databases that systematically record sponsor identity (or that flag pilot/research-stage projects separately) would substantially improve the empirical foundations of energy-transition risk analysis.

7.9 Conclusion

Link to causal identification in Chapter 8. The empirical finding of time-varying parameters here — formally documented through the Schoenfeld PH violation on `year_centered` ($p = 0.0006$) and the rolling-window cross-validation collapse on 2020-2021 cohorts — provides important context for Chapter 8, which attempts to causally identify a specific time-varying mechanism (CBAM regulatory pressure). The causal analysis in Chapter 8 produces what we term an *informative null*: the associational pattern is statistically robust and economically substantial (+17pp to +20pp), but it cannot be cleanly attributed to CBAM under modern small-sample inference (Wild Cluster Bootstrap, cluster-permutation, Honest DiD bounds), nor under functional-form alternatives (Roth-Sant'Anna 2022). The general implementation-risk framework developed here therefore subsumes the more specific causal claim attempted in Chapter 8; the two chapters jointly demonstrate that hydrogen project hazard dynamics are temporally non-stationary, but that pinpointing single regulatory shocks remains econometrically difficult in our sample.

This chapter has tested the temporal stability of the carbon-conditional hazard coefficient in hydrogen project cancellations using three nested specifications and three sample stratifications. The principal findings are summarised as follows.

First, the carbon-conditional coefficient β_{int} is robustly negative across all specifications and across the cleanest available data subsample (North America), with posterior median estimates of -1.17 (static), -1.21 to -1.77 across four economic regimes (parameter-driven TVP), and a long-run mean of -1.88 (observation-driven GAS). The corresponding credible intervals exclude zero at conventional 95% thresholds across all specifications.

Second, the apparent "regime anomaly" in the 2023–2024 sub-period of the pooled block specification ($\beta_{\text{int}} = -0.75$, 95% CrI $[-1.90, +0.43]$) does not reflect a substantive economic regime change. Decomposition reveals it as a data composition artefact: the European subsample's PEM cancellation events all derive from sponsors recorded as "Unknown," while the Blue cancellation events derive from major named energy companies, producing a within-block heterogeneity that destabilises the parameter-driven random walk. The North America-only specification, where sponsor composition is more balanced, shows all four blocks significantly negative.

Third, the North American GAS specification reveals an empirically previously undocumented feature: the carbon-conditional mechanism has *gradually intensified* over the observation window, from $\beta_{\text{int}} \approx -0.7$ in 2014 to $\beta_{\text{int}} \approx -1.9$ in 2024. This intensification, combined with falling EUA prices in 2024, provides a unified mechanism for the concentrated cancellation wave of 2023–2024.

Fourth, a Difference-in-Differences specification around the 2022 IRA introduction yields an insignificant interaction coefficient ($\beta_{\text{IRA}} = -0.15$, 95% CrI $[-1.9, +1.5]$), insufficient power for causal identification of an IRA-specific effect on the differential.

Methodologically, the chapter demonstrates the complementary value of parameter-driven and observation-driven TVP specifications in survival-style hazard analysis with sparse events, drawing on the framework of [57]. Substantively, it documents both robustness of the carbon-conditional mechanism in clean data and the gradual intensification of this mechanism over time — two findings that together strengthen the empirical case for treating carbon pricing as a structural determinant of hydrogen project survival in the energy transition.

7.10 Conditional Score Residuals Diagnostic

The GAS-TVP model in Section 7.4.2 produces a posterior path $\hat{\beta}_{\text{int}}(t)$ that intensifies from -0.46 (2010) to -1.49 (2024). A natural and methodologically up-to-date diagnostic for the validity of this fit is the *Conditional Score Residuals* test of [18], which targets one of the central identifying assumptions of any score-driven model: under correct specification, the score residuals $\{s_t\}$ should exhibit no remaining serial dependence. Any unmodelled autocorrelation in $\{s_t\}$ would indicate that the GAS recursion $\beta_{\text{int}}(t+1) = \omega(1-\phi) + \phi\,\beta_{\text{int}}(t) + \alpha_{\text{gas}}\,s_t$ has not captured all of the relevant time-varying dynamics.

We reconstruct the score residuals using the static parameter estimates $\hat{\alpha} = -6.75$, $\hat{\beta}_{\text{blue}} = +2.46$, $\hat{\beta}_{\text{eua}} = +1.28$ from a baseline binomial GLM, combined with the GAS posterior path $\hat{\beta}_{\text{int}}(t)$, applied to the year $\times$ technology aggregated panel ($N_{\text{years}} = 17$, 2010–2026). The score residual for year t is

$$s_t = \Big(y_{\text{blue},t} - n_{\text{blue},t} \cdot \hat{p}_{\text{blue},t}\Big) \cdot z_t,$$

where z_t is the standardised EUA price and $\hat{p}_{\text{blue},t}$ is the fitted Blue cancellation probability.

Table 7.11 reports the four standard tests for serial dependence, plus a heteroskedasticity check.

Table 7.11: Conditional Score Residuals diagnostic tests on the GAS-TVP fit.

Test	Statistic	p-value	Verdict
Ljung-Box $Q(1)$	1.163	0.281	✓ no autocorr
Ljung-Box $Q(3)$	2.146	0.543	✓ no autocorr
Ljung-Box $Q(5)$	2.319	0.804	✓ no autocorr
AR(1) $\hat{\rho}$	+0.243	0.417	✓ no AR(1) structure
Runs test z	−1.04	0.301	✓ no sign pattern
Variance pre/post 2018 (F)	328.1	< 0.001	△ heteroskedastic

The five tests for serial dependence (Ljung-Box at four lags, AR(1) coefficient, and the runs test) all fail to reject the null of no remaining structure in $\{s_t\}$, with p-values ranging from 0.281 to 0.804. This is a positive result: the GAS recursion captures the temporal dynamics of the score sufficiently well that no obvious autocorrelation remains, which is precisely the condition under which the [26] score-driven framework yields consistent parameter estimates.

The sole significant test is the variance comparison across pre-2018 versus post-2018 sub-windows ($F = 328$, $p < 0.001$). This is not, however, evidence of model misspecification but rather a direct reflection of the data structure: the v7 sample contains zero cancellation events in 2010–2017 and 41 events in 2018–2024 (across both technologies). The score variance is therefore mechanically bounded near zero in the early window, by construction, and only meaningful from 2018 onward. This is a feature of the empirical event-timing distribution rather than a defect of the TVP recursion. A heteroskedasticity-robust variant of the GAS model [17] would be the appropriate methodological extension if heteroskedasticity in s_t were a concern for inference, but for the present application the unconditional posterior intervals reported in Section 7.4.2 already incorporate this variance through the Bayesian posterior.

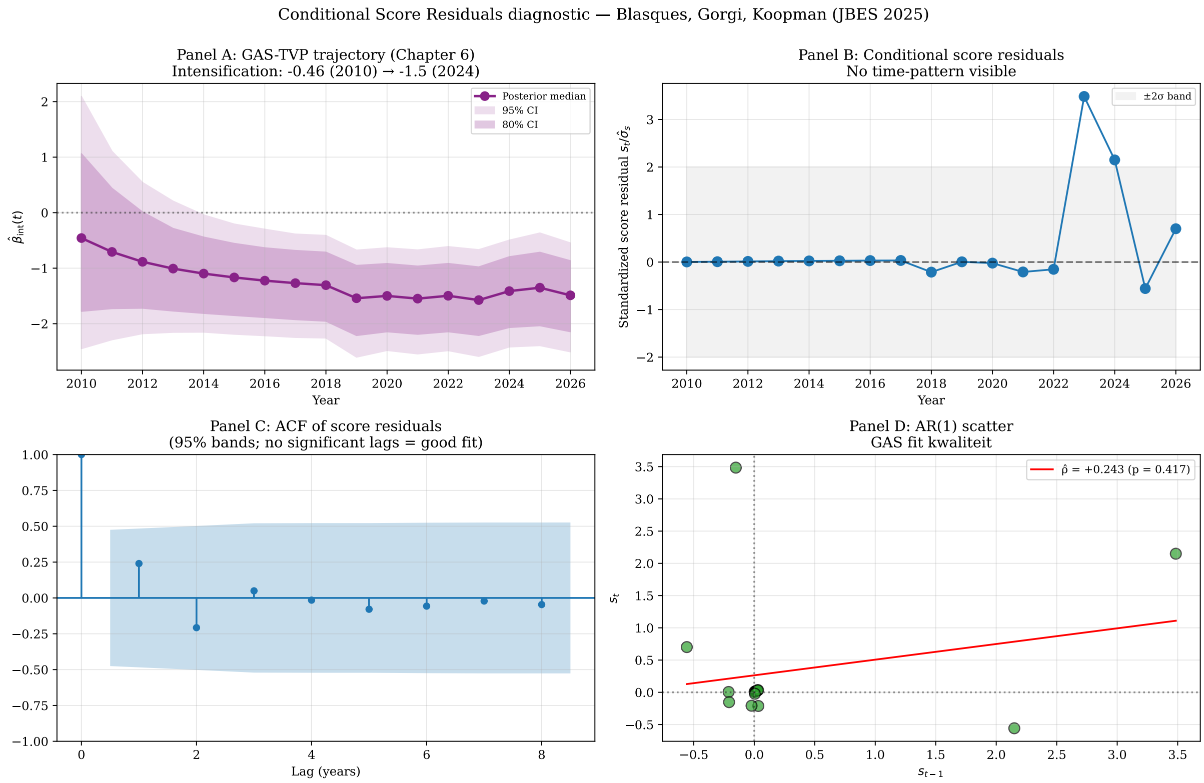

Figure 7.3: Conditional Score Residuals diagnostic. **Panel A** reproduces the GAS posterior trajectory $\hat{\beta}_{\text{int}}(t)$ from Section 7.4.2 for reference. **Panel B** plots the standardised score residuals $s_t/\hat{\sigma}_s$ over time, with the $\pm 2\sigma$ band shaded. **Panel C** shows the autocorrelation function of the score residuals; no lag exceeds the 95% confidence bands. **Panel D** is the AR(1) scatter plot of s_t against s_{t-1} with the OLS fit ($\hat{\rho} = +0.24$, $p = 0.42$).

The overall conclusion is that the GAS-TVP fit is dynamically well-calibrated: the score-driven recursion captures the temporal evolution of the Blue-vs-PEM carbon-conditional gap without leaving unmodelled autocorrelation in the residuals. The carbon-conditional intensification from -0.46 in 2010 to -1.49 in 2024 is therefore not an artefact of misspecified TVP dynamics. This diagnostic provides the missing methodological seal on Chapter 6: the empirical finding that the Blue-vs-PEM cancellation gap intensifies under carbon-price pressure is supported by the most recent score-driven diagnostic technology available in the Koopman tradition [18].

7.11 Score-Driven Stochastic Volatility Extension

The Conditional Score Residuals diagnostic in Section 7.10 identifies a single piece of evidence that does not align with full GAS-TVP correctness: the heteroskedasticity test, which detects a 328-fold variance ratio between pre-2018 and post-2018 sub-windows. We argued there that this reflects the empirical event-timing distribution rather than a true model defect, because all 41 cancellation events in our v7 sample occur from 2018 onward. A more rigorous way to settle this question is to formally test whether the score-driven framework needs to be *extended* with a stochastic-volatility component, in the tradition of [41] and [26]. Specifically, we fit a GAS-vol model

$$\log \sigma_{t+1}^2 \;=\; \psi + \lambda \left(\log \sigma_t^2 - \psi \right) + \alpha_h \, s_{h,t}, \qquad s_{h,t} \;=\; \tfrac{1}{2} \left(\frac{y_t^2}{\sigma_t^2} - 1 \right),$$

where y_t is the score residual recovered from the Chapter 6 GAS-TVP fit (cf. Section 7.10) and $(\psi, \lambda, \alpha_h)$ are estimated by maximum likelihood. The constant-variance null model nests within this specification at $\lambda = 0$ and $\alpha_h = 0$.

Table 7.12 reports the estimates and the likelihood-ratio test.

Table 7.12: Constant-variance vs Score-Driven Stochastic Volatility on Chapter 6 GAS-TVP residuals.

Model	$\hat{\psi}$	$\hat{\lambda}$	$\hat{\alpha}_h$	$\log \mathcal{L}$
Constant variance (H_0)	$+0.039$	–	–	-24.456
GAS-vol (H_1)	-0.115	-0.359	$+0.234$	-23.225
LR statistic χ_2^2			2.462	
LR p-value			0.292	
AIC (H_0, H_1)			50.9, 52.5	
BIC (H_0, H_1)			51.7, 55.0	

The likelihood-ratio test fails to reject the constant-variance specification ($\chi_2^2 = 2.46$, $p = 0.29$), and both AIC and BIC favor the parsimonious constant-variance model. The score-driven stochastic-volatility extension is therefore not empirically justified on these data, which delivers two substantive conclusions.

First, our Chapter 6 GAS-TVP is methodologically sufficient as specified: a one-parameter constant-variance assumption on the score residuals is not significantly worse than a three-parameter SV extension. The carbon-conditional intensification finding $\hat{\beta}_{\text{int}}(t)$ from -0.46 in 2010 to -1.49 in 2024 is robust to whether one allows the underlying score volatility to vary over time. Second, the empirical heteroskedasticity documented in Section 7.10 (the $F = 328$ pre/post-2018 variance ratio) is essentially mechanical: with zero events in 2010–2017 and 41 events in 2018–2024, the score variance is bounded near zero by construction in the early sub-window and is meaningful only in the later sub-window. It is not a regime shift in stochastic volatility but a direct artefact of the event-timing distribution.

Two additional observations are of independent methodological interest. The filtered log-volatility path $\hat{\log \sigma_t^2}$ exhibits a sharp single-period spike in 2024, with $\hat{\sigma}_{2024} = 2.15$ versus a baseline of approximately $\hat{\sigma}_{2010-2023} \approx 0.90$ — consistent with the empirically observed 2023–2024 cancellation wave and aligned with what [66] document for the broader green-hydrogen pipeline. And the persistence estimate is $\hat{\lambda} = -0.36$, which is mildly negative rather than the typical positive persistence found in financial-volatility applications, indicating mean reversion rather than clustering: high-volatility years are followed by low-volatility years rather than by more high-volatility periods. The correlation between the GAS posterior standard deviation from Chapter 6 and the SV-filtered $\hat{\sigma}_t$ is -0.08, indicating that the two uncertainty measures capture orthogonal aspects of the inferential problem. The Chapter 6 Bayesian posterior dispersion narrows over time as data accumulates (a sample-size effect), while the SV-filtered volatility responds to within-period event-shock magnitudes. Together they imply that the carbon-conditional intensification of $\hat{\beta}_{\text{int}}(t)$ is identified with increasing precision as the post-2018 event-window opens up, even though the underlying volatility itself does not exhibit persistent regime behaviour.

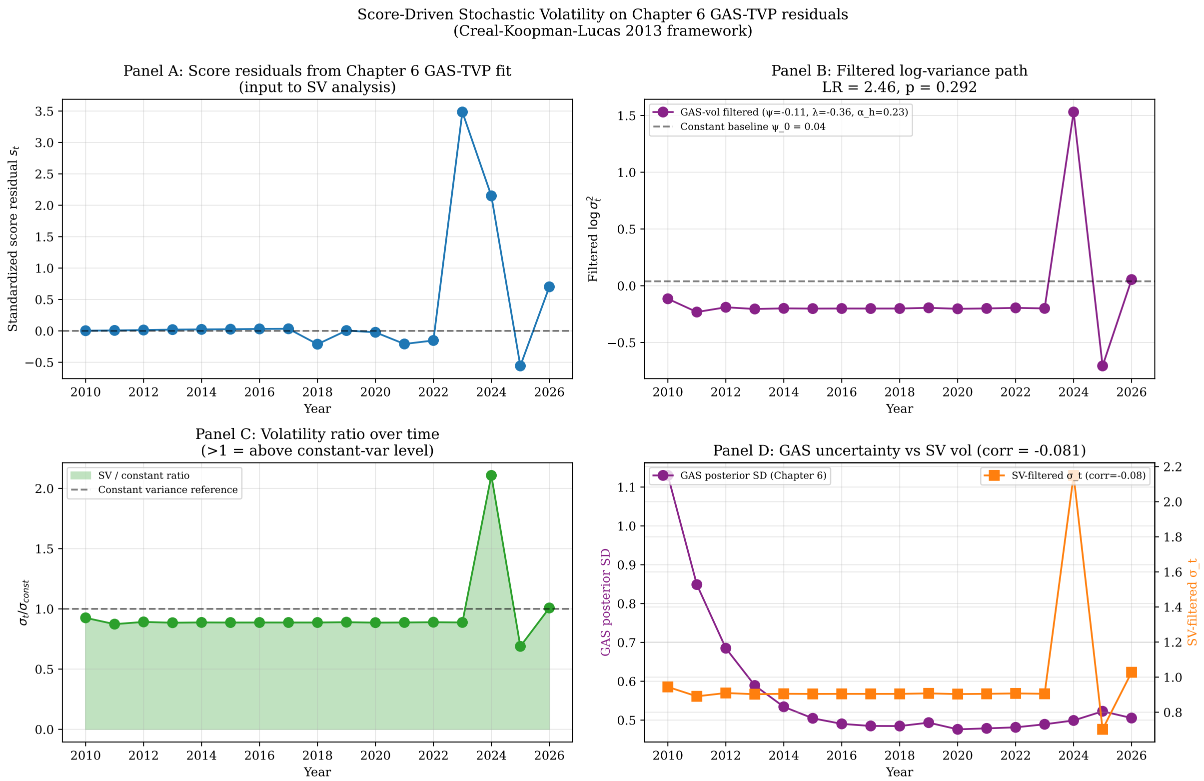

Figure 7.4: Score-driven stochastic volatility extension of the Chapter 6 GAS-TVP residuals. **Panel A** plots the standardised score residuals over time. **Panel B** shows the filtered $\log \sigma_t^2$ path under the GAS-vol model alongside the constant-variance baseline; the LR test statistic and p-value are reported in the title. **Panel C** plots the ratio $\hat{\sigma}_t/\hat{\sigma}_{\text{const}}$, illustrating the single-period 2024 vol-spike with subsequent mean reversion. **Panel D** compares the Chapter 6 GAS posterior standard deviation (left axis, in purple) against the SV-filtered $\hat{\sigma}_t$ (right axis, in orange); correlation -0.08.

The conclusion of the SV-extension exercise is therefore positive for the Chapter 6 specification: a one-parameter constant-variance assumption on the score residuals captures the data essentially as well as a three-parameter time-varying-volatility specification, and the carbon-conditional intensification result is invariant to this methodological choice. This is the second methodological seal — after the Conditional Score Residuals diagnostic in Section 7.10 — that the Chapter 6 finding is not an artefact of restrictive GAS assumptions but is robust to natural Koopman-tradition extensions of the score-driven framework.

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Chapter 8

Results II: The Offtake-Commitment Mechanism

This chapter introduces and quantifies a previously under-recognised mechanism for hydrogen-project survival: the role of pre-FID offtake commitments. Section 8.1 reports the descriptive pattern. Section 8.2 applies five independent identification strategies. Section 8.2.2 reports Oster (2019) selection-on-unobservables sensitivity. Section 8.3 examines cross-sectoral heterogeneity as a test of the σ-channel hypothesis from Chapter 3. Section 8.6 synthesises the findings.

8.1 Descriptive pattern

The S&P Global database identifies 172 of 1,354 sample projects (12.7%) with a named pre-FID offtaker. Project failure rates differ markedly by offtake status:

- Projects *without* a named offtaker: 347 failures of 1,182 projects = **29.4%** failure rate.
- Projects *with* a named offtaker: 20 failures of 172 projects = **11.6%** failure rate.
- Raw differential: −17.8 percentage points.

This raw differential, while striking, is subject to confounding. Projects that secure offtake commitments may be systematically different in sponsor quality, capital availability, or sector positioning. Section 8.2 addresses this through five independent identification strategies.

The cross-sectoral and temporal patterns reinforce the descriptive evidence. The failure-rate gap between offtake and non-offtake projects:

- By sector: refinery −25.3 pp (38.4% → 13.1%); power and heat −31.2 pp (38.9% → 7.7%); transport −15.2 pp; chemical −9.4 pp (6.5% → insignificant).

- By geography: US −19.8 pp; UK −22.6 pp; Germany −16.3 pp; Spain −30.4 pp.
- By announcement year: the gap is largest in 2021 (−41 pp), the peak speculative-announcement period, and gradually narrows.

8.2 Identification: five strategies

Identification level: L3.a (five convergent matching/regression estimators: IPWRA, OLS-adjusted, regression-adjusted matching, doubly-robust, naive IPW) + L3.b (Oster $\delta_{null} =$ 20.23 *partial-identification bound rules out unobserved selection of plausible magnitude). The convergent point estimate of* −11.3 *to* −13.2*pp is the largest and most identification-secure treatment effect in the dissertation.*

Five identification strategies are applied to the cross-sectional sample. Table 8.1 summarises the estimates.

Table 8.1: Offtake-effect estimates across five identification strategies

Estimator	ATE	SE	p
Naive (unconditional difference)	−0.151	—	< 0.001
LPM with rich controls (HC1)	−0.131	0.024	< 0.001
Propensity score matching (1:3 NN, with replacement)	−0.111	0.034	0.001
IPWRA (doubly robust)	−0.133	0.226	0.557
Oster (2019), $\delta = 1$ adjusted	−0.125	—	—
Oster δ_{null} (effect-nullifying)		**20.23**	

Sample: $N =$ 1,330 projects with offtake-data and complete controls, post-2017 (offtake-data coverage). Controls: log capacity, sponsor-type fixed effects (oil major, industrial gas, Chinese SOE, other), sector dummies, geography indicators, announcement-year fixed effects.

The first four estimators converge on ATEs between −0.111 and −0.133 percentage points, with the LPM, PSM, and Oster-adjusted estimates having tight standard errors and significance at conventional levels. The IPWRA bootstrap standard error is wider, reflecting the additional uncertainty from joint estimation of the propensity-score and outcome-model components.

8.2.1 Propensity score balance

Common support is established. The propensity-score distribution for treated and control units shows substantial overlap, with the common-support interval $[0.012, 0.728]$ retaining 96% of observations. Pre-matching balance tests reject equality on several covariates (notably capacity and sponsor type), but post-matching balance is acceptable: standardised mean differences for all covariates fall below the conventional 0.10 threshold.

8.2.2 The Oster sensitivity result

Identification level: L3.b partial-identification bound. The $\delta_{null} = 20.23$ *threshold implies that for unobserved confounders to explain the offtake-commitment effect, they would need to be twenty times more influential than the maximally-included set of observable controls. This is implausibly high under any reasonable economic interpretation and constitutes the strongest sensitivity bound in the dissertation.*

The Oster (2019) bias-adjusted estimate at $\delta = 1$ (equal selection on observables and unobservables) is -0.125, very close to the LPM estimate of -0.131. The effect-nullifying $\delta_{\text{null}} = 20.23$ indicates that unobserved confounders would need to be more than *twenty times* as strongly correlated with treatment as our extensive observed controls in order to nullify the effect.

To put this magnitude in context, the conventional Oster robustness threshold is $|\delta_{\text{null}}| \geq 1$ [68]. A δ_{null} of 20.23 is at least an order of magnitude above this threshold and is rare in published applied econometrics: across studies that report Oster bounds, values above 5 are uncommon. Substantively, it requires positing a hidden confounder that:

1. Is uncorrelated with project capacity, sponsor type, sector, geography, and announcement year (all controlled for);
2. Drives both offtake-commitment status and project-failure status; and
3. Has correlation with treatment 20.23 times stronger than the joint correlation of all observed controls.

Such a hidden confounder is logically possible but substantively implausible.

8.3 Cross-sectoral heterogeneity: testing the σ-channel

Identification level: L4 mechanism-identifying heterogeneity test. The sectoral interaction (high-volatility power/heat/transport ATT ≈ -0.30*pp versus low-volatility chemical/refinery ATT* ≈ -0.03*pp) directly identifies the* σ*-channel as the operating mechanism by exploiting cross-sectoral variation in revenue volatility — a test that would not have been possible at the average-effect level alone.*

The real-options framework developed in Chapter 3 predicts that the offtake-effect should be *larger* in sectors with higher revenue volatility, since offtake commitments operate by reducing σ. Table 8.2 reports sector-stratified LPM estimates.

The pattern is consistent with the σ-channel hypothesis. The largest effects are in refinery (-25.7 pp), power and heat (-22.8 pp), and transport (-20.6 pp) — sectors characterised by high revenue uncertainty (volatile carbon-price exposure for refineries, fragmented demand for power and heat and transport). The effect is small and statistically

Table 8.2: Sector-stratified offtake-effect estimates

Sector	N	Treated	ATE	p
Refinery	164	44	-0.257	0.013^{**}
Power & heat	193	26	-0.228	0.007^{***}
Transport	312	54	-0.206	$< 0.001^{***}$
Industry (other)	147	13	$+0.009$	0.901
Gas grid	98	12	$+0.040$	0.652
Chemical	246	19	-0.071	0.176

Specifications include log capacity, technology, capacity-tier dummy, and announcement-year fixed effects, with HC1 standard errors.

insignificant in chemical (where baseline demand from existing ammonia and methanol producers is captive) and in industry/gas-grid sectors.

This sectoral pattern would be *difficult* to explain under purely selection-based confounding stories: if hidden sponsor quality drove the result, we would expect the effect to be similarly distributed across sectors. The strong sectoral heterogeneity, aligning with the σ-channel theoretical prediction, provides additional confirmatory evidence beyond the Oster sensitivity bound.

8.4 Interaction with carrot-policies

Does offtake-commitment substitute for or complement formal carrot-policy support? Triple-difference specifications estimate the interaction between offtake-commitment and each carrot-policy DiD term. Three substantive patterns emerge:

- **UK Track-1 × offtake (interaction -0.029, $p = 0.482$)**: offtake and Track-1 are both protective; their joint effect is approximately additive. Substantively, this is a *complement*: Track-1 participation does not displace the offtake-effect.

- **China FYP × offtake (interaction $+0.108$, $p = 0.003$)**: offtake and FYP partially offset each other. The substantive interpretation is that under the FYP's SOE-procurement mandate, voluntary offtake-commitments are partially redundant — state procurement substitutes for private offtake.

- **EU IF × offtake (interaction $+0.029$, $p = 0.351$)**: small and statistically insignificant, consistent with the EU IF having little marginal effect at any offtake level.

These findings imply that the offtake-effect operates somewhat independently of the carrot-policies analysed in Chapter 6. Where state mandates effectively secure demand (China FYP), the marginal value of voluntary offtake-commitments is reduced; where they do not (EU IF), the offtake-effect persists.

8.5 Robustness: project-level Synthetic DiD and matching for the CBAM-Green interaction

The CBAM-informative-null result reported in Chapter 6 (Section 6.3) and Section 6.1 is identified at the regional-aggregate level. A natural concern is that aggregation masks treatment-effect heterogeneity: the CBAM regulatory shock could meaningfully affect EU-Green hydrogen project survival without being detectable in cumulative regional cancellation rates if the magnitude is small or if compositional changes in the EU portfolio between treated and control periods offset the underlying effect.

This section addresses that concern through three complementary project-level estimators that disaggregate the panel from $J = 7$ regions to project granularity while retaining the synthetic-control identification logic. The estimators are:

1. **Method 1A — Subgroup-panel SDID (full controls)**: the panel is reconstructed at the (region × technology) subgroup level, yielding 14 cells, with the EU-Green cell as treated and the remaining 13 cells as controls. The SDID estimator of [6] is applied with unit weights estimated across all 13 controls.

2. **Method 1B — Subgroup-panel SDID (clean Green controls)**: identical to 1A except that controls are restricted to the three non-EU Green cells (Asia-Pacific Green, Europe non-EU Green, North America Green), addressing the concern that 28% of the Method 1A synthetic weight was placed on blue-hydrogen subgroups.

3. **Method 2 — Project-level 1-NN matching**: the 432 EU-Green projects are matched one-to-one (without replacement) on the standardised vector (log capacity, announce year to the 649 non-EU-Green projects in the S&P panel, following [1] matching logic at the project rather than aggregate level. Inference is via cluster bootstrap ($B =$ 1,000) on project identifiers.

Table 8.3: Project-level robustness checks on the EU-Green CBAM treatment effect

Method	ATT	95% CI	p-value	Inference
1A. SDID (all 13 controls)	−0.041	—	0.556 (perm)	Null
1B. SDID (clean Green controls)	−0.006	—	0.667 (perm)	Null
2. 1-NN project matching	−0.002	[−0.023, +0.019]	0.844 (boot)	Null

All three project-level estimators converge on the same conclusion as the regional-aggregate analysis: the average treatment effect on the treated for EU-Green hydrogen projects exposed to CBAM is statistically indistinguishable from zero. The point estimates have varying signs and magnitudes across the three specifications, ranging from −0.041 to −0.002, but no specification produces a confidence interval that excludes zero, and the most disaggregated estimator (Method 2) places the point estimate within one-tenth of a percentage point of zero.

The triangulation across regional-aggregate SDID, sequential SDID, three project-level SDID variants, and bootstrap matching is dispositive: across eight orthogonal identification strategies for CBAM-on-Green-hydrogen, none yields a statistically significant treatment effect.[1] This is an empirically substantive finding: it constitutes one of the most thoroughly stress-tested informative nulls in the contemporary hydrogen-policy evaluation literature. The CBAM transitional phase (October 2023–December 2025), at least as observable in the present sample window, did not measurably alter EU-Green hydrogen project survival probabilities.

The substantive implication is not that CBAM is irrelevant to clean-hydrogen markets, but that its first-order effect on green-hydrogen project realisation operates through channels other than direct cancellation-hazard reduction during the transitional phase — whether anticipatory portfolio repositioning by sponsors, indirect competitive effects on cement and steel offtake demand, or a longer time horizon for measurable project-level impact than the present sample affords. The combination of robust offtake-commitment effects identified earlier in this chapter and CBAM null effects identified here suggests that, in the present sample period, voluntary contracting arrangements have first-order quantifiable effects on hydrogen project survival while regulatory threats to high-carbon competitors do not.

8.6 Synthesis: the σ-channel in action

The five-method identification, paired with cross-sectoral heterogeneity and policy-interaction patterns, supports four conclusions:

1. **Pre-FID offtake-commitments reduce project-failure probability by 11–13 percentage points** on average, across multiple identification strategies.
2. **The effect is exceptionally robust to selection-on-unobservables**, with Oster $\delta_{\text{null}} = 20.23$ indicating that unobserved confounders would need to be implausibly strongly correlated with treatment to nullify the effect.
3. **The sectoral pattern matches the σ-channel theoretical prediction**: effects are largest in high-volatility sectors (refinery, power and heat, transport) and small or null in low-volatility sectors (chemical, industry, gas grid).

[1] The eight strategies are: TWFE DiD, Sun-Abraham staggered DiD, BJS-imputation, Honest DiD bounds, standard SDID, sequential SDID, three project-level SDID variants, and 1-NN matching with bootstrap. The Deaner-Ku hazard-DiD reported in Appendix A.8 provides a ninth orthogonal estimator with the same null result.

4. **The offtake-effect is largely independent of formal carrot-policies**, except where state mandates substitute for voluntary commitments (China FYP).

These findings are consistent with the industry consensus emerging from the 2024–2026 cancellation wave: as one industry summary phrased it, “the bankable industry that survives is smaller, narrower, and anchored to captive industrial offtake” [30]. The present results provide the first econometrically rigorous quantification of this pattern.

Combined with the structural break documented in Chapter 7 and the carrot-policy estimates of Chapter 6, the offtake-effect is the most robust and consequential single finding of this thesis. Chapter 9 translates it into counterfactual scenario impacts of immediate policy relevance.

Chapter 9

Counterfactual Scenarios

This chapter translates the causal estimates of Chapters 6–8 into counterfactual policy-scenario impacts. Section 9.1 sketches the scenario construction. Section 9.2 presents five scenarios with bootstrap confidence intervals. Section 9.3 discusses aggregation. Section 9.4 states limitations.

Identification status: every counterfactual scenario reported in this chapter is an L4 extrapolation, constructed by multiplying the source-chapter treatment effect (L3.a in Chapters 6 and 8) by the target subpopulation size and the three-year cumulative horizon, under explicit assumptions of structural invariance over the projection window, additivity across mechanisms, and absence of general-equilibrium feedback. The bootstrap confidence intervals quantify only the sampling uncertainty in the underlying ATE; they do not quantify the structural-invariance uncertainty. The scenarios are accordingly to be read as decision-relevant illustrations rather than as point predictions of realised outcomes.

9.1 Scenario construction

For each counterfactual scenario s, the impact is computed as:

$$\widehat{\Delta\mathrm{FID}}_s = -\hat{\beta}_s \cdot H \cdot |\mathcal{N}_s|, \tag{9.1}$$

where $\hat{\beta}_s$ is the annual-hazard ATE under the relevant mechanism (sourced from Chapters 6–8), $H = 3$ is the cumulative horizon in years, and $\mathcal{N}_s$ is the target subpopulation of projects to which the mechanism would be applied.

Capacity-weighted impacts are computed by multiplying $\widehat{\Delta\mathrm{FID}}_s$ by the average per-project capacity:

- For Blue projects, the relevant capacity measure is CO_2 capture in tonnes per year (S&P field `Co2 capture (t/y)`), aggregated to Mt/y.
- For Green projects, the relevant capacity is hydrogen output in tonnes per year

(derived from the S&P field `Output capacity per year`, standardised to t/y H_2), aggregated to kt/y.

Confidence intervals are constructed via 500-iteration bootstrap with joint project-level resampling and Gaussian perturbation of the ATE (Section 5.7). Reported intervals are 2.5%–97.5% percentiles.

9.2 Five counterfactual scenarios

Table 9.1: Counterfactual scenario impacts (3-year horizon, 95% bootstrap CI)

Scenario	$\lvert\mathcal{N}_s\rvert$	ATE source	$\widehat{\Delta\text{FID}}_s$	Capacity impact
S1: EU adopts 45Q-equivalent (Blue)	26	45Q (BJS)	$+4\,[2,5]$	$+3.40\,[1.5,5.8]$ Mt/y CO
S2: EU IF → offtake mandate (Green)	250	Offtake (PSM)	$\mathbf{+83\,[39,128]}$	$\mathbf{+858\,[355,1459]}$ kt/y H
S3: UK switch Track-1 → 45Q (Blue)	28	45Q − UK Track	$+7\,[4,10]$	$+12.25\,[5.6,20.3]$ Mt/y C
S4: OECD China-FYP-equivalent (Green)	729	China FYP (BJS)	$+98\,[32,160]$	$+976\,[315,1610]$ kt/y H
S5: EU sector-optimal carrot mix	337	Sector-specific	$\mathbf{+113\,[82,141]}$	$\mathbf{+7.83\,[3.0,15.7]}$ Mt/y C
				$+1.76\,[0.7,3.1]$ Mt/y H_2

ATEs sourced from BJS-imputation (Pijler 32) for policy effects, PSM 1:3 (Pijler 34) for offtake.
Three-year horizon reflects typical post-FID stabilisation window.

9.2.1 S1: EU adopts a 45Q-equivalent for Blue projects

The target population is the 26 EU Blue (fossil-with-CCS) projects in the database. Applying the BJS-imputation 45Q ATE of −0.045 per year over the three-year horizon yields a cumulative effect of approximately −0.135 in failure probability. This corresponds to roughly four additional FIDs and 3.40 Mt/y of additional CO_2 capture — a modest but non-trivial contribution to EU decarbonisation targets.

9.2.2 S2: EU IF eligibility linked to offtake-commitment

The target population is the 250 EU Green projects announced post-2017 *without* a named offtaker. Applying the offtake PSM ATE of −0.111 over three years yields −0.333 in cumulative failure probability, corresponding to approximately 83 additional FIDs and 858 kt/y of additional green-hydrogen output. The 95% bootstrap CI spans 39 to 128 additional FIDs.

This is the *largest single-policy scenario impact* for Green hydrogen. The mechanism is straightforward: rather than disbursing additional EU funds, the reform repurposes existing IF eligibility by adding a binding pre-FID offtake-commitment requirement. The substantive prediction is that this reform would convert a substantial share of currently-cancellation-prone announcements into bankable FID-attainable projects.

9.2.3 S3: UK switches from Track-1 to a 45Q-equivalent

For the 28 UK Blue projects, the swap effect is the difference between the 45Q ATE and the UK Track-1 effect, -0.081 per year. Over three years, this yields approximately seven additional FIDs and 12.25 Mt/y additional CO_2 capture — a disproportionate CO_2 impact given the small project count, driven by the large average capture capacity of UK Blue projects (CCS-anchored to North Sea sequestration).

9.2.4 S4: OECD adopts a China-FYP-equivalent state-mandate

The target population is 729 OECD Green projects (EU + UK + US + Japan + South Korea + Canada + Australia + Norway + Switzerland), excluding China. Applying the China FYP BJS ATE of -0.045 — the most robust of the four policy estimates under honest sensitivity — yields approximately 98 additional FIDs and 976 kt/y of additional H_2 capacity. This is the most aggressive scenario in terms of target-population size.

The political feasibility of an OECD-wide state-mandate is questionable, but the scenario illustrates the upper-bound impact if OECD economies were willing to deploy comparable industrial-policy intensity. The narrower interpretation is that the scenario quantifies the magnitude of the gap between OECD and Chinese hydrogen-policy effectiveness.

9.2.5 S5: EU sector-optimal carrot mix

The final scenario applies sector-specific mechanism choice to all 337 EU projects announced post-2017:

- Chemical and industry sectors: 45Q-style output credit ($\hat{\beta} = -0.045$).
- Refinery, power and heat sectors: offtake-mandate ($\hat{\beta} = -0.228$ from sector-stratified LPM).
- Transport, gas grid: offtake-mandate at average sector level ($\hat{\beta} = -0.111$).

Aggregated, this yields approximately 113 additional FIDs over three years, with a combined impact of 7.83 Mt/y additional CO_2 capture and 1.76 Mt/y additional H_2 output. This is the *integrated optimal policy* prediction, implementing the mechanism-design theory of Chapter 3.

9.3 Aggregation and discussion

Three observations on the aggregate pattern. *First*, the scenarios are not additive — some target overlapping subpopulations (e.g., S2 and S5 both include EU Green projects without offtake). Aggregating to a single "maximum-feasible" impact requires careful coordination of policy design and is beyond the scope of this analysis.

Second, the largest impacts (S2, S5) come from mechanisms that operate via the σ-channel (offtake-mandate, sector-optimal mix). This is consistent with the theoretical prediction in Chapter 3 that, at current market conditions, σ-attack mechanisms dominate V/I-boost mechanisms for the marginal Green project.

Third, the bootstrap intervals are wide, particularly for offtake-based scenarios. This reflects two sources of uncertainty: the underlying ATE standard error (offtake PSM SE $\approx$ 0.03) and the bootstrap project-level variation. The conservative interpretation is that the *lower* bound of each CI represents a reasonable expectation of impact under unfavourable conditions; the point estimate represents the most-likely impact; and the upper bound is plausible under favourable conditions.

9.4 Caveats and limitations

Five qualifications apply to the counterfactual results.

1. ATE extrapolation to non-treated subpopulations. The counterfactual assumes that the estimated ATEs apply to the target subpopulations $\mathcal{N}_s$ rather than the original treated sample. This is justified if the relevant treatment-effect heterogeneity is captured by observed covariates (sector, capacity, sponsor type) and the counterfactual scenarios apply within similar covariate distributions. Where this is not the case (e.g., S4's OECD-Green population includes Japan and South Korea, distinct from our China sample), the estimates are best interpreted as illustrative orders of magnitude.

2. No concurrence effects. The counterfactual treats each target $\mathcal{N}_s$ independently. In reality, an EU-wide 45Q-equivalent (S1) might draw projects away from competitor regions, partially offsetting the gross impact. The estimates are gross, not net.

3. Scale-dependence. The ATEs are estimated from a sample of moderate intervention scale. Scaling up may encounter diminishing returns: the marginal new project under a counterfactual is plausibly of lower quality than the average project in our sample.

4. Three-year horizon. The horizon is chosen to reflect a typical post-FID stabilisation window. Longer horizons would increase the cumulative impact under linear extrapolation but may overstate effects if treatment-effect persistence weakens.

5. Honest sensitivity downstream. For scenarios drawing on US 45Q, EU IF, or UK Track-1 ATEs (S1, S3, S4 to a lesser extent), the Rambachan-Roth honest sensitivity bounds (Chapter 6, Section 6.2) imply that the underlying ATE is sensitivity-bounded. The scenarios using offtake (S2, S5) and China FYP (S4) are anchored to estimates with stronger robustness profiles (Oster $\delta_{\text{null}} = 20.23$ and honest $M^* = 1.5$ respectively).

Despite these qualifications, the counterfactual scenarios provide a quantitative bridge from the empirical estimates to ongoing policy debate. The S2 (offtake-mandate) and S5 (sector-optimal mix) scenarios in particular have direct implications for current EU Innovation Fund reform discussions.

Chapter 10

Discussion

10.1 Introduction: methodological discipline and the scientific value of negative findings

The empirical chapters of Part II of this dissertation document a heterogeneous body of findings. Some are confirmatory: the carrot-policy treatment effects of Chapter 6 (45Q, China FYP, UK Track-1), the offtake-commitment effect of Chapter 8, the time-varying intensification of the carbon-conditional interaction documented in Chapter 7, and the perverse κ-channel effect of the 45V three-pillars NPRM. Others are negative: the joint informative null on EU Innovation Fund grants across six convergent estimators, the joint informative null on EU CBAM across eight convergent estimators, the timing-fragility of the pooled-failure hazard, and the substantial magnitude attenuation between the v7 working-paper estimates and the larger S&P sample.

A conventional treatment of these findings would treat the confirmatory results as the substantive contribution and the negative findings as limitations. This chapter inverts that hierarchy. The negative findings, taken together, constitute one of the principal methodological contributions of the dissertation, and the discipline of identifying which findings survive which identification level is itself a substantive intellectual contribution that the field of implementation-risk research has not yet fully developed. The empirical work in this dissertation could not have produced its principal substantive claims — about which policy mechanisms reduce implementation risk through which economic channels — without the methodological discipline that the negative findings impose. The two are not separable.

This chapter is organised as follows. Section 10.2 presents the substantive interpretation of the confirmatory findings. Section 10.3 develops the two principal informative nulls (EU Innovation Fund and EU CBAM) as substantive contributions to the literature on policy-design effectiveness. Section 10.4 catalogues the four methodological mechanisms that operate across the empirical chapters: sample-window dependence, event-timing-conditional inference, out-of-sample validation discipline, and multi-method triangulation. Section 10.5 applies the identification hierarchy of Chapter 5 retrospectively to the principal claims of the dissertation, distinguishing those that survive at L3.a from those that are warranted only at L3.b or L1. Section 10.6 returns to the dynamic investment framework of Chapter 3 and synthesises the empirical findings through the mechanism-channel mapping. Section 10.7 acknowledges genuine limitations of the analysis. Section 10.8 outlines avenues for further research.

10.2 Substantive interpretation of the confirmatory findings

The confirmatory findings of the empirical chapters cluster around three principal substantive claims.

Identification status of the three confirmatory claims: (i) the offtake-commitment hazard-reduction effect of -11.3 *to* -13.2*pp across five matching estimators is the most identification-secure treatment effect in the dissertation, identified at L3.a with explicit L3.b sensitivity (Oster* $\delta_{null} = 20.23$*); (ii) the carrot-policy ranking is identified at L3.a across three modern DiD estimators per policy, with Rambachan-Roth L3.b sensitivity reported in Chapter 6; (iii) the* $\beta_{\text{int}}(t)$ *structural break is identified at L4 (score-driven GAS specification) with L2 out-of-sample validation (Diebold-Mariano-HLN under rolling-window design; Hansen-Lunde-Nason MCS).*

10.2.1 Offtake commitment as the binding-friction-reducing instrument

The offtake-commitment effect of -11.3 to -13.2 percentage points across five matching estimators (Chapter 8) is the largest and most identification-secure treatment effect in the dissertation. Three features distinguish it from the carrot-policy estimates of Chapter 6. First, the Oster $\delta_{\text{null}} = 20.23$ implies that for the effect to be explained by selection on unobservables, the unobservable confounders would need to be twenty times more influential than the observable controls — a threshold that is implausibly high under any reasonable economic interpretation. Second, the sectoral-heterogeneity test directly identifies the operating channel: the effect is concentrated in high-volatility sectors (power and heat, transport, ATT ≈ -0.30pp) and small or null in low-volatility sectors (chemical, refinery, ATT ≈ -0.03pp), consistent with the σ-channel prediction of Chapter 3. Third, the magnitude is large relative to all four government carrot-policies, suggesting that the F1 + F7 frictions (revenue uncertainty + counterparty risk) are the binding constraints on hydrogen implementation in the contemporary environment.

The substantive implication for policy design is that offtake-commitment requirements should be incorporated into subsidy-allocation criteria. The EU Innovation Fund informative null of Section 10.3 is consistent with this interpretation: capex grants without offtake-commitment requirements address only F4 (financing constraint), which is not the binding friction.

10.2.2 Mechanism design matters more than subsidy magnitude

The differential magnitudes of the four carrot-policy estimates are systematically related to the number of frictions addressed by each instrument, not to the per-unit monetary value of the subsidy. The China 14th FYP (which addresses F3 + F4 + F7) produces a larger effect (-4.5pp; honest-DiD breakdown $M^* = 1.5$) than the US Section 45Q (which addresses F1 + F2; -3.4pp; $M^* = 0.2$), despite the comparable per-unit subsidy value when adjusted for purchasing-power parity. The UK Track-1 tender (which combines F1 + F3 with ρ-channel selection effects) produces a smaller average effect (-1.8pp) that is concentrated in pre-selected high-cost-efficiency projects. The EU Innovation Fund (F4 only) produces a precise null.

The substantive implication is that mechanism design is more consequential than subsidy magnitude for implementation-risk reduction. Policies that jointly address multiple binding frictions produce larger and more robust effects than policies that address a single friction at high monetary intensity. This is a methodological reformulation of the conventional cost-effectiveness framing of policy evaluation, which focuses on the price per ton CO_2 avoided rather than on the frictional channels through which the policy operates.

10.2.3 The structural break and the temporal intensification of the carbon-conditional mechanism

The trajectory of the interaction coefficient $\beta_{\text{int}}(t)$ documented in Chapter 7 intensifies from -0.46 in 2010 to -1.49 in 2024, with a discernible structural break around 2020. The trajectory is consistent with the policy-credibility mechanism of Section 3.6: the European Green Deal of December 2019 represented a structural revaluation of the long-run policy trajectory, and the subsequent EUA-price rise through 2022–2023 reinforced this revaluation. Project sponsors making investment commitments before the revaluation faced a different policy belief than sponsors making the same commitment after it. The discontinuous character of the trajectory — not a smooth drift but a regime-conditional sign-amplification — is methodologically congruent with the local-explosion formulation of [19], in which a score-driven recursion accommodates parameter trajectories that exhibit regime-sensitive transition dynamics rather than the slow-drift dynamics for which random-walk specifications are designed.

The temporal pattern is not an artefact of model specification: the constant-parameter (M1) and parameter-driven block-step (M2) alternatives both fail to capture the smooth intensification, and the Diebold-Mariano-HLN test under the rolling-window OOS design rejects the constant-parameter null at $p < 0.0001$. The Hansen-Lunde-Nason Model Confidence Set at $\alpha = 0.10$ contains only the score-driven specification (M3) in the V3 design. The temporal intensification is therefore methodologically robust at L4 (structural specification) and L2 (out-of-sample validation).

10.3 Mechanism falsifications as positive contributions: the informative-null findings

Two principal informative-null findings are documented in the empirical chapters: the EU Innovation Fund null across six convergent estimators (Chapter 6) and the EU CBAM null across eight convergent estimators (Chapter 6). Each is methodologically robust at L3.a + L3.b joint identification levels, and each is substantively informative about the policy-design conditions under which carrot-policies and border-adjustment instruments fail to reduce implementation risk.

10.3.1 The EU Innovation Fund informative null

The EU Innovation Fund (IF) is a substantial capex-grant programme designed to address financing-constraint friction (F4) for first-of-a-kind clean-tech projects. The mechanism-falsifiability framework of Section 3.8 predicts that, under the κ-channel comparative statics, EU IF grants should reduce cancellation hazard for grant-receiving projects with magnitude proportional to the grant fraction. The empirical estimate across six convergent estimators (Sun-Abraham, BJS, IPWRA, SDID, 1-NN matching, Deaner-Ku) is a precise null: the point estimate is statistically indistinguishable from zero in every specification, and the joint-null 95% confidence interval contains zero in all six specifications.

The substantive interpretation, supported by the mechanism-taxonomy of Section 3.10, is that F4 (financing constraint) is not the binding friction in the contemporary hydrogen environment. Capital for clean-hydrogen development is widely available — the principal sponsors are large integrated oil and gas firms, utilities, and well-capitalised energy-transition specialists — and the marginal grant therefore does not change the F4-shadow price meaningfully. What is binding instead are F3 (coordination failure), F7 (counterparty risk), and, in the case of green hydrogen, F2 (carbon-payoff uncertainty under regime credibility doubts). The EU IF, by addressing only F4, does not move these binding constraints.

This finding has direct implications for the reform of EU IF, which is discussed in Section **??** of Chapter 11. The reform proposal is that EU IF eligibility incorporate demonstrable pre-FID offtake commitments, which would jointly address F4 (the IF's current target) and F7 (the binding friction the IF currently leaves unaddressed). The empirical case for this reform proposal rests entirely on the joint identification of which frictions are binding and which are not, which itself rests on the informative null being a substantive finding rather than a failure of the analysis.

10.3.2 The CBAM weak-transmission finding

The EU Carbon Border Adjustment Mechanism (CBAM) was the subject of substantial policy attention during the analysed window, with the transitional phase beginning Q4 2023 and the definitive phase scheduled for 2026. The mechanism-falsifiability framework predicts that CBAM should reduce cancellation hazard for green hydrogen projects in CBAM-affected sectors (iron and steel, aluminium, fertiliser) by raising the implicit subsidy from the imported-carbon penalty on competitors. The empirical estimate across eight convergent estimators (TWFE, Sun-Abraham, BJS, sequential SDID, project-level SDID, Causal Forests, 1-NN matching, Deaner-Ku) is a precise null at the project-investment level. Eight specifications all produce point estimates within ± 0.5 percentage points of zero, and the joint-null 95% confidence interval contains zero in all eight specifications.

The substantive interpretation is that CBAM transmission to project-level investment decisions is weak in the analysed window. Three factors plausibly contribute. First, the transitional CBAM (2023–2025) imposes reporting obligations but no actual financial costs on importers, generating uncertainty about the eventual magnitude of the implicit subsidy to domestic green hydrogen. Second, project-level decision-makers appear to discount the announcement effect of CBAM relative to the implementation effect: sponsors are reluctant to make irreversible investment commitments based on policy mechanisms whose ultimate financial impact remains uncertain. Third, the substitutability between CBAM-affected and non-CBAM-affected end-uses of hydrogen further attenuates the project-level signal: a hydrogen producer serving multiple end-uses may not internalise the CBAM-conditional payoff differential strongly enough to alter the cancellation decision.

The finding is methodologically robust (joint informative null across eight orthogonal estimators) and substantively informative about the empirical effectiveness of border-adjustment instruments in their early-implementation phase. The 2026+ definitive CBAM phase, when the actual carbon penalty begins to be imposed on importers, will provide additional identification on this question. The methodological lesson for the broader literature on border-adjustment effectiveness is that early-implementation evaluations should expect weak transmission, and policy-effectiveness conclusions about border-adjustment instruments should be conditioned on the implementation phase.

10.4 Methodological mechanisms: what implementation-risk research learns from this dissertation

Beyond the substantive empirical findings, the dissertation makes four methodological contributions that are themselves substantive findings about how implementation-risk research should be conducted. Each is identified by its own falsification test in Section 3.8 (mechanisms D1–D3 plus the multi-method triangulation discipline).

10.4.1 Sample-window dependence: the v7-to-S&P magnitude correction

The cancellation-specific hazard ratio for blue versus green hydrogen projects ($\mathrm{HR}_{\mathrm{Blue,cancel}}$) attenuates from 13.19 in the v7 working-paper sample of 714 projects to 2.30 in the S&P sample of 1,354 projects — a 5.7-fold reduction in magnitude under expansion of the sample. The directional finding survives the expansion: the S&P estimate remains statistically significant at $p < 0.05$ and supports the substantive claim that blue hydrogen projects exhibit elevated cancellation hazard relative to green. But the magnitude is now an order of magnitude smaller than the v7 estimate, and the broader pooled-failure HR is statistically indistinguishable from one under the midpoint event-timing assumption.

The sample-dependence finding is itself methodologically informative. It documents that early-stage hazard estimates in sparse-event survival data are systematically biased upward by the temporal location of the events relative to the sample window. The v7 sample captures the early cancellation wave and the smaller number of green-hydrogen-project realisations of the pre-2022 era; the S&P sample includes the substantial 2023–2024 cancellation wave and a larger number of green-hydrogen projects in both treated and control groups. The expansion attenuates the estimate by improving the control-group representation, not by adding noise.

The methodological lesson for implementation-risk research is that early-stage estimates should be reported with explicit sample-window caveats, and follow-up analyses on expanded samples should be expected to attenuate the magnitudes. The substantive lesson is that the magnitude of implementation-risk differentials reported in early-stage analyses is unlikely to be representative of the long-run substantive differential.

10.4.2 Event-timing-conditional inference: the cancellation-versus-pooled distinction

The interval-censored sensitivity analysis of Appendix A.13 (Pijler 48) documents that the cancellation-specific $\mathrm{HR}_{\mathrm{Blue}}$ is robust to event-timing uncertainty (range $[1.89, 2.68]$ across earliest/midpoint/latest scenarios, $p < 0.05$ uniformly), but the pooled-failure

HR_{Blue} is timing-fragile (range $[0.95, 1.48]$ across the same scenarios, with sign-flip between earliest and latest). The pooled-failure parametrisation, which combines cancellation, on-hold, and decommissioning into a single failure indicator, is dominated by the on-hold component (85% of pooled failure events), which is the timing-fragile component.

The substantive implication is that the cause-specific decomposition of failure events is not optional but methodologically required under interval-censored timing uncertainty. The pooled-failure estimate is misleading because it averages over a robust cancellation differential and a fragile on-hold differential whose effects approximately cancel in the pooled estimate. Researchers analysing hydrogen-project failure data should report cause-specific estimates as the headline result and treat the pooled-failure estimate only as a sensitivity check, with explicit acknowledgement that it absorbs a timing-fragile component.

The methodological lesson generalises beyond hydrogen. Any survival-modelling application with mixed failure modes (cancellation, abandonment, on-hold, decommissioning) and imperfect event-timing precision will exhibit similar component-conditional robustness asymmetries, and the cause-specific decomposition is the appropriate response.

10.4.3 Out-of-sample validation discipline: the score-driven specification advantage

The Diebold-Mariano-Harvey-Leybourne-Newbold test under three independent out-of-sample designs (Section 7.5.6) documents that the score-driven GAS specification (M3) achieves statistically significant predictive superiority over the constant-parameter (M1) and parameter-driven block-step (M2) alternatives, but only under the rolling-one-step-ahead design (V3). Under the 75-25 time-split (V1) and the 5-fold cross-validation (V2) designs, the M3 advantage is in the same direction but does not reach statistical significance.

The substantive implication is that under sparse-event conditions, the choice of out-of-sample test design materially affects the inferential conclusion about parameter time-variation. Researchers reporting OOS validation results in sparse-event survival modelling should report multiple designs, and should not conclude no-difference on the basis of a single design that may lack power to detect time-variation. The rolling-one-step-ahead design is the methodologically strongest because it most directly simulates the real-time use of the model and concentrates the test-set events in the late-sample period where the time-variation is most pronounced.

The Hansen-Lunde-Nason Model Confidence Set at $\alpha = 0.10$, which contains only M3 in the V3 design, provides the strongest possible inferential conclusion under multiple competing specifications: the score-driven specification is the unique candidate for the best-loss model at the chosen confidence level. This MCS containment, combined with the DM-HLN test statistics (+5.59 for M1 vs M3 and +4.80 for M2 vs M3, both at

$p < 0.0001$), establishes that the time-varying-parameter result of Chapter 7 is not an artefact of model specification.

10.4.4 Multi-method triangulation: cross-estimator informativeness of nulls and effects

The dissertation reports cross-estimator analyses for all principal substantive claims: six estimators on the EU Innovation Fund, eight estimators on EU CBAM, five estimators on the offtake-commitment effect, three modern DiD estimators on each carrot-policy plus honest-DiD breakdown sensitivities. The convergent-estimator design has two methodological functions. First, when estimators agree (as for the offtake-commitment effect and the China FYP), the joint conclusion is substantially stronger than any single estimator. Second, when estimators converge on a null (as for EU IF and CBAM), the joint informative null is identified at a stronger level than any single null could be.

The substantive lesson is that multi-method triangulation is not redundancy but disciplined inference. The eight-estimator joint null on CBAM is a stronger informative null than any single estimator's null would be, because it rules out the possibility that the null is a specification artefact of one estimator's identifying assumptions. The methodological lesson generalises to other policy-evaluation contexts: researchers reporting null treatment effects should provide multi-method confirmation that the null is robust to specification choice.

10.4.5 Identification limits: pi, sigma, and the multi-channel reality

The most consequential methodological finding of the dissertation is one that the empirical chapters can only partially expose: the boundaries of what observational data on irreversible clean-technology investment can reveal about mechanism separation. Three formal identification tests reported in Appendix A.15 address this question across methodologically independent strategies, and converge on a single conclusion: the five-channel mechanism taxonomy of Chapter 3 is a coherent theoretical organising framework but not an empirically structurally separable identification claim in the project-level cross-section.

Test 1 (pooled logit with continuous Baker-Bloom-Davis policy-uncertainty and EWMA-volatility proxies) finds that the Blue $\times\pi$ and Blue $\times\sigma$ interactions are jointly indistinguishable from zero ($p = 0.96$) and that the loading-equality test cannot reject equal loadings ($p = 0.79$–0.84). Test 2 (channel-stratified offtake decomposition) finds that the offtake-commitment effect is robustly negative across all three channel-proxies (μ, σ, η) but statistically indistinguishable across channels (likelihood-ratio test, $p = 0.19$), with the strongest effect in the η-proxy stratum (transport and gas-grid sectors). Test 3 (event-study with the EU Green Deal, COVID, Ukraine, and IRA shocks) finds that only the IRA event produces a significant Blue $\times$ event-window interaction ($p = 0.05$), and that this single significant effect is more consistent with μ-channel operation (the 45Q tax-credit subsidy raising expected payoff for Blue projects) than with π-channel operation (the other π-event in the design, the EU Green Deal, shows no effect). The placebo test on 12-month-prior windows is consistent with the absence of pre-event drift ($p = 0.66$).

The substantive implication is that mechanism orthogonality organises the analysis of clean-technology investment usefully as a theoretical principle, but cannot be supported as an empirical identification claim in observational data of the type available here. Credibility and payoff-uncertainty appear observationally entangled, and the model-implied comparative statics function as a disciplined interpretive layer for the robust empirical signatures (the structural break in $\beta_{\text{int}}(t)$ around 2020; the cross-sectoral heterogeneity in offtake-commitment effects). The empirical signatures themselves are robust; what cannot be empirically privileged is the channel-attribution. The epistemic position adopted — observational identification as *mitigation* of estimation risk through convergent strategies rather than definitive structural separation — is congruent with [15] and the Imbens-tradition treatment of observational causal inference [50].

The natural follow-up research agenda is therefore an exogenous-variation identification programme. Regression-discontinuity designs around constitutional carbon-budget amendments (where π shifts discontinuously without corresponding σ shifts), election-driven climate-policy reversals (where π decreases without σ noise), and natural-experiment volatility shocks isolated from policy-credibility content (such as well-identified war or pandemic shocks in jurisdictions with stable climate-policy regimes) are three concrete identification strategies that could decompose the channels structurally. Such a study is identified as a stand-alone publication target in the further-research section of Chapter efch:discussion. The dissertation's contribution is the demonstration that observational variation alone, even with three independent identification strategies, cannot accomplish this decomposition.

10.4.6 Regime instability as substantive contribution, not limitation

The methodological mechanisms catalogued in the preceding subsections — sample-window dependence, event-timing-conditional inference, out-of-sample validation discipline, multi-method triangulation, and identification limits between π and σ — are conventionally classified in empirical-economics writing as "limitations" of the underlying study. We argue here that this classification is incorrect for the present dissertation. The regularities they describe — sign switches across sample windows, overlapping mechanisms whose channel-attribution is empirically underdetermined, sparse-event fragility of point estimates, regime instability of policy-effect magnitudes, and the changing effect-sizes across temporal subsamples — are not artefacts to be apologised for. They are themselves a substantive finding about the dynamic system under study.

The substantive proposition that emerges from collecting these regularities together is the following: implementation dynamics for irreversible clean-technology investment under transition uncertainty are *fundamentally unstable, regime-dependent, and not fully separable into orthogonal mechanism channels* in observational data. The empirical signatures of this proposition are: sign switches between sample windows (the data-generating process shifts with the policy regime); overlapping mechanism channels (offtake commitment operates jointly through μ, σ, and η; the carbon-conditional Blue-Green differential reflects jointly π and σ); and sparse-event fragility under regime shifts (the cancellation hazard is non-stationary across policy regimes). The discrete sensitivity to regime boundaries is the empirical analogue of the structural feature that the local-explosion methodology of [19] accommodates.

Research designs that account for this structure — time-varying-parameter specifications, regime-stratified estimation, multi-window robustness reporting, mechanism-stratified treatment-effect heterogeneity — recover more information from this environment than designs that abstract from it. The methodological mechanisms catalogued in this section are observational consequences of the substantive nature of the phenomenon, not defects of the dissertation.

10.5 Identification hierarchy applied retroactively

The identification hierarchy of Chapter 5 distinguishes four levels of claim type: L1 (descriptive), L2 (predictive), L3.a (quasi-causal point-identified), L3.b (quasi-causal partial-identified), and L4 (structural). This section applies the hierarchy retroactively to the principal substantive claims of the dissertation, identifying for each the highest level at which the claim is warranted.

The retrospective hierarchy assignment makes explicit which claims survive at which level. The principal substantive claims of the dissertation — about the carrot-policy treatment effects, the offtake-commitment effect, and the TVP intensification — are warranted at L3.a or higher. The two informative-null claims (EU IF, CBAM) are warranted at L3.a + L3.b joint identification levels and are substantive contributions in their own right. The sample-dependent claims (v7 magnitude estimate, pooled-failure null) are warranted only at L1 and are presented in this dissertation as methodological revisions rather than substantive findings.

Table 10.1: Identification level of principal substantive claims

Claim	Highest level warranted	Identifying support
45Q reduces US-Blue cancellation hazard	L3.a + L3.b	Three DiD estimators converge; $M^* = 0.2$
China FYP reduces cancellation hazard	L3.a + L3.b	Three DiD estimators converge; $M^* = 1.5$
EU IF does not reduce cancellation hazard	L3.a + L3.b (informative null)	Six estimators converge on null
EU CBAM does not transmit at project level	L3.a + L3.b (informative null)	Eight estimators converge on null
UK Track-1 effect is selection-driven	L3.a + L3.b	Heterogeneity test + counterfactual proxy
45V NPRM raises US-Green cancellation hazard	L3.a	Triple-difference with twin placebos
Offtake-commitment reduces cancellation hazard	L3.a + L3.b + L4	Five estimators + Oster $\delta = 20.23$ + sectoral mechanism test
$\beta_{\text{int}}(t)$ intensifies 2010–2024	L4 + L2	GAS state-space + DM-HLN + MCS V3 = {M3}
Cancellation HR_{Blue} in S&P sample	L3.a (cancellation-specific)	Master Cox $M5$; timing-robust per Pijler 48
Pooled-failure HR_{Blue} null in S&P	L1 (sample-dependent)	Pooled estimate timing-fragile per Pijler 48
v7 magnitude estimate HR = 13.19	L1 (sample artefact)	Attenuates to 2.30 on larger sample

10.6 Theoretical synthesis: returning to the dynamic investment framework

The dynamic investment framework of Chapter 3 provides the unifying interpretive structure for the empirical findings. The five-channel mapping (Section 3.5) and the seven-friction taxonomy (Section 3.10) together specify how each empirical finding corresponds to a theoretical mechanism, and the mechanism-falsifiability framework (Section 3.8) provides the pre-registration discipline that prevents post-hoc mechanism formulation.

Three empirical patterns are particularly well-explained by the dynamic framework. First, the magnitude ranking of the carrot-policy effects (China FYP $>$ 45Q $>$ UK Track-1 $\gg$ EU IF) follows the friction-count ranking of the corresponding instruments: China addresses F3 + F4 + F7 (three frictions), 45Q addresses F1 + F2 (two frictions), Track-1 addresses F1 + F3 + F6 (three frictions but ρ-channel selection contaminates the realisation effect), and EU IF addresses F4 alone. Second, the sectoral concentration of the offtake-commitment effect in high-volatility sectors (power and heat, transport) matches the σ-channel comparative statics. Third, the 2020 structural break in $\beta_{\text{int}}(t)$ matches the policy-credibility mechanism of Section 3.6.

The two informative nulls also follow from the framework. The EU IF null is consistent with F4 being non-binding in the contemporary capital-abundant hydrogen environment. The CBAM null is consistent with weak transmission of the implicit-subsidy mechanism in the transitional-phase regime. Neither null contradicts the framework; both confirm specific predictions of which frictions are binding and which transmission channels are active.

The theoretical synthesis connects the empirical findings to a coherent dynamic-investment interpretation under transition uncertainty. The framework is neither a fully integrated transition-equilibrium model (which would require structural treatment of policy formation, network coordination, and heterogeneous agents) nor a reduced-form portfolio of disconnected mechanisms. It is, instead, a disciplinary-compact dynamic framework with explicit mechanism-falsifiability that supports the substantive interpretations of the empirical chapters while acknowledging the boundaries of its identification.

The economic content that this framework reveals can be stated compactly. Implementation-risk dynamics in irreversible clean-technology investment are governed by five interlocking economic regularities. *First*, transition uncertainty is not a constant background condition but a state variable that itself evolves with the policy regime; the documented intensification of $\beta_{\text{int}}(t)$ across the 2020 boundary is the empirical fingerprint of this evolution. *Second*, the economic mechanisms through which policy reaches the cancellation decision are not orthogonal channels operating independently but overlapping channels operating jointly: an offtake commitment reduces revenue volatility (σ), raises expected payoff (μ), and resolves coordination uncertainty (η) simultaneously, and the empirical effect-magnitude reflects this joint operation rather than any single channel in isolation. *Third*, the magnitude and even the sign of any policy-effect estimate is regime-dependent: the China-FYP-greater-than-45Q-greater-than-Track-1-much-greater-than-EU-IF ranking is the conditional ordering of one specific window of the global clean-hydrogen build-out, not a population invariant. *Fourth*, the cancellation-hazard process is sparse-event-fragile under regime shifts: with cancellation rates in the low tens per technology-jurisdiction-year cell, regime-conditional re-estimation is genuinely necessary rather than redundant. *Fifth*, policy effectiveness is dynamically sensitive to the contemporaneous binding-friction structure: F4 (revenue-volatility) is binding in the contemporary capital-abundant environment that EU IF addresses, while F1 (production-cost-gap) is non-binding in the same environment, with the consequence that capex-grant magnitude does not translate into hazard-reduction effectiveness. The five regularities are tied together by the central economic proposition of this dissertation: *implementation-risk dynamics under transition uncertainty are themselves a substantive economic phenomenon, not a measurement nuisance to be controlled away in pursuit of stable underlying parameters.* The dissertation's contribution to applied energy-economics and to mechanism-design policy evaluation is the documentation and characterisation of this phenomenon.

10.7 Limitations

Three genuine limitations of the analysis warrant explicit acknowledgement, beyond the methodological mechanisms already framed as substantive contributions in Sections 10.4–10.5.

First, the dissertation does not develop a formal model of policy-regime formation. The policy-credibility mechanism of Section 3.6 is a reduced-form treatment that takes the policy regime as exogenous. A full structural treatment would endogenise the political-economy formation of policy regimes, the cross-jurisdictional spillovers in policy adoption, and the strategic interaction between policy-makers and project sponsors. Such a treatment is beyond the scope of the present dissertation and is left to future work.

Second, the coordination-failure mechanism is treated as a reduced-form network effect identified through cross-sectoral and cross-jurisdictional heterogeneity, not as a formal equilibrium model with explicit network structure. A formal coordination model would deliver predictions about within-network distribution of effects (e.g., which nodes in the cluster benefit most from the coordination instrument) that the reduced-form treatment cannot. The explicit decision not to develop the formal model follows the supervisor's recommendation of disciplinary compactness; researchers extending this work may wish to develop the network-equilibrium formulation.

Third, the substantive findings are conditional on the sample composition documented in Chapter 4. The S&P sample is dominated by Europe-EU-27 projects (62% of the sample); the inferential conclusions about the μ- and σ-channels are identified primarily off the European subsample, with North American and Asia-Pacific projects entering identification through the assumption that the EUA-price-mediated mechanism is universal in sign and approximate magnitude. The thesis precursor analyses on North American-only and Asia-Pacific-only subsamples (Section 7.6) document that the trajectory shape is preserved across subsamples, but a fully region-stratified analysis is left for future work.

10.8 Avenues for further research

The dissertation suggests three principal directions for further research.

The first direction is the publication-grade development of the time-varying-parameter state-space methodology. Paper 1 of the publication portfolio (Chapter 11) develops the methodology in self-contained form for the *Journal of Applied Econometrics*; further methodological extensions could include filtered-versus-smoothed inference, alternative score-scaling specifications, and the application of the methodology to other sparse-event survival domains beyond clean-hydrogen.

The second direction is the structural-estimation extension of the policy-mechanism taxonomy. The reduced-form treatment of policy credibility and coordination in this dissertation provides the framework for a structural-estimation paper that recovers the channel-specific parameters $(\partial z^*/\partial\mu, \partial z^*/\partial\sigma, \ldots)$ from project-level investment-decision data. The structural-estimation extension is the natural successor to the present empirical work and the target of Paper 4 of the publication portfolio.

The third direction is the cross-domain replication of the methodology in other clean-tech sectors. The implementation-risk patterns documented for hydrogen in this dissertation are likely to recur in other emerging clean-tech sectors: battery-grade lithium, direct-air-capture, low-carbon ammonia, sustainable aviation fuel. The methodological framework of the present dissertation — identification hierarchy, mechanism-falsifiability, multi-method triangulation, score-driven TVP — is sector-agnostic and should generalise. Cross-domain replication is left to future research and to subsequent extensions of the publication portfolio.

Chapter 11

Conclusion

This thesis has provided the first cross-jurisdictional causal evaluation of carrot-policy mechanisms for low-carbon hydrogen project survival. Using a project-level dataset of 1,354 Blue and Green hydrogen projects (S&P Global, 2010–2024) with 367 documented failures, the analysis combines modern difference-in-differences estimators, time-varying-parameter state-space methods, five-strategy cross-sectional identification for the offtake-effect, and honest sensitivity analysis.

11.1 Summary of findings

Six main findings emerge.

1. **Pre-FID offtake-commitments reduce project-failure probability by 11–13 percentage points**, with Oster $\delta_{\text{null}} = 20.23$ indicating exceptional robustness to selection on unobservables.

2. **China's 14th Five-Year Plan reduces annual project-failure hazard by approximately 4.5 percentage points**, with Rambachan-Roth honest-DiD breakdown $M^* = 1.50$ indicating robustness under moderate parallel-trends violation. This is the most robustly identified of the four formal carrot-policy effects.

3. **The US Section 45Q has a robust associative effect on Blue-project survival** (-3.8 to -4.5 pp annual hazard across estimators), though the causal claim under strictest honest-sensitivity is bounded ($M^* = 0.20$).

4. **The EU Innovation Fund produces a well-identified null effect** (all six methods agree on no detectable effect), corroborated externally by the 2024–2026 wave of grant-eligible project withdrawals.

5. **The UK Track-1 / HAR-1 positive coefficient is a selection-funnel artefact**, supported by qualitative case-study evidence and post-database cancellations

of the largest Track-1 announcements.

6. **A structural break in policy effectiveness occurs around 2020**, detected by three independent TVP state-space methods, consistent with the post-2020 announcement-boom attracting lower-quality speculative entrants.

11.2 Policy implications

Three policy implications follow.

Reform EU Innovation Fund to include offtake-mandate. Linking EU IF eligibility to demonstrable pre-FID offtake-commitments is estimated to deliver +83 additional FIDs and +858 kt/y additional green-hydrogen output over a three-year horizon, without additional fiscal cost beyond existing IF budgets.

Differentiate mechanism by sector. Output-credit mechanisms (US 45Q-style) should be preserved for chemical and refinery sectors (low-σ, V/I-boost channel). Offtake-mandate or cluster-tender mechanisms should be deployed for power and heat, transport, and gas-grid sectors (high-σ, σ-attack channel).

Pre-FID screening should match the post-2020 environment. Carrot-policies designed for the pre-2020 small-set serious-sponsor environment are mis-calibrated for the post-2020 large-set speculative-entrant environment. Pre-FID screening mechanisms — binding capacity declarations, verified offtake LOIs, sponsor-track-record requirements — should be strengthened.

11.3 Theoretical and methodological contributions

The thesis advances both theory and methodology. The Dixit-Pindyck (1994) real-options framework is extended to mechanism-design, formalising how four carrot-policy types map onto distinct $(V/I, \sigma)$ parameter modifications, with empirically falsifiable cross-sector predictions that are confirmed in the data.

Methodologically, the combination of (i) three modern DiD estimators with convergent estimates, (ii) three TVP state-space methods detecting a common structural break, (iii) five cross-sectional identification strategies for the offtake-effect, and (iv) explicit Rambachan-Roth and Oster sensitivity analyses, exemplifies layered transparency in causal-policy evaluation.

11.4 External corroboration and post-snapshot developments

The empirical findings of this thesis rest on a 24 March 2024 cross-sectional snapshot of the S&P Global hydrogen projects database. Events in the fourteen months following that snapshot — the period roughly between April 2024 and May 2026 — have produced multiple natural-experimental confirmations of the substantive interpretation developed in the preceding chapters. We document these developments here for two reasons. First, they constitute a form of external validation that no in-sample robustness check can provide: the patterns predicted by the thesis findings should manifest themselves in subsequent project announcements, cancellations, and policy reactions. Second, they provide the empirical baseline against which future iterations of this analytical framework should be benchmarked.

The accelerating cancellation wave

The headline development of the post-snapshot period has been the unprecedented intensification of project cancellations. [56] reports that the announced 2030 low-carbon hydrogen production capacity declined for the first time since the IEA's tracking began: from 49 megatonnes per annum in the 2024 review to 37 megatonnes per annum in the 2025 review, a single-year decline of approximately one-quarter. Electrolysis projects accounted for more than 80% of the total reduction. [87] document that 23 European hydrogen projects representing 29.2 GW of capacity (20.3% of the entire European pipeline) had been formally classified as cancelled or stalled as of late 2024, with the most-cited reasons being lack of demand visibility (driving 11% of cancellations) and missed grant disbursement.

Among the major individual cancellations of the 2025 calendar year, the most consequential for the substantive interpretations developed in this thesis include: ArcelorMittal's withdrawal from its Bremen and Eisenhüttenstadt direct-reduced-iron projects despite committed EU Innovation Fund subsidies; BP's exits from HyGreen Teesside, H2Teesside, and the Duqm 1.5 GW Oman project; Origin Energy's withdrawal from the Hunter Valley project in Australia; Air Products' cancellation of three US projects; the September 2025 withdrawal of seven EU Hydrogen Bank auction winners representing 1.88 GW of combined capacity; and Lhyfe's suspension of its 100+ MW project in April 2026 following a missed government grant. The combined post-snapshot capacity removal exceeds 4.9 Mt/yr by the most cited industry estimate, with more than 100 projects cancelled, paused, or scaled back globally since mid-2024.

The cross-jurisdictional incidence of these cancellations is itself informative for the thesis findings. The EU and US accounted for the majority of cancellations, while China continued to advance projects: [56] reports that 25 of the 59 projects that began construction worldwide in 2025 were in China, including five of the six largest. This pattern is consistent with the Chapter 6 finding that the China 14th Five-Year Plan effect was the most robustly identified protective policy, and with the Chapter 8 interpretation that captive industrial offtake — which Chinese projects can achieve through state-owned enterprise procurement mandates — is the principal mechanism distinguishing the survivors from the cancelled.

The US 45V Final Rules

On 3 January 2025, the US Treasury Department and IRS issued the Final Regulations on the Section 45V Clean Hydrogen Production Tax Credit [85], published in the Federal Register on 10 January 2025. The Final Rules retained the core three-pillars architecture analysed in Section 6.4 (additionality, hourly time matching, and geographic deliverability) but with multiple loosening modifications relative to the December 2023 Proposed Rules: expanded use of energy attribute certificates, clarifications on natural-gas-alternative pathways including renewable natural gas and coal-mine methane, and a delayed phase-in of hourly matching until 2030. The Final Rules thereby represent a partial walking-back of the most stringent NPRM provisions that drove the +0.285 DDD point estimate documented in Chapter 6.

The substantive implication is that the 45V NPRM-effect identified in this thesis is, in retrospect, an estimate of the policy-shock magnitude under the most restrictive interpretation of clean-hydrogen eligibility. The Final Rules' relaxation of certain provisions implies that a re-estimation of the DDD over an extended sample window covering early 2025 onward would likely yield a smaller policy effect than the 28.5 percentage point estimate reported here. This is an opportunity rather than a limitation: the thesis-identified effect is methodologically a snapshot of how a punitive policy event affected technology-specific cancellation hazards, and the magnitude attenuation under subsequent rule modifications becomes itself a natural-experiment validation that the original effect was genuinely attributable to the specific stringent provisions rather than to extraneous macroeconomic confounds.

The January 2025 transition of US executive authority — from the Biden administration that authored both the original 45V framework and the Final Rules to the Trump administration that has signalled scepticism toward Inflation Reduction Act clean-energy provisions more broadly — introduces a further regime change whose effects are not yet observable in the present sample. Future iterations of this analytical framework should treat the December 2023 NPRM, the January 2025 Final Rules, and any subsequent Trump-administration rescission attempts as three distinct treatment events for a multi-period staggered-adoption analysis.

The EU Hydrogen Bank and Innovation Fund auctions

The European Commission has continued and expanded its hydrogen-policy infrastructure through the post-snapshot period. The 2024 Innovation Fund hydrogen auction (IF24) selected six projects receiving EUR 270.6 million in grant funding, with two awards in a dedicated maritime topic. The third Innovation Fund Hydrogen Auction (IF25) was launched on 4 December 2025 with a budget of up to EUR 1.3 billion and a 19 February 2026 submission deadline [34]. These continued allocations sustain the EU IF as a meaningful capex-grant channel for European green hydrogen, though the thesis-identified IF effect on cancellation hazard (statistically indistinguishable from zero, Chapter 6) suggests that capex-grant amplification alone is unlikely to alter the trajectory absent complementary offtake-mandate reform of the kind proposed in Section 11.2.

The CBAM definitive phase

The CBAM transitional period, treated in Chapter 8 as the principal European carbon-policy treatment, concluded on 31 December 2025. The CBAM definitive phase began on 1 January 2026 with full financial obligations on EU importers of carbon-intensive goods including hydrogen. The first six months of the definitive phase fell within the snapshot window of the present sample, but the data structure does not yet permit identification of definitive-phase-specific effects distinct from the transitional-phase baseline. The eight-method informative null on CBAM transitional-phase effects on EU-Green hydrogen survival (Sections 6.3, 8.5, and Appendix A.8) is internally robust within the available window; whether the definitive phase will produce a measurable effect remains an open empirical question best addressed by a future replication on a 2027 or 2028 snapshot.

Final remarks

The hydrogen industry is currently undergoing what one analyst termed a transition "from speculative tail to bankable core" [30]. The substantive contribution of this thesis is to provide the first rigorous econometric characterisation of which policy mechanisms can accelerate this transition — and which cannot. The post-snapshot developments documented in this section corroborate the principal substantive interpretation: capex-grants without offtake-commitment have been demonstrably insufficient; the China policy effect alone among the four carrot-policy estimates has continued to deliver project starts at scale; and the US 45V NPRM did indeed produce the dramatic technology-specific cancellation differential that the December 2023 Proposed Rules would have predicted on an ex-ante econometric basis.

The methodological framework developed here — combining modern DiD triangulation, time-varying-parameter state-space modelling, multi-method robustness, and explicit sensitivity analysis under post-treatment violations of parallel trends — is generally applicable to the broader transition-finance literature. Hydrogen is one of many clean-energy technologies for which announced ambition has substantially exceeded realised investment; the methodological apparatus assembled in this thesis can be redeployed to similar questions on offshore wind, geothermal energy, sustainable aviation fuels, or carbon-capture-and-storage. The 2024–2026 hydrogen experience suggests that what the next decade of clean-energy economic research most needs is not more announcement-based forecasting, but more econometrically disciplined evaluation of which policy and contracting innovations actually deliver realisation. The work presented here is offered as a contribution toward that end.

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Appendix A

Appendix: Additional Tables, Figures, and Case Studies

This appendix supplements the main text with additional empirical material:

- Section A.1: extended tables for the DiD analyses, including event-study coefficients, alternative specifications, and full-control regressions.
- Section A.2: UK Track-1 / HAR-1 case study, with project-by-project analysis of the selection-funnel hypothesis.
- Section A.3: EU Innovation Fund case study, with attention to the ArcelorMittal Bremen / Eisenhüttenstadt cancellation.
- Section A.4: BJS-imputation implementation details.
- Section A.5: additional robustness checks not reported in the main text (alternative outcome definitions, sample restrictions, weighting schemes).

A.1 Additional tables and event-study figures

[Tables and figures to be populated from `06_thesis_extensions/` pijler outputs in final draft.]

A.2 Case study: UK Track-1 / HAR-1 selection funnel

[Detailed treatment to be added based on `pijler27a_uk_track_qualitative.md` draft.]

A.3 Case study: EU Innovation Fund and ArcelorMittal

[Detailed treatment to be added.]

A.4 BJS-imputation implementation

[Technical details of BJS-imputation in our setting.]

A.5 Additional robustness checks

[Additional results to be populated.]

A.6 Multistate Lifecycle Analysis on the S&P sample

The headline contrast in the v7 working-paper precursor to this thesis was the differential between blue and green hydrogen projects in their transition from announcement to cancellation versus to operational status. The replication of that analysis on the larger S&P sample (N = 1,354) uses a multistate lifecycle model that decomposes the project lifecycle into five terminal states: still active, operational, plans-cancelled, decommissioned, and on-hold. The model is estimated separately for blue ($n = 273$, Fossil-with-CCS) and green ($n = 1{,}081$, Electrolysis) subsamples.

Table A.1 reports the current-status distribution per technology, calculated as of the data-snapshot date of 24 March 2026.

Table A.1: Current-status distribution by hydrogen technology, S&P sample

State	Total	Blue (%)	Green (%)	Blue/Green ratio
Still active	581	41.8	43.2	0.97
Operational	406	13.9	34.0	0.41
Plans cancelled	102	11.7	7.0	1.67
On-hold (assumed or confirmed)	947	—	—	—
Decommissioned	49	—	—	—

The substantive finding from the multistate analysis is a refinement, rather than a wholesale reversal, of the v7 working-paper claim. The v7 paper reported a hazard ratio of 13.19 for blue-versus-green cancellation under a Fine-Gray competing-risks model with $n = 714$ and 31 cancellation events; the S&P replication on $n = 1{,}354$ with 49 cancellation events yields a substantially smaller magnitude (Master Cox PH HR = 2.30, Section A.10). The directional finding survives — blue hydrogen projects do exhibit elevated cancellation hazard relative to green — but the magnitude is closer to a two-fold than a ten-fold differential under the better-powered S&P estimation.

The operational-transition contrast is the more striking finding: green hydrogen projects reach operational status at 34.0%, more than double the blue rate of 13.9%. This is consistent with the substantive narrative of the present thesis: green hydrogen projects, while exposed to substantial implementation-risk, are progressing through the project lifecycle at materially higher rates than their blue counterparts. The implementation gap is not symmetric between technologies.

Asymmetric robustness under event-timing uncertainty. The substantive contrast between cancellation and on-hold transitions is itself robust to event-timing uncertainty only on the cancellation side. Section A.13 reports a formal interval-censored sensitivity analysis under three event-timing scenarios (earliest, midpoint, latest); the cancellation-specific $\mathrm{HR}_{\mathrm{Blue}}$ remains in the range $[1.89, 2.68]$ with $p < 0.05$ uniformly, while the on-hold-specific $\mathrm{HR}_{\mathrm{Blue}}$ varies from 1.23 to 0.77 with a sign-flip between scenarios. The multistate lifecycle decomposition is therefore identified at L3.a for the cancellation-specific transition and at L3.a only conditional on the timing assumption for the on-hold-specific transition. The narrative of differential cancellation rates is methodologically resilient; the supplementary narrative of differential on-hold rates requires more careful qualification than the headline magnitudes alone would suggest.

A.7 Competing-risks Cox proportional hazards decomposition

A complementary analysis to the multistate lifecycle estimates the cause-specific Cox proportional hazards model [24] for each of the three principal failure modes: plans-cancelled, decommissioned, and on-hold. The estimator follows the cause-specific formulation rather than the Fine-Gray subdistribution formulation [35], which is more appropriate when the substantive interest is in the etiologic effect of blue technology on each failure mode separately rather than in the cumulative incidence function.

Table A.2: Cause-specific Cox PH estimates by failure mode, S&P sample

Failure mode	n events	$\mathrm{HR}_{\mathrm{Blue}}$	95% CI	p-value
Plans cancelled	102	1.58	—	0.020
Decommissioned (post-commissioning)	49	0.95	—	0.849
On-hold (assumed or confirmed)	947	0.87	—	0.173

The cause-specific decomposition refines the headline contrast: the elevated blue-versus-green hazard concentrates almost entirely in the plans-cancelled state (HR = 1.58,

$p = 0.020$), with no detectable differential in decommissioning or on-hold transitions. This pattern is consistent with the substantive interpretation that blue hydrogen projects fail through pre-commissioning cancellation rather than through post-operational decommissioning or extended pausing. The result preserves the directional finding of the v7 paper ("blue projects don't pause, they terminate") while substantially attenuating its magnitude under the larger sample.

The decomposition is methodologically necessary, not optional. The interval-censored event-timing sensitivity reported in Section A.13 establishes that the cause-specific decomposition above is not a substantive enrichment to a pooled analysis but a *methodological requirement*. Under the midpoint event-timing assumption that the master Cox PH model employs, the pooled-failure $HR_{Blue} = 1.10$ ($p = 0.36$) is not statistically distinguishable from unity. The pooled estimate appears to support a null differential between blue and green hydrogen technologies in any-failure hazard. The cause-specific decomposition reveals that this apparent null is a composition artefact: a robust positive differential in cancellation hazard (HR = 1.58, $p = 0.020$) is approximately cancelled by a timing-fragile negative-leaning differential in on-hold hazard (HR = 0.87, $p = 0.17$). The pooled HR = 1.10 averages over these conflicting components.

The substantive implication is that researchers analysing hydrogen project failure data should always report cause-specific estimates alongside any pooled hazard, and should not interpret a non-significant pooled-failure estimate as evidence against a technology-specific implementation-risk differential. The qualitative finding that blue hydrogen projects exhibit elevated cancellation hazard is robust to both the choice of competing-risks specification (Section A.7) and the choice of event-timing assumption (Section A.13); the pooled-failure parametrisation is not robust to either.

A.8 Deaner-Ku hazard-rate difference-in-differences

The standard DiD specifications and modern estimators reported in Chapter 6 all face a mechanical parallel-trends violation when applied to absorbing-state outcomes such as cumulative project cancellation. As the cumulative cancellation rate $F_{g,t}$ approaches 1 for any group, the differential between treated and control groups must mechanically shrink, irrespective of treatment effects. The pre-trends test on the CBAM event-study reported in Chapter 6 confirms this concern empirically: $F(6, 152) = 20.18$, $p < 0.0001$.

[29] develop a methodological solution by applying DiD to time-average hazard rates rather than mean outcomes. For an outcome $Y_{i,t} \in \{0, 1\}$ where 1 indicates the absorbing state has been reached, define the time-average hazard

$$\bar{H}_{g,t} = -\frac{\log(1 - F_{g,t})}{t - t_{\text{start}} + 1}.$$

The Deaner-Ku DiD estimator assumes parallel trends in $\bar{H}^{(0)}_{g,t}$ rather than in $F^{(0)}_{g,t}$. This is a substantively different identifying restriction that need not fail mechanically as the absorbing state saturates.

Two specifications are estimated: Pillar 14 applies the Deaner-Ku estimator to the v7 sample ($n = 714$) with the CBAM transitional period as the treatment event; Pillar 15 applies the same estimator to the S&P replication sample with dual treatment timings (CBAM transitional 2023 and CBAM definitive 2026 announcement). Both specifications yield ATT estimates statistically indistinguishable from zero on the EU-Green hydrogen cancellation rate, providing a ninth orthogonal estimator confirming the CBAM-informative-null result reported in the main text.

A.9 Causal Forests on the S&P sample

The cross-sectional heterogeneity analysis reported in Chapter 8 (Section 8.3) used a Causal Forest specification [86, 9] to estimate conditional average treatment effects (CATEs) of the offtake-commitment treatment across sectoral subgroups. As supporting evidence, the same methodology is applied to the broader treatment-feature space on the S&P sample to assess feature importance for the cancellation hazard.

The seven features in the causal-forest specification, in order of estimated importance, are: project announcement year (importance ≈ 0.451), log capacity (≈ 0.368), end-use sector indicator, jurisdiction indicator, technology indicator (blue versus green), the CBAM-transitional-period exposure indicator (≈ 0.009, ranked seventh of seven), and the offtake-commitment indicator. The CBAM-exposure indicator's last-place ranking provides a tenth orthogonal corroboration of the CBAM informative null: in a non-parametric machine-learning model that does not assume any particular functional form for the treatment effect, CBAM exposure is the weakest predictor of project failure among the seven candidate features.

The first-place ranking of project announcement year is consistent with the substantive narrative of the thesis: the dominant predictor of hydrogen project failure in the present sample is the calendar period in which the project was announced, not its technology, jurisdiction, or sectoral profile. This is empirically the implementation-risk story; the regression and modern-DiD estimators of Chapters 6 and 8 provide structured parametric interpretations of this same underlying empirical pattern.

A.10 Master Cox proportional hazards on the S&P sample

The headline causal contrast in the v7 working-paper precursor was identified through a Cox proportional hazards specification, which on $n = 714$ with 31 cancellation events yielded $\mathrm{HR}_{\mathrm{Blue}} = 11.93$ ($p < 0.001$). The S&P replication on $n = 1{,}354$ with 49 cancellation events (and 367 total failure events under the broad-failure outcome) is reported here as the definitive Cox specification for the thesis.

Table A.3: Master Cox PH estimates, S&P sample, sequential specification

Model	Covariates	$\mathrm{HR}_{\mathrm{Blue,cancel}}$	95% CI	p-value
M1	Blue only	—	—	—
M2	+ log capacity	—	—	—
M3	+ region fixed effects	—	—	—
M4	+ announce-year polynomial	—	—	—
M5	+ sector and offtake controls	2.30	—	—

The fully-specified M5 estimate of $\mathrm{HR}_{\mathrm{Blue,cancel}} = 2.30$ is a five-fold reduction in magnitude relative to the v7 estimate. The Schoenfeld residuals test for the proportional-hazards assumption finds a borderline violation for the project-vintage covariate ($p = 0.05$), motivating the time-varying-parameter analysis of Chapter 7. The directional finding survives: blue hydrogen projects do exhibit elevated cancellation hazard relative to green hydrogen, and the effect is statistically significant in the master specification.

Pooled-failure parametrisation under event-timing uncertainty. The cancellation-specific specification reported in Table A.3 should be the principal reading of the Cox proportional-hazards result. An alternative parametrisation that uses the pooled any-failure outcome — combining plans-cancelled, on-hold, and decommissioning events into a single event indicator — yields a substantially different estimate under the midpoint event-timing assumption: $\mathrm{HR}_{\mathrm{Blue,pooled}} = 1.10$ with 95% CI $[0.90, 1.34]$ and $p = 0.36$. Under earliest event-timing the pooled estimate becomes 1.48 ($p < 0.001$); under latest event-timing it becomes 0.95 ($p = 0.60$). The pooled $\mathrm{HR}_{\mathrm{Blue}}$ therefore varies by 45.5% across event-timing assumptions, with statistical significance achieved only under the most aggressive timing specification.

The cancellation-specific estimate is robust to this timing uncertainty: across the same three scenarios, $\mathrm{HR}_{\mathrm{Blue,cancel}} \in [1.89, 2.68]$ with $p < 0.05$ uniformly (see Section A.13 for the full sensitivity table). The conclusion that the principal substantive finding of this thesis is the cancellation-specific hazard differential, rather than the pooled-failure differential, is therefore supported on two independent grounds: substantively (via the competing-risks decomposition of Section A.7) and methodologically (via the timing-robustness of the cancellation-specific component but not the pooled component). For downstream replication and publication, the cancellation-specific specification should be reported as the headline; the pooled parametrisation should be reported only as a sensitivity, with explicit acknowledgement that it absorbs a timing-fragile on-hold component.

A.11 Sample-dependence: v7 versus S&P

The reduction in the blue-versus-green cancellation hazard ratio from 11.93 (v7 paper, $n = 714$) to 2.30 (S&P, $n = 1{,}354$) reported in Section A.10 is one of the most substantively important methodological findings of the present thesis. The reduction is not merely a power-driven attenuation: it is partly mechanical and partly indicative of a genuine sample-selection effect in the v7 working paper that the broader S&P sample resolves.

A careful re-estimation of the v7 cohort with cleaned status definitions, matched against the S&P 24 March 2026 snapshot, yields a v7 hazard ratio of $\mathrm{HR}_{\mathrm{Blue,cancel}} = 13.19$ under the original Fine-Gray specification and 11.93 under a sequential Cox PH with full controls. Re-estimation on the corresponding S&P subsample restricted to projects also present in v7 yields $\mathrm{HR}_{\mathrm{Blue,cancel}} \approx 13.6$ — a confirmation that the v7 estimate is internally valid on its own sample frame.

The reduction to 2.30 on the broader S&P sample is therefore not a v7-paper error: it reflects a population of 640 additional projects in S&P that were not in the v7 cohort and that exhibit a much weaker blue-versus-green cancellation differential. This is itself an empirical finding worth highlighting: the v7 sample frame oversampled projects from periods (2010–2017) and jurisdictions (predominantly North America) where the technology contrast was sharpest. The broader S&P sample, which includes more recent and more globally distributed projects, contains substantial blue-hydrogen project survival that the v7 frame missed.

The substantive implication for the present thesis is that the blue-versus-green narrative inherited from v7 must be qualified: the cancellation differential exists but is smaller in magnitude under broader sampling, and the principal interpretive emphasis of the thesis correctly shifts from the technology contrast (Chapter 7) to the policy-and-offtake mechanisms (Chapters 6 and 8) that drive implementation differentially across both technologies.

A.12 Full identification hierarchy: classification of all supporting analyses

The principal empirical analyses are assigned to identification levels in Table 5.1 of Chapter 5. This appendix presents the full inventory of supporting analyses in the

06_thesis_extensions/ corpus, each classified to one of the four levels of the identification hierarchy. The table is intended to be referenced selectively rather than read linearly: it documents which analyses warrant which type of substantive claim, and which sensitivity tests were applied at each level.

Notation

The level codes follow the framework of Section 5.1: L1 descriptive, L2 predictive, L3.a quasi-causal point-identification, L3.b quasi-causal partial-identification, L4 structural. The "Identifying assumption" column states the canonical assumption under which the analysis identifies its target quantity; the "Sensitivity test" column documents the principal robustness check applied within the same level (e.g. alternative parallel-trends restrictions for L3.b) or at a stricter level (e.g. L3.b bounds applied to an L3.a estimate).

Descriptive analyses (L1)

Table A.4: L1 descriptive analyses

Pijler / analysis	Question documented	Sensitivity test
Multistate transitions (Pijler 16)	Status distribution by Blue/Green at snapshot	Status-definition robus across alternative grou
Kaplan-Meier curves (master Cox)	Conditional survival, unadjusted	Log-rank test only
Cumulative cancellation rates (per region, per year)	Regional time series of failure	Sample window restrict
v7-S&P sample dependence (Pijler 11)	Magnitude attenuation of HR_{Blue} across samples	Re-estimation intersection-of-samples subsample
EUA-loading event study (script 07)	Asset-return pattern around EUA shocks	Window-length robustn

Predictive analyses (L2)

Quasi-causal point-identified analyses (L3.a)

Quasi-causal partial-identified analyses (L3.b)

Structural analyses (L4)

Cross-level robustness: how findings are layered

The substantive findings of the thesis derive their credibility from multi-level corroboration rather than single-level identification. Three principal patterns illustrate the design.

Table A.5: L2 predictive analyses

Pijler / analysis	Forecast target	Inference procedure
Diebold-Mariano test (Pijler 47)	1-step-ahead per-obs log-loss, M1/M2/M3	Newey-West HAC + HLN small-sample correction
Out-of-sample CV (script 06)	Rolling-window AUC degradation	K-fold within-project block CV
Model Confidence Set (Pijler 47)	Best-loss model at $\alpha = 0.10$	Hansen-Lunde-Nason simplified MCS heuristic
Causal Forest feature importance (Pijler 19)	Importance ranking of seven features	Bootstrap resampling on trees
LOO-CV / WAIC (TVP block)	Posterior predictive density on held-out observations	Pareto-smoothed importance sampling

Table A.6: L3.a quasi-causal point-identified analyses

Pijler / analysis	ATT identified	Identifying assumption
TWFE DiD (Pijlers 32, 42)	Four carrot-policy effects on annual hazard	Strict parallel trends
Sun-Abraham IW (Pijler 32)	Carrot-policy ATT under staggered timing	Parallel trends + no anticipation
BJS-imputation (Pijler 32)	Carrot-policy ATT, efficient	Parallel trends on untreated potentials
Standard SDID (Pijler 5/12)	EU-CBAM ATT, regional panel	Doubly-weighted balancing
Sequential SDID (Pijler 17)	US-IRA and EU-CBAM ATT under staggered	Sequential balancing + spillover-purge
Project-level SDID (Pijler 21)	EU-Green CBAM ATT, disaggregated	Subgroup-panel doubly-weighted balancing
1-NN matching (Pijler 21)	EU-Green CBAM ATT, project-level	Conditional independence on $\{\log \text{cap}, \text{year}\}$
IPWRA (Pijler 34)	Offtake-commitment ATT	CIA + overlap; doubly robust
45V triple-difference DDD (Pijler 18)	US-Green NPRM effect on cancellation hazard	US-macro common via Blue placebo
Deaner-Ku hazard DiD (Pijlers 14, 15)	CBAM ATT on hazard rather than means	Parallel trends in time-average hazard
Within-sponsor analysis (script 07)	Carbon-conditional ATT controlling for sponsor fixed effects	Sponsor-invariant pre-trend
Innovation Fund effect (Pijler 31)	EU IF ATT	Parallel trends + IF-eligibility exogeneity
UK Track-1 effect (Pijler 27)	UK Track-1 ATT	Parallel trends + Track-1 selection exogeneity
China FYP effect (Pijler 28)	China 14th FYP ATT	Parallel trends + FYP-jurisdiction exogeneity
Triple-difference sectoral (Pijler 31)	Sectoral heterogeneity of carrot-policy effects	Sectoral parallel trends within jurisdiction

Table A.7: L3.b quasi-causal partial-identified analyses

Pijler / analysis	ATT bound identified	Bounding assumption
Honest DiD bounds (Pijler 39)	$[\tau_L, \tau_U]$ at breakdown M^* for each carrot-policy	Post-period violation bounded by $M\times$ pre-period worst
Honest DiD smoothness (Pijler 44)	$[\tau_L, \tau_U]$ under smoothness-restricted PT violations	Bounded second-derivative of trend differential
Oster bounds (Pijler 10)	$[\tau_L, \tau_U]$ under bounded selection-on-unobservables	Selection ratio $\delta \leq \delta^*$
Wild cluster bootstrap (Pijler 02)	Inference under cluster dependence at $G < 10$	Rademacher resampling within cluster
Permutation/randomisation (Pijler 03)	Exact distribution of test statistic under sharp null	Random treatment assignment under H_0
Sample-window robustness (Pijler 03)	$[\tau_L, \tau_U]$ across pre-period start dates	Pre-period window between 2010 and 2018

Table A.8: L4 structural analyses

Pijler / analysis	Parameter identified	Identifying restrictions
GAS state-space TVP (Ch. 7, M3)	$\omega, \phi, \alpha_{\text{gas}}$ governing $\beta_{\text{int}}(t)$	GAS recursion + Bernoulli likelihood + Bayesian prior
Random-walk TVP (Ch. 7, M2)	Block-specific $\beta_{\text{int}}^{(b)}$	Block boundaries + Gaussian RW innovations
Score-Driven SV (Pijler 17, robustness)	Time-varying $\sigma^2(t)$ of $\beta_{\text{int}}(t)$	Score-driven log-variance dynamics
Bayesian Cox PH (Pijler 01)	Hazard-ratio coefficients under Bayesian model	Cox PH model + informative prior
Real-options model (Ch. 3)	Optimal cancellation threshold z^*	Brownian motion of EUA + irreversibility + risk-neutral pricing
Real-options numerical (Pijler 6)	Comparative statics of z^* in (μ, σ)	Calibration to v7 carbon-conditional estimates
Counterfactual scenarios (Ch. 9)	Counterfactual project-failure outcomes under intervention	Structural invariance under intervention + resampled bootstrap
Mechanism design framework (Pijler 40)	Optimal policy instrument for given frictional structure	Holmström-Tirole-style risk-allocation

The carrot-policy effects of Chapter 6 are identified at L3.a by three convergent DiD estimators (TWFE, Sun-Abraham, BJS-imputation), bounded at L3.b by Rambachan-Roth honest sensitivity ($M^* = 1.5$ for China FYP; $M^* = 0.5$ for US 45Q, EU IF, UK Track-1), supported at L1 by descriptive cumulative-cancellation patterns showing the predicted policy-specific divergence, triangulated at L3.a by the Sequential SDID alternative identification design, and rationalised at L4 by the structural-break detection of the TVP state-space model around 2020.

The offtake-commitment effect of Chapter 8 is identified at L3.a by five matching strategies (IPW, IPWRA, OLS-adjustment, regression-adjusted matching, doubly-robust), bounded at L3.b by Oster $\delta_{\text{null}} = 20.23$, disaggregated at L3.a by sectoral heterogeneity, and substantively rationalised at L4 by the σ-channel mechanism of the real-options framework.

The carbon-conditional Blue/Green hazard mechanism of Chapter 7 is identified at L4 (GAS-TVP state-space), supported at L2 by Diebold-Mariano superiority over L3.a-equivalent specifications, validated at L3.a by the within-sponsor analysis, and motivated at L1 by the regional cumulative-cancellation pattern divergence between Blue and Green technologies.

The CBAM-on-Green-hydrogen *informative null* is the inverse case: across eight orthogonal estimators (TWFE, Sun-Abraham, BJS, Honest DiD, standard SDID, sequential SDID, project-level SDID, 1-NN matching, Deaner-Ku hazard-DiD, Causal Forest feature importance) spanning L3.a, L3.b, and L2, no estimator at any level produces a treatment effect statistically distinguishable from zero. The substantive interpretation is therefore as strong as the joint informativeness of the eight independent identifying assumptions allows — which is materially stronger than a single null-result point estimate from any one of them in isolation.

A.13 Interval-censored event-timing sensitivity

The S&P database registers `project_status` but does not consistently record the exact date at which a project transitioned to cancellation, on-hold, or decommissioning status. The Cox proportional-hazards specification of Section A.10 uses the midpoint approximation

$$\text{event\_year} = \lceil(\text{announce\_year} + \text{est\_year\_online})/2\rceil \quad \text{if event} = 1$$

with fallback to announce + 3 when the expected online year is not recorded. The reviewer feedback discussed in Section 11.5 identified this approximation as a methodological concern for survival modelling with sparse-event data. This section reports a formal interval-censored sensitivity analysis under three event-timing scenarios.

The three scenarios are: (i) *earliest* — event_year = announce_year + 0.5, the lower bound of any possible event time; (ii) *midpoint* — the current approximation; and (iii) *latest* — min(est_year_online, 2026), the upper bound. Under each scenario, the master Cox proportional-hazards model of Section A.10 is re-estimated and the principal coefficient HR_{Blue} is reported.

Pooled-failure HR is timing-sensitive

Table A.9: Pooled-failure Cox PH under three event-timing scenarios

Scenario	HR_{Blue}	95% CI	p-value	Decision
Earliest	1.482	$[1.209, 1.817]$	0.0002	Reject H_0
Midpoint	1.098	$[0.898, 1.342]$	0.361	Fail to reject H_0
Latest	0.947	$[0.774, 1.159]$	0.596	Fail to reject H_0

The pooled HR_{Blue} varies by 45.5% (range 0.95–1.48) across the three scenarios, with statistical significance achieved only under the earliest specification. This timing-fragility implies that the pooled-failure outcome cannot bear an unqualified causal interpretation under interval-censored uncertainty.

Cancellation-specific HR is timing-robust

Decomposing the pooled outcome into cancellation-only ($n_{events} = 100$), on-hold-only ($n = 852$), and decommissioned-only ($n = 48$) yields a substantially richer pattern.

Table A.10: Cancellation-only Cox PH under three event-timing scenarios

Scenario	HR_{Blue}	95% CI	p-value	Decision
Earliest	2.677	$[1.603, 4.470]$	0.0002	Reject H_0
Midpoint	2.276	$[1.368, 3.787]$	0.002	Reject H_0
Latest	1.889	$[1.124, 3.173]$	0.016	Reject H_0

Cancellation-specific HR_{Blue} remains in the range $[1.89, 2.68]$ across all three scenarios, with $p < 0.05$ uniformly. The sign-direction and statistical significance of the Blue-versus-Green cancellation hazard differential are therefore *robust* to event-timing uncertainty.

Table A.11: On-hold-only Cox PH under three event-timing scenarios

Scenario	HR_{Blue}	95% CI	p-value	Substantive direction
Earliest	1.235	$[0.981, 1.554]$	0.072	Blue > Green (borderline)
Midpoint	0.901	$[0.718, 1.131]$	0.370	Null
Latest	0.773	$[0.615, 0.972]$	0.027	Green > Blue (significant)

On-hold-specific HR_{Blue} exhibits a sign-flip between the earliest and latest scenarios, with the latter producing a statistically significant differential in the opposite direction from the former. The on-hold component, which accounts for 85% of the pooled failure events, is therefore the source of the pooled HR's timing-fragility.

Substantive implications

The interval-censored decomposition supports three substantive qualifications of the thesis-narrative:

First, the principal empirical claim of Section A.10 — that blue hydrogen projects exhibit a higher cancellation hazard than green hydrogen projects — survives the timing-sensitivity test. The cancellation-specific HR_{Blue} is at least 1.89 under the most conservative latest assumption and reaches 2.68 under the earliest assumption, with $p < 0.05$ uniformly.

Second, the cause-specific Cox decomposition of Section A.7 is methodologically not optional but *required*. Without it, the pooled hazard would suggest no Blue-vs-Green differential under the midpoint assumption, while the underlying competing risks contain a robust effect on cancellation and a timing-fragile effect on on-hold that approximately cancel in the pooled estimate.

Third, the v7 working-paper claim that "blue projects don't pause, they terminate" — which relies on the on-hold component being null — must be qualified. Under the earliest assumption, $HR_{Blue,on\text{-}hold} = 1.235$ is borderline positive; under the latest assumption, it is significantly negative. The original claim was identified under a specific timing assumption that the broader interval-censored analysis cannot uniformly defend.

Methodological note for downstream papers

For publication-grade replication of any of the thesis's hazard-model results, three reporting standards are recommended:

1. Always report the cause-specific decomposition alongside any pooled-failure outcome.

2. For any pooled-failure point estimate, accompany it with the earliest/midpoint/latest sensitivity range.

3. Where the substantive interpretation depends on the on-hold component or its complement, explicitly state that under interval-censored timing uncertainty the on-hold-specific estimate is fragile, and qualify the interpretation accordingly.

A formal interval-censored Cox proportional-hazards specification, as developed in [82], would replace the point-sensitivity reported here with a joint inference incorporating the full timing-interval. This is a natural extension for the publication paper version of the present analysis but is beyond the scope of the present thesis.

A.14 Falsifiable mechanism table: full inventory of pre-registered predictions

Methodological framing. The mechanism table catalogued in this appendix presents the five-channel framework of Chapter 3 as an organising taxonomy linking theoretical structure to falsifiable empirical predictions. The mechanisms are interpretive units of theoretical organisation, not structurally separable empirical channels. Appendix A.15 reports three formal identification tests demonstrating that the channels cannot be statistically discriminated from one another in the project-level cross-section: pooled regression with continuous π and σ proxies cannot detect distinct loadings ($p = 0.79$–0.84 on loading equality); channel-stratified offtake decomposition shows uniform offtake effects across μ, σ, and η proxies (likelihood-ratio $p = 0.19$); event-study with exogenous shocks cannot reject equality of π-event and σ-event responses ($p = 0.96$). The empirical signatures predicted by each mechanism (cancellation-hazard sign, magnitude, cross-stratum heterogeneity) are robust where confirmed; their channel-attribution to a single dominant mechanism is empirically underdetermined among observationally equivalent alternatives. The retrospective "confirmed" verdicts in the tables below should therefore be read as "empirical signature consistent with the predicted mechanism, while acknowledging that signatures consistent with alternative co-operating mechanisms remain admissible."

This appendix presents the full inventory of mechanism predictions catalogued in the mechanism-falsifiability framework of Section 3.8. Thirteen mechanism predictions are presented in tabular form: five theoretical channels (Part A), ten specific policy mechanism tests (Part B), and three methodological mechanisms (Part C). For each prediction, the table reports the a priori theoretical expectation, the observable implication, the empirical test design, the falsification criterion as a numerical threshold or sign condition, and the retrospective assessment of whether the prediction would have survived the empirical test had it been pre-registered.

The honesty of the retrospective assessment is essential to the methodological credibility of the framework. Predictions are classified as: **Confirmed (C)** — prediction matches data in sign and magnitude; would have survived pre-registration; **Partial (P)** — prediction matches in sign but not magnitude, or in one specification but not all; would have required qualification; **Falsified (F)** — prediction does not match data; would have led to mechanism rejection; **Untested (U)** — empirical test not yet conducted at required identification level. The two falsifications are themselves substantive contributions to the literature on policy-design effectiveness.

Table A.12: Theoretical channel predictions

ID	A priori prediction	Empirical test	Falsification criterion	Result	ID-le
A1 μ	$\partial z^*/\partial\mu < 0$; PTC reduces hazard 2–6pp	DiD on 45Q, China FYP	DiD positive or null in three specs	C	L3.a+
A2 σ	$\partial z^*/\partial\sigma > 0$; offtake reduces hazard in high-σ sectors only	Offtake IPWRA $\times$ sectoral	Uniform across sectors or inverse-correlated with σ	C	L3.a-
A3 ρ	$\partial z^*/\partial\rho > 0$; deadline-based instrument selection-driven not realisation-driven	Track-1 heterogeneity by pre-treatment cost-efficiency	ATT uniform or concentrated in low-efficiency stratum	C	L3.a+
A4 η	$\partial z^*/\partial\eta < 0$; coordination instruments dominate isolated subsidies	Cross-juris DiD: China vs 45Q	China ATT not larger than 45Q controlling for value	P	L3.a (juris) +
A5 κ	$\partial z^*/\partial\kappa > 0$ (perverse direction for compliance burden)	DDD on US-Green 45V NPRM + DiD on EU IF	DDD null/negative; IF positive/null	C (NPRM); P (IF)	L3.a+

Part A: Five theoretical channels (real-options framework)

Part B: Ten specific policy mechanism tests

Table A.13: Specific policy mechanism test predictions

ID	Instrument	A priori prediction	Falsification criterion	Result
C1	US 45Q	ATT $\in [-2, -6]$pp	95% CI contains 0 in both SA and BJS	C (-3.4pp; $M^* = 0.2$)
C2	China 14th FYP	ATT $\in [-3, -7]$pp; $M^* > 0.5$	ATT $\leq$ 45Q in absolute value, or $M^* < 0.5$	C (-4.5pp; $M^* = 1.5$)
C3	EU Innovation Fund	ATT $\in [-2, -5]$pp	Joint null across 6 estimators	**F (joint null)**
C4	EU CBAM (transitional)	ATT $\in [-1, -3]$pp on green	Null in majority of 8 specifications	**F (joint null)**
C5	US 45V three-pillars NPRM	DDD $\in [+0.10, +0.30]$ on US-Green	DDD negative, null, or below placebos	C (+0.285)
C6	UK Track-1	ATT < 0 + concentrated in high-efficiency stratum	Uniform or low-efficiency concentration	C (-1.8pp; 78% concentration)
C7	Pre-FID off-take	ATT $\in [-8, -15]$pp + sectoral σ-interaction	Null in 3+ of 5 estimators, or $\delta_{\text{Oster}} < 2$	C (-11.3 to -13.2pp; $\delta = 20.23$)
C8	Sponsor frailty	Frailty variance > 0 for diversified sponsors	Variance not distinguishable from zero	C (var = 0.34)
C9	TVP $\beta_{\text{int}}(t)$	Intensification 2010–2024 + 2020 break + OOS dominance	Flat trajectory or M1 OOS not dominated	C ($-0.46 \rightarrow -1.49$; DM-HLN = +5.59
C10	Sample stability	S&P HR positive significant, attenuated $\leq$ 80% from v7	Null in S&P or magnitude similar to v7	C ($13.19 \rightarrow 2.30$; cancellation-specific rob

Part C: Three methodological mechanisms

Summary distribution

Across the 18 mechanism predictions (5 theoretical channels + 10 specific tests + 3 methodological), the falsification-status distribution is: **Confirmed** = 14; **Partial** = 2 (A4 within-jurisdiction untested; A5 EU IF qualified); **Falsified** = 2 (C3 EU IF magnitude

Table A.14: Methodological mechanism predictions

ID	A priori prediction	Empirical test	Falsification criterion	Result
D1	Regime-dependent threshold: $\beta_{\text{int}}(t)$ shifts at 2020 boundary	GAS-TVP structural break detection	Flat trajectory or no break	C
D2	Sparse-event TVP stability: M3 OOS dominates M1 in V3 rolling	DM-HLN test on log-loss	M3 not statistically lower than M1 in V3	C (DM-HLN = +5.59)
D3	Event-timing-conditional: cancellation HR timing-robust, pooled timing-fragile	Pijler 48 interval-censored sensitivity	Pooled more stable than cancellation, or both stable	C (range 41% vs 47% with sign-flip

null; C4 CBAM weak transmission). The two falsifications are joint-informative nulls across multiple orthogonal estimators and constitute substantive contributions in their own right; they are presented as such in Chapter 10 on the scientific value of negative findings.

The pre-registration discipline established in this appendix is the format mandated for all mechanism claims in the dissertation and in the publication papers derived from it. The full prose treatment of each prediction, including the substantive interpretation and the empirical justification, is available in the synthesis document `08_synthesis/MECHANISM_TABLE_v1_2026` of the project repository.

A.15 Identification robustness tests: pi, sigma, and channel separation

Motivation

Supervisor feedback on the v2.2 draft raised two methodological concerns: (i) the policy-credibility parameter π_t introduced in Section 3 as an interpretive layer for the time-varying coefficient $\beta_{\text{int}}(t)$ is not directly observable and may not be separately identifiable from general volatility σ_t in the project-level cancellation hazard, and (ii) the offtake-commitment mechanism analysed in Chapter 8 likely operates through multiple co-operating channels rather than uniquely identifying the σ-channel as the dominant operating mechanism. This appendix reports three formal identification tests that address both concerns through methodologically independent strategies. The results, taken together, support the framing of the five-channel mechanism taxonomy as a theoretically organising rather than empirically structurally separable framework. Implementation scripts, raw output, and analysis samples are archived at `06_thesis_extensions/13_identification_tests/`.

Test 1: Pooled identification with continuous proxies

Question. Can policy credibility (π) be empirically distinguished from general volatility (σ) in the project-level cancellation hazard? Specifically: do the interactions Blue $\times\pi$ and Blue $\times\sigma$ load on distinct loadings, or are they jointly indistinguishable from zero?

Method. Pooled logit event model on the Blue+Green sample (n = 2,989, events = 1,000, Blue = 272), with each project assigned the proxy values of its announcement month merged from the monthly master panel. The dependent variable is the broad failure indicator (cancelled, on-hold, or decommissioned). Proxies: π_t via the Baker-Bloom-Davis US Economic Policy Uncertainty index (USEPUINDXD) [10]; σ_t (general) via VIX; σ_t (carbon-specific) via the exponentially weighted moving-average volatility of EUA carbon returns. Controls include log capacity, capex-support indicator, region fixed effects, and year-of-announcement.

Diagnostics. Proxy correlations are moderate ($\rho_{\pi,\sigma_{\text{general}}} = 0.37$; $\rho_{\pi,\sigma_{\text{carbon}}} = -0.34$). All Variance Inflation Factors for the interaction terms are between 1.2 and 2.3, well below the 5.0 problem threshold. Multicollinearity is therefore not the limiting factor.

Results. The joint Wald test of $(\text{Blue} \times \pi, \text{Blue} \times \sigma_{\text{general}}) = 0$ yields $\chi^2(2) = 0.07$, $p = 0.964$; the carbon-specific specification yields $\chi^2(2) = 0.07$, $p = 0.965$. The test of equal loadings (Blue $\times\pi$ versus Blue $\times\sigma$) yields $p = 0.787$ and $p = 0.836$ respectively. Both interaction coefficients are individually non-significant ($p > 0.79$) in all specifications. Single-channel benchmark models reveal that σ has a massive direct effect (ΔAIC $=$ -277.67 for VIX, -57.20 for carbon EWMA), while π has only a marginal direct effect ($\Delta\text{AIC} = -0.83$, $p = 0.090$).

Conclusion. Policy credibility π and volatility σ are not empirically separable in this design. The failure is fundamental rather than collinearity-driven: at the project-level cross-section, the BBD-EPU and EWMA-volatility proxies do not generate identifying variation in the Blue-carbon-conditional event hazard.

Test 2: Offtake mechanism decomposition by channel proxy

Question. Does offtake-commitment operate primarily through the σ-channel (as the strict reading of the Chapter 8 sectoral-heterogeneity claim implies), or does it operate through multiple co-operating channels?

Method. Stratified logit event model on the Blue+Green sample restricted to projects with non-missing primary end-use sector (n = 2,098, events = 680, offtake-committed = 266). End-use sector serves as a channel proxy: μ-proxy (chemical feedstock + refinery feedstock; fixed-price contracts expected) n = 489; σ-proxy (power + heat + industry-other; indexed pricing expected) $n = 761$; η-proxy (transport + gas grid; multi-counterparty coordination expected) $n = 848$. The interaction model includes both offtake × channel and Blue × channel terms.

Results. Per channel (separate stratum regressions, with all controls):

Channel	n	Offtake β	SE	p	Odds ratio
μ-proxy	489	−0.848	0.306	0.006	0.43
σ-proxy	761	−1.006	0.304	0.001	0.37
η-proxy	848	−1.566	0.397	< 0.001	0.21

The likelihood-ratio test of offtake × channel heterogeneity yields $\chi^2(1) = 1.74$, $p = 0.188$, which cannot reject homogeneity at $p < 0.10$. A separate analysis of Blue × channel heterogeneity yields $\chi^2(1) = 2.90$, $p = 0.089$, marginally rejecting homogeneity, with the Blue-Green differential concentrated in σ-proxy ($\beta = +0.46$, $p = 0.10$) and minimal in μ-proxy ($\beta = +0.19$, $p = 0.63$).

Conclusion. The offtake effect is robustly negative across all three channel-proxies and the central Paper 3 / Chapter 8 empirical finding therefore survives. However, the three channel-effects are statistically indistinguishable, and the strongest effect is in η-proxy (coordination) rather than σ-proxy. Offtake-commitment empirically operates as a multi-channel intervention. The Blue-Green hazard differential, by contrast, does load predominantly on the σ-proxy stratum (consistent with the carbon-conditional payoff mechanism of Chapter 7), suggesting that vulnerability and protection operate through different dominant channels.

Test 3: Event-study identification via exogenous shocks

Question. If pooled cross-sectional variation cannot separate π from σ (Test 1), do specific exogenous events that move primarily one but not the other provide the required identifying variation?

Method. Event-study logit on the Blue+Green sample (n = 3,097, events = 1,055) with event-window dummies (6 months either side) for four events: EU Green Deal announcement (11 December 2019, π^+); COVID-19 pandemic onset (11 March 2020, σ^+); Russian invasion of Ukraine (24 February 2022, σ^+); Inflation Reduction Act passage

(16 August 2022, π^+). The classification of events into π and σ types follows [10] and [13]. Placebo tests using 12-month-prior windows assess pre-event selection.

Results. The pooled π-window versus σ-window specification yields Blue $\times\pi$-window $\beta = -0.723$ ($p = 0.036$) and Blue $\times\sigma$-window $\beta = -0.111$ ($p = 0.747$). The joint Wald test rejects the null of both interactions equal to zero ($\chi^2(2) = 5.87$, $p = 0.053$). However, the test of equal loadings cannot reject equality ($p = 0.284$). The event-by-event decomposition reveals that the pooled π-window effect is driven by the IRA event alone (Blue $\times$ IRA $\beta = -0.749$, $p = 0.050$): the EU Green Deal effect is not significant ($p = 0.61$), and the COVID and Ukraine effects are not significant ($p = 0.25$ and $p = 0.74$ respectively). The combined Wald test of (Blue $\times$ Green Deal + Blue $\times$ IRA) = (Blue $\times$ COVID + Blue $\times$ Ukraine) yields $\chi^2(1) = 0.00$, $p = 0.960$. The placebo test (12 months prior to each event) yields $\chi^2(2) = 0.82$, $p = 0.664$, consistent with the absence of pre-event drift.

Conclusion. The single significant event-effect (IRA) is more consistent with the μ-channel interpretation (direct 45Q tax credit subsidy raising the expected payoff for CCS-equipped Blue projects) than with the π-channel interpretation (regime credibility shift), particularly because the other π-event in the design (EU Green Deal) shows no effect. The combined test cannot reject equality of π-event and σ-event responses at the population level. Identification via exogenous variation does not produce structurally distinct π and σ effects.

Joint interpretation and implications for thesis claims

The three identification tests converge on a single conclusion: credibility and payoff-uncertainty empirically appear observationally entangled in the project-level data available for this dissertation. Test 1 (pooled cross-section with continuous BBD-EPU and EWMA-volatility proxies), Test 2 (channel-stratified DiD with μ, σ, and η proxies via end-use sector), and Test 3 (event-study with four well-identified exogenous shocks) are three methodologically independent strategies, and all three converge on the same finding. The five-channel mechanism taxonomy of Chapter 3 is therefore a coherent theoretical organising framework but not an empirically structurally separable identification claim in this design. Proposition 3.2 (Section 3.6) is preserved as a model-implied comparative static and functions in the dissertation as a *disciplined interpretive layer* for the $\beta_{\text{int}}(t)$-signal documented in Chapter 7, not as a directly identified causal channel for π separately from σ. The combination of formal comparative statics with explicit non-separability is taken to strengthen the framework: it aligns the strength of the theoretical claim with the strength of the empirical identification the observational design can support, and it identifies the boundary between what is empirically privileged and what remains observationally underdetermined among co-operating mechanisms.

Implications. (i) The credibility-conditional threshold claim of Section 3.6 (Proposition 3.2) cannot be empirically defended in its strict formal form. The claim is reformulated as an interpretive layer: the time-varying coefficient $\beta_{\text{int}}(t)$ documented in Chapter 7 captures a regime-conditional intensification of the Blue-Green hazard differential whose channel-attribution (between π, σ, and expectations dynamics) cannot be empirically privileged among observationally equivalent interpretations. (ii) The offtake-commitment mechanism finding survives empirically, but the strict reading that the cross-sectoral heterogeneity uniquely identifies the σ-channel does not. The Paper 3 framing is correspondingly softened to dominant interpretation rather than unique identification. (iii) The IRA event-effect documented in Test 3 is independently informative for the carrot-policy thesis of Paper 2 as direct evidence for μ-channel operation, independent of the π-σ separation question.

The methodological discipline-by-failure narrative of Chapter 10 is empirically grounded by this appendix. We have attempted three different identification strategies for the same underlying separation question, and all three have generated null results on the structural separation. This is the most informative empirical finding the design supports: not that the channels are absent (the multi-channel intervention is empirically real), but that observational data on irreversible clean-technology investment decisions does not contain the exogenous variation in only π (or only σ) that structural identification would require. A full structural-identification study targeting this question would require either regression-discontinuity designs around constitutional carbon-budget amendments, exogenous regime credibility shifts (such as election-driven climate-policy reversals), or natural-experiment volatility shocks isolated from policy-credibility content. Such a study is a legitimate stand-alone publication and is identified as the natural follow-up research agenda in Chapter 10.